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Asset Management Techniques under ESG constraints: Efficient Frontier Analysis and Portfolio Weights Forecasting with a Kalman Filter

Tesi di Laurea Magistrale in
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Abstract

This thesis analyses the integration of Environmental, Social and Governance (ESG) factors into portfolio optimization, combining traditional asset management frameworks with sustainability constraints. The aim is to provide a quantitative assessment of how different ESG integration methods influence portfolio efficiency and robustness over time.

The empirical study is based on the S&P 500 constituents from 2015 to 2024, with ESG data from Refinitiv applied across several frameworks: Constrained optimization, where ESG scores act as portfolio restrictions; Negative screening, excluding low-rated sectors; Best-in-Class selection, favoring top ESG performers within each sector and shorting the lowest; and Linear limits on low rated assets, a softer constraint imposing upper caps on low-score assets. Each configuration is analyzed under long-only and limited short-selling regimes to evaluate diversification trade-offs. A forecasting extension introduces a Kalman Filter (KF) to model the dynamic evolution of minimum-variance portfolio weights, assessing predictive accuracy and out-of-sample performance over 2023–2024.

Results from the first chapter show that short-selling enhances frontier efficiency and diversification, especially in higher-return regions, but increases computational instability. Across all models, stronger ESG constraints generate a negative efficiency premium, with tighter rules reducing Sharpe and Calmar ratios and raising volatility. The Best-in-Class with short-selling and Constrained optimization with mild ESG limits achieve the best balance between return and stability.

The second chapter shows that the Kalman Filter captures real portfolio dynamics, producing smoother and more coherent weight trajectories with lower volatility but slightly weaker returns. Its adaptive smoothing mitigates short-term shocks but lags in fast-changing markets. Overall, the research finds that rigid ESG mandates entail measurable efficiency losses, while adaptive, data-driven and moderately flexible approaches, especially those combining dynamic filtering and selective short exposure, can align sustainability with robust portfolio behavior.

Keywords: ESG, Sustainable Finance, Portfolio Optimization, Efficient Frontier, Short-Selling, Kalman Filter, Linear Constraints.

Abstract in Italiano

La tesi analizza l'integrazione dei fattori Ambientali, Sociali e di Governance (ESG) nei processi di ottimizzazione di portafoglio, combinando modelli tradizionali di asset management con vincoli di sostenibilità. L'obiettivo è fornire una valutazione quantitativa e sistematica dell'impatto delle diverse tecniche di integrazione ESG sull'efficienza, la diversificazione e la robustezza del portafoglio nel tempo.

L'analisi empirica si basa sui titoli dell'indice S&P 500 nel periodo 2015–2024, utilizzando dati ESG di Refinitiv applicati a diversi approcci: ottimizzazione vincolata, in cui i punteggi ESG fungono da restrizioni di portafoglio; esclusione negativa, che elimina interi settori a basso rating; selezione Best-in-Class, che privilegia le imprese con punteggi più elevati e consente la vendita allo scoperto delle peggiori; e limiti lineari, versione più flessibile con un tetto massimo sui titoli a basso punteggio ESG. Ogni configurazione è analizzata sia in regime *long-only* sia con *short-selling* limitato. Una sezione successiva introduce un'estensione previsionale basata sul Filtro di Kalman (KF), applicato all'evoluzione dinamica dei pesi dei portafogli a varianza minima, per valutare l'accuratezza predittiva e la performance fuori campione (2023–2024).

I risultati del primo capitolo mostrano che la vendita allo scoperto migliora l'efficienza e la diversificazione della frontiera, ma aumenta l'instabilità computazionale. Vincoli ESG più stringenti generano un "premio negativo di efficienza", riducendo gli indici di Sharpe e Calmar e incrementando la volatilità. Tra i modelli statici, l'approccio Best-in-Class con *short-selling* e le ottimizzazioni con vincoli moderati ottengono il miglior equilibrio.

Il secondo capitolo evidenzia che il Filtro di Kalman riproduce le principali dinamiche di portafoglio, producendo traiettorie più stabili e coerenti, con volatilità ridotta ma rendimenti leggermente inferiori. Nel complesso, la ricerca dimostra che vincoli ESG rigidi generano costi misurabili in termini di performance, mentre strategie adattive, *data-driven* e flessibili, soprattutto quelle che combinano filtraggio dinamico e posizioni di *short-selling*, possono conciliare sostenibilità e stabilità del portafoglio.

Parole chiave: ESG, Finanza sostenibile, Ottimizzazione di portafoglio, Frontiera efficiente, Vendita allo scoperto, Filtro di Kalman, Vincoli lineari.

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Introduction

The growing importance of Environmental, Social and Governance (ESG) considerations in financial markets has stimulated a profound transformation in asset management practices. In recent years, investors, asset managers and regulators have progressively aligned portfolio construction processes with sustainability principles, recognizing that ESG factors can materially influence long-term performance, risk profiles and reputational integrity. This paradigm shift reflects the increasing awareness that financial value and sustainable behavior are not mutually exclusive, but can instead be complementary dimensions of responsible investment and risk management.

At the same time, the rapid evolution of quantitative finance has made possible to translate qualitative sustainability objectives into measurable and operational portfolio constraints. Traditional portfolio optimization techniques, such as the mean–variance framework introduced by Markowitz, are now being adapted to include ESG metrics as decision variables, allowing investors to explicitly account for sustainability preferences alongside expected returns and risks. This integration, however, raises fundamental questions about the efficiency and stability of such portfolios: to what extent can ESG objectives coexist with financial optimization? Do sustainability constraints compromise diversification, or can they instead enhance resilience and long-term robustness? These are the guiding questions that motivate this thesis.

This work aims to explore and compare various asset management strategies that integrate ESG scores into the portfolio optimization framework. Drawing inspiration from the methodologies proposed by Barucci, Azzone, and Stocco [1], the study systematically investigates the implications of ESG inclusion from multiple quantitative perspectives, combining traditional mean–variance theory with modern sustainability constraints. The thesis is entirely empirical: all models are implemented, tested and compared using real market data, with the objective of producing concrete evidence on the performance, risk and predictive power of ESG-integrated portfolios.
Both the static frontier analysis and the forecasting weights computations are carried out in **Matlab**.

The empirical analysis is based on the S&P 500 constituents observed between 2015 and 2024, applying a rigorous data-filtering process to ensure quality, liquidity and comparability of the sample. ESG scores, sourced from Refinitiv, are incorporated into the optimization process as:

- **constraints**, enforcing minimum ESG thresholds at portfolio level;
- **preference weights**, to tilt the allocation toward sustainable firms;

Models are estimated on the in-sample period 2015–2022 and validated out of sample over 2023–2024 using a dynamic forecasting approach based on the Kalman Filter.

The research compares different ESG integration techniques that represent distinct philosophies of sustainable investing:

- **Constrained optimization**, where ESG scores act as formal portfolio restrictions;
- **Negative screening**, which excludes low-ESG sectors to quantify the cost of exclusionary policies;
- **Linear limits on low-score assets**, representing soft and flexible version of the Best in Class selection portfolio;
- **Best-in-Class selection**, prioritizing the top ESG performers within each sector and allowing limited short-selling on the lowest-rated firms.

Each configuration is analyzed under both long-only and limited short-selling regimes to evaluate diversification trade-offs, risk exposure and efficiency losses or gains associated with sustainability constraints.

Key questions addressed in this thesis include:

- How do ESG constraints affect the efficiency of the portfolio frontier?
- What is the trade-off between sustainability integration and risk-adjusted performance?
- Can limited short-selling mitigate the costs of stringent ESG mandates?
- Do dynamic models like the Kalman Filter improve the stability and predictive accuracy of portfolio weights?

The first part of the thesis focuses on static portfolio construction, analyzing how different ESG integration techniques reshape the efficient frontier, alter diversification, and impact key performance indicators such as the Sharpe ratio, volatility and maximum drawdown.

Special attention is given to short-selling policies, which often enhance performance but may increase instability and concentration risk. This section provides a comprehensive assessment of how sustainability constraints influence portfolio efficiency under different configurations.

The second part introduces a dynamic component by applying a **Kalman Filter (KF)** to the minimum-variance portfolio weights. This extension allows to forecast the temporal evolution of optimal weights and to evaluate whether the predictive model can reproduce real portfolio trajectories with economic coherence and reduced volatility. By comparing the out-of-sample performance of the Kalman-based forecasts with the actual portfolios, the analysis assesses the practical value of adaptive filtering techniques for sustainable asset management.

The thesis serves two complementary purposes.
First, it quantifies the empirical impact of ESG integration on portfolio efficiency, identifying the conditions under which sustainability constraints are compatible with performance.
Second, it evaluates whether a dynamic, model-based approach can enhance the stability and forecasting capability of ESG-aware portfolios over time. Together, these two perspectives provide a unified view of sustainability integration in both static and predictive settings, offering insights that are relevant for asset managers.

All analyses are conducted on two main portfolio sets: the **standard frontier**, which includes the top 100 S&P 500 stocks by market capitalization, and a **restricted frontier**, composed of the top 50 stocks with the highest ESG scores. This dual structure allows for an additional layer of comparison, isolating the effects of universe concentration and the role of ESG quality on efficiency.

Ultimately, this thesis contributes to the ongoing academic and professional debate on sustainable finance by combining theoretical rigor with empirical validation. It demonstrates that ESG integration—if properly designed and supported by adaptive quantitative tools—can balance ethical responsibility with financial efficiency, offering a viable path toward data-driven and resilient sustainable investing.

1 | Asset management techniques considering ESG scores

1.1. Asset Management

Asset management is the professional discipline devoted to the allocation of capital across financial instruments with the aim of achieving a balance between expected return, risk and liquidity.
It spans a wide range of approaches, from traditional security selection to quantitative optimization and ultimately seeks to construct portfolios that respect investor risk–return preferences while using information efficiently.
The field builds on **Modern Portfolio Theory (MPT)**, which demonstrates how diversification optimizes the risk–return trade-off. Subsequent advances include the **Capital Asset Pricing Model (CAPM)**, multifactor models and robust optimization techniques that adapt to market conditions and mandate constraints.

Over time, asset management has evolved from discretionary and fundamental analysis towards increasingly data-driven and systematic approaches. Quantitative methods, including machine learning and econometric forecasting, now support dynamic portfolio rebalancing and the identification of latent market factors.
Institutional investors such as pension funds, sovereign wealth funds and insurance companies play a central role, as they often act as both long-term capital providers and key drivers of innovation in portfolio construction methodologies.

From policy to portfolio. Operationally, the asset management process follows a well-defined loop:

- **Define policy:** specify objectives, benchmarks, investable universe and risk limits such as tracking error, concentration, liquidity or turnover;
- **Prepare data:** collect historical prices, fundamental ratios, risk factors and alternative datasets;

- **Construct the portfolio:** apply optimization criteria (mean–variance, minimum variance, downside risk, factor exposure) and translate policies into investable allocations;
- **Monitor and report:** assess ex-post performance, attribution and compliance with risk limits and investment guidelines.

This structured cycle ensures the alignment between investor mandates and market dynamics, while also facilitating transparency and accountability in the investment process.

1.2. ESG Scores and Sustainable Investing

In recent decades, the scope of asset management has expanded beyond purely financial metrics. The integration of *Environmental, Social and Governance (ESG)* considerations has become a structural feature of global investment practice. ESG factors capture non-financial aspects of corporate behaviour that can shape long-run performance, cost of capital and reputational risk, connecting investment choices to broader societal outcomes.

- **Environmental (E):** climate risk, carbon emissions, energy efficiency, waste and resource use;
- **Social (S):** human rights, labour standards, diversity and inclusion, product responsibility, community impact;
- **Governance (G):** board independence, shareholder rights, transparency, ethics, and internal controls.

ESG's rise is driven by three converging forces:

- investors increasingly view ESG as a source of risk and opportunity with measurable financial materiality;
- regulation and disclosure frameworks (such as the EU Taxonomy and SFDR) promote standardization and transparency;
- market demand catalyses the development of innovative products, including ESG indices, decarbonized benchmarks, and impact investment funds.

In practice, ESG integration can be achieved through several levers: **Best-in-class selection, Score-based optimization, Exclusionary screening** as explained in the Introduction, represent the possible ways to integrate these scores.

Risk channels and horizons. ESG factors affect both expected returns and risks through transition dynamics (policy, technology and demand shifts), physical impacts (acute events and chronic trends) and legal or reputational exposure. Their materiality depends on the investment horizon: shorter horizons emphasize event-driven and sentiment risk, while longer horizons amplify structural and compounding effects.

Finally, ESG scores themselves are not static measures but dynamic reflections of corporate behavior and external conditions. Differences across rating providers highlight the methodological challenges of quantifying sustainability, underscoring the need for transparency, comparability and continuous data quality improvement in sustainable finance.

1.2.1. ESG Refinitiv Scores

In this thesis, sustainability is proxied *exclusively* by composite ESG *scores* from Refinitiv. In practice, the provider's ESG scores are taken for the investable universe and keep them fixed when constructing frontiers and running the out-of-sample evaluations.[1] This choice mirrors common practice in asset management—where a single, comparable scalar is often required for mandates—and ensures coverage, simplicity and straightforward integration into quadratic programs.
The original scores range from 0 to 100, where lower values indicate poor performance across all three ESG dimensions, while higher values reflect stronger sustainability performance: in this thesis they are normalized as in "Asset management with an ESG mandate" [1] removing the mean of the available ESG scores, since ESG data include heterogeneous variables (binary indicators, percentages, financial ratios, qualitative scores) whose magnitudes differ widely.

Benefits and limitations. On the one hand, scores summarize large indicator sets into a bounded, comparable ranking that can be embedded as thresholds, constraints or selection rules in mean–variance problems.
On the other hand, scores inherit well-known opacity and methodological drift: rating agencies employ proprietary, sector/country-specific transformations; the final ESG value is typically a percentile rank; and historical series can be rewritten as methodologies evolve. Recent evidence (Del Vitto, Marazzina, Stocco, 2023) [6] shows that machine-learning models can *partially* reconstruct Refinitiv's ESG assessment rules with high accuracy, yet a residual, *unlearnable* noise remains and periodic methodology updates imply that scores may change even ex post.

[1] See the data filtration pipeline described later in this chapter.

Methodological stance. Given these features, ESG are treated as a summary, benchmark-relative score suitable for constraints and universe design; they perform all comparisons within a single-provider framework to avoid cross-provider divergence effects and emphasize robustness across universes, both full universe and Top-ESG subsets.

1.3. Literature Review

ESG as a portfolio constraint. Research that embeds an ESG target directly in the optimizer shows that efficiency can be preserved when the target is calibrated to the benchmark and to how markets price ESG.
A key reference is Azzone, Barucci and Stocco, “Asset Management with an ESG Mandate” [1]. Their practical guidance is clear:

- treat the ESG target as a formal constraint with explicit tolerances;
- calibrate it to the benchmark to avoid unintended and persistent active bets;
- manage implementation frictions (turnover, caps, liquidity) so that benefits survive trading costs.

Their work is developed in a Tracking Error Variance (TEV) setting, but the same logic applies to mean–variance.
Once the ESG constraint binds, the frontier may shift; with proper calibration and guardrails (sector neutrality, caps), the loss in efficiency stays small.
This thesis adopts a mean–variance setup (not TEV) and inserts the ESG target as a portfolio-level score constraint.
It then observes how the frontier moves when the target is tightened or relaxed, when the investable universe changes (full sample versus a Top-50 subset by score), and when short selling is allowed or forbidden.
In short, the Azzone, Barucci and Stocco framework is used for design principles; the objective function differs, but the principles carry over directly.

Best-in-class selection. The Best-in-Class strategy developed in this thesis extends traditional ESG selection by introducing long positions on high-rated issuers and short exposure to low-rated ones, amplifying the ESG signal rather than merely reweighting within the universe.
De Spiegeleer et al., "ESG: A New Dimension in Portfolio Allocation" [5], highlights that portfolio outcomes depend on design specifics such as sector-relative versus absolute rankings, coverage and liquidity filters, rebalancing frequency and that, when these are sensibly calibrated, the efficient frontier remains close to the baseline.

In this work, these principles are applied through two consistent universes (full sample and Top-50 ESG subset) under the same mean-variance setup, so that performance differences can be traced to universe design and ESG polarity rather than to optimizer changes.
The reference is thus used to justify the sector-aware construction, the attention to coverage and rebalancing, and the long - short interpretation of Best-in-Class, which extends the Azzone–Barucci–Stocco framework to an active sustainability tilt.

Negative Screening Evidence on exclusions is mixed. Blitz and Swinkels, "Does Excluding Sin Stocks Cost Performance?" and "Is Exclusion Effective?" [2, 3], document that broad exclusion lists tend to raise tracking error, reduce diversification and may remove exposure to rewarded factors (value, profitability, low risk), while narrower and well-implemented rules limit these side effects and engagement can sometimes be more effective than general exclusions.
In this thesis, the negative screening approach is applied at the sector level, excluding entire industries characterized by structurally low ESG standards to evaluate the aggregate effect of sectoral bans on portfolio efficiency.
This setting allows to measure how diversification, volatility and active risk evolve when sustainability restrictions affect specific segments rather than single issuers.
In parallel, a complementary analysis is carried out using the Top-50 ESG subset in every portfolio analyzed, where only the highest-rated firms are retained.
This design isolates the role of universe concentration and evaluates how the efficient frontier and the stability of returns change when the investable universe is progressively restricted to the most sustainable assets. Together, these exercises reproduce the trade-offs discussed by Blitz and Swinkels between ESG purity, factor exposure and tracking error, providing a structured comparison across increasingly stringent sustainability filters.
Earlier in the analysis, different ESG score thresholds are tested to see how the portfolio reacts to changes in constraint strength and universe composition.
These preliminary tests provide the basis for the later studies on sector exclusions and the Top-50 ESG subset.
The references are mainly used to support the interpretation of results, especially regarding list breadth and factor exposure.

Adjacent but out of scope: climate metrics and alignment. Two carbon-focused approaches are relevant but not implemented here.
Andersson, Bolton and Samama, "Hedging Climate Risk" [7], design decarbonized indexes that cut the portfolio's carbon footprint (often by about half) while keeping tracking error very low—the "free option on carbon." Bolton, Kacperczyk and Samama, "Net-Zero

Carbon Portfolio Alignment" [4], align portfolios to science-based carbon budgets and forward-looking pathways with negligible tracking error.
Roncalli et al., "Portfolio Construction with Climate Risk Measures" [8], show that enforcing Scope 3 and alignment metrics increases model uncertainty and can materially change allocations.
In this thesis these strands are not used operationally: it does not optimize on footprint or intensity, does not impose alignment trajectories and does not add Scope 3-based constraints.
They are used conceptually to define scope (composite ESG scores are the chosen proxy) and to motivate future developments; no data or algorithms from these papers enter the empirical exercises.

Ratings opacity: implications for score-based designs. Del Vitto, Marazzina and Stocco, "ESG Ratings Explainability Through Machine Learning" [6], show that Refinitiv ratings are largely explainable by transparent models, but residual noise remains; methodologies differ by sector and country and historical series can be revised over multi-year windows.
In this thesis, the paper is used partially: the machine-learning pipeline is not replicated, but its findings support three choices:

- a single-provider setup for methodological consistency,
- a fixed score snapshot for the main experiments,
- a cautious reading of frontier shifts as conditional on one evolving methodology.

The contribution is therefore to interpretation and data policy, not to the optimization engine itself.

1.4. Data Filtration

The initial dataset consisted of all stocks included in the S&P 500 Index, focusing on the period from 2015 to 2019. To enhance the analysis, the ESG scores for these assets, available at the end of 2019, were included.

Similarly to the approach adopted by Barucci, Azzone and Stocco [1], the first filtering step involved selecting only those stocks for which ESG scores were available. The ESG values were then normalized by subtracting the mean score across all assets.

Subsequently, illiquid assets were removed: specifically, stocks that had not been traded for over a year, as well as those with prices equal to zero or constant over extended periods. This ensured the inclusion of only actively traded and liquid assets.

Then, only the assets with highest capitalization are considered: indeed, a portfolio frontier should be computed on a small amount of assets, otherwise most of the assets would have not significative weights. So, the minimum between 100 and the available filtered assets is taken.

After applying these filtration criteria, the dataset was reduced from the original 500 stocks to a final selection of **93 assets.**



Figure 1.1: Monthly Returns

1.4.1. Winsorization

Similarly to the approach proposed by Barucci, Azzone and Stocco [1], a winsorization procedure is applied to the returns data. Given a returns matrix $X \in \mathbb{R}^{N\times M}$ (assets in rows, dates in columns), the routine applies a per–asset winsorization with tail level α: for each row i, it computes the empirical percentiles at $\alpha/2$ and $1-\alpha/2$ and replaces values below (above) these cutoffs with the corresponding percentile. The output is a winsorized matrix of the same size as the input, where extreme observations have been clipped asset-by-asset. The parameter α controls the aggressiveness: in this case it is used $\alpha = 0.05$ and the outer 2.5% of each tail is clipped out. This improves the stability of the estimated moments on the 2015–2019 sample and prevents isolated extremes from dominating the covariance structure.

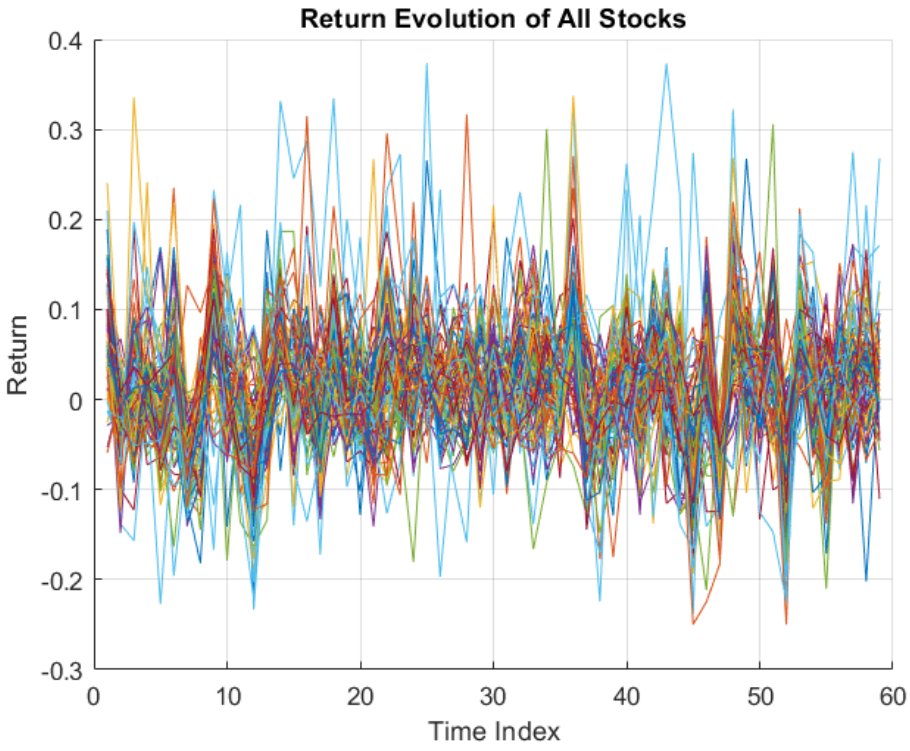


Figure 1.2: Monthly Winsorized Returns

Before vs. after winsorization. In the raw plot, several series display occasional sharp spikes (both positive and negative), producing thicker tails and wider dispersion. After winsorization, these extreme observations are clipped asset-by-asset, yielding visibly narrower bands around the center and smoother trajectories, while the bulk pattern of returns is preserved.

In practice, winsorization reduces the influence of outliers on moment estimates (means, variances and covariances) without altering the central tendency of each series.

1.5. Standard Frontier

1.5.1. Mean–Variance Frontier

We consider n risky assets with expected returns $\mu \in \mathbb{R}^n$ and covariance matrix $\Sigma \in \mathbb{R}^{n\times n}$, assumed positive definite. A portfolio is represented by a weight vector $w \in \mathbb{R}^n$ such that

$$\mathbf{1}^\top w = 1,$$

where $\mathbf{1}$ is the n-dimensional vector of ones. Depending on the specification of lower bounds, two cases are considered:

- a **long-only** configuration ($w_i \geq 0$ for all i);
- a **relaxed** configuration with limited short-selling ($w_i \geq -0.1$).

In both cases, portfolio return and variance are given by

$$\mu_p = \mu^\top w, \qquad \sigma_p^2 = w^\top \Sigma w.$$

1.5.2. Standard Frontier

The baseline optimization problem for a given target return μ_p is

$$\begin{aligned} \min_{w\in\mathbb{R}^n} \quad & w^\top \Sigma w \\ \text{s.t.} \quad & \mathbf{1}^\top w = 1, \\ & \mu^\top w = \mu_p, \\ & w_i \geq 0 \quad \forall i. \end{aligned}$$

If the non-negativity constraint is temporarily ignored, the analytical solution yields the familiar Markowitz coefficients:

$$A = \mathbf{1}^\top \Sigma^{-1}\mathbf{1}, \quad B = \mathbf{1}^\top \Sigma^{-1}\mu, \quad C = \mu^\top \Sigma^{-1}\mu, \quad D = AC - B^2 > 0,$$

and the unconstrained (reference) frontier:

$$\sigma_p^2(\mu_p) = \frac{C - 2B\mu_p + A\mu_p^2}{D}.$$

However, under the realistic constraint $w_i \geq 0$, the solution becomes a convex quadratic

program, solved numerically for each target return μ_p.
The resulting efficient frontier considering the long case is:

- **Piecewise quadratic**: as μ_p increases, some assets reach $w_i = 0$, leaving only the active set.
- **Smooth and stable**: no negative weights, ensuring economic interpretability and no leverage.
- **Dominated in variance** by the unconstrained case, since shorting could, in theory, improve diversification.

The long-only frontier thus represents the feasible benchmark for practical portfolio management — consistent with long-only mandates, mutual funds, and regulatory constraints. All subsequent ESG frontiers are benchmarked against this baseline.

1.5.3. Short-Selling

To test robustness, a relaxed configuration is introduced by allowing limited short positions up to 10% of total wealth per asset:

$$\begin{aligned} \min_{w \in \mathbb{R}^n} \quad & w^\top \Sigma w \\ \text{s.t.} \quad & \mathbf{1}^\top w = 1, \\ & \mu^\top w = \mu_p, \\ & w_i \geq -0.1 \quad \forall i. \end{aligned}$$

This configuration maintains overall feasibility while allowing small negative weights. The optimization remains a convex quadratic program, and the frontier preserves the same parabolic shape but expands slightly outward:

- The relaxation increases diversification and smoothness near high-return regions.
- Small short exposures allow the optimizer to hedge correlated assets more efficiently, slightly improving the risk–return trade-off.
- However, larger negative bounds (e.g., -0.2) tend to cause instability and concentration, while smaller constraint (e.g. -0.05) are too strict. So -0.1 is chosen as a more stable compromise.

Remarks

- The analytical expression

$$\sigma_p^2(\mu_p) = \frac{C - 2B\mu_p + A\mu_p^2}{D}$$

  serves only as a theoretical benchmark.

- All empirical results in this thesis are computed numerically under either the $w_i \geq 0$ or $w_i \geq -0.1$ constraint.

- The two configurations are directly compared to assess the marginal impact of limited shorting on frontier efficiency, stability and concentration.

1.5.4. Key Metrics

Let $\{r_t\}_{t=1}^T$ be portfolio returns sampled at frequency freq (e.g., freq = 12 for monthly, freq = 252 for daily) and let r_f be the *annual* risk–free rate.

Annualized Return (AR).

$$\bar{r} = \frac{1}{T}\sum_{t=1}^{T} r_t, \qquad \text{AR} = (1 + \bar{r})^{\text{freq}} - 1. \tag{1.1}$$

Interpretation. The annualized return measures the average rate of capital growth over a one-year horizon, assuming compounding rate. It captures the long-term profitability of the portfolio, abstracting from short-term volatility.

Annualized Volatility (AVol).

$$s_r = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}(r_t - \bar{r})^2}, \qquad \sigma_{\text{ann}} = s_r\sqrt{\text{freq}}. \tag{1.2}$$

Interpretation. Volatility quantifies the standard deviation of returns and serves as a proxy for total risk. Annualization allows consistent comparison across assets or strategies with different data frequencies.

Sharpe Ratio (SR). Convert the annual risk–free rate r_f to its per-period equivalent:

$$r_f^{(\text{per})} = (1 + r_f)^{1/\text{freq}} - 1, \tag{1.3}$$

then compute

$$\text{Sharpe} = \frac{\bar{r} - r_f^{(\text{per})}}{s_r} \sqrt{\text{freq.}} \tag{1.4}$$

Interpretation. The Sharpe ratio expresses the excess return earned per unit of total risk. Higher values indicate a more efficient risk–return trade-off. A ratio above 1 is generally considered good, while values above 2 are exceptional.
For simplicity, $r_f^{(\text{per})} = 0$ is considered.

Maximum Drawdown (MDD).

$$\text{NAV}_t = \prod_{i=1}^{t}(1 + r_i), \qquad \text{Peak}_t = \max_{1 \le s \le t} \text{NAV}_s, \tag{1.5}$$

$$\text{DD}_t = \frac{\text{NAV}_t - \text{Peak}_t}{\text{Peak}_t}, \qquad \text{MDD} = -\min_{1 \le t \le T} \text{DD}_t. \tag{1.6}$$

Interpretation. Maximum Drawdown measures the largest historical loss from a peak to a subsequent trough of the equity curve. It reflects the worst cumulative decline experienced by the portfolio and indicates downside risk exposure.

Calmar Ratio (CR).

$$\text{Calmar} = \frac{\text{AR}}{\text{MDD}}. \tag{1.7}$$

Interpretation. The Calmar ratio evaluates performance adjusted for drawdown risk, expressing how much annualized return is achieved per unit of maximum loss. It complements the Sharpe ratio by focusing on downside rather than symmetric volatility.

Limitations. Markets are non-stationary and regimes change. Metrics based on historical data assume stability of mean and variance, which may not hold in practice. Results should therefore be interpreted as indicative rather than predictive and complemented with robustness checks, sensitivity tests and out-of-sample validation.

1.6. Top ESG Scores Frontiers Analysis

In this section, the investable universe is restricted by ESG rank. Specifically, I construct a sequence of efficient frontiers by selecting the K assets with the highest ESG scores, with K progressively increased (e.g., $K \in \{10, 19, 28, \ldots, 93\}$). For each K, the classical mean–variance frontier is computed under the standard budget constraint.

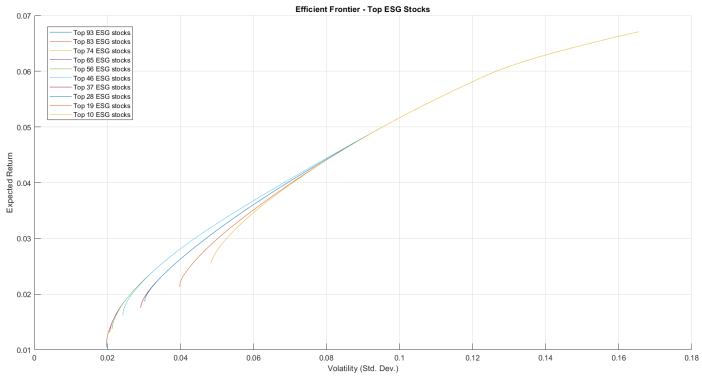


Figure 1.3: Standard Frontier considering Top ESG scores

As expected, frontiers obtained with smaller K are dominated by those with larger K, since enlarging the investable set expands the feasible region in (σ_p, μ_p) space and weakly improves efficiency.

Interpretation. When portfolios are built only from top-ESG names, the additional selection filter can entail a (potential) efficiency loss relative to the broader universe, as lower-ESG assets that could improve the risk–return trade-off are excluded. Empirically, the frontiers obtained with fewer assets are indeed inward with respect to those based on richer universes, corroborating the dominance result above.

1.6.1. Short Selling

To examine the role of short positions, I relax the no-short constraint $w_i \geq 0$ and allow limited short-selling by imposing a lower bound $w_i \geq -0.1$ for all assets.

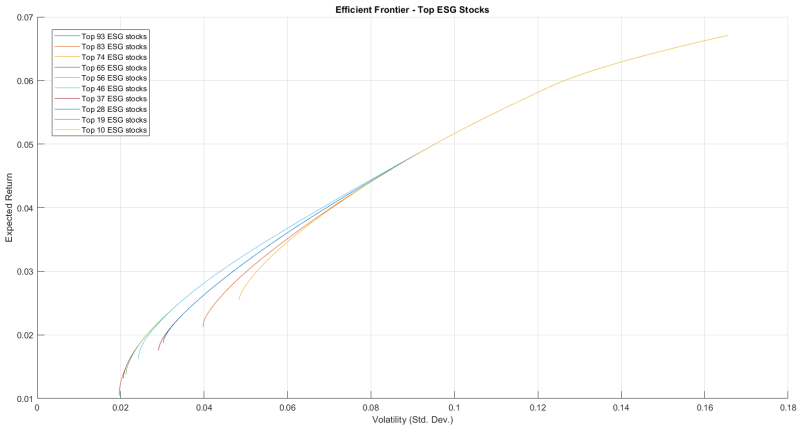


Figure 1.4: Standard Frontier considering Top ESG scores - Short selling

In practice, removing non-negativity entirely caused numerical instabilities in tracing the frontier; the limited-short bound restores well-posedness while preserving the economic intent of permitting modest short exposure.

1.6.2. Results

- Frontiers constructed from smaller ESG-selected sets remain dominated by those from larger sets, even when limited short-selling is permitted. This is consistent with the expansion of the feasible set when the number of assets increases.
- Relative to the no-short case, the limited-short frontier typically weakly dominates (or matches) the constrained one, reflecting the added flexibility from short positions.

Implication. The evidence suggests a possible "ESG restriction premium" in this setting: when the universe is narrowed to high-ESG names only, the attainable frontier may shift inward relative to the broader (standard) universe.

This section serves as a robustness check aimed at introducing a potential negative ESG impact effect. Reducing the number of assets entails a loss of diversification, making it difficult to determine whether observed effects arise from the smaller asset universe or from the ESG constraints themselves. Only the efficient frontier constructed with 50 assets will be considered in the next sections of this chapter in order to show a portfolio with less assets, but with high ESG Scores.

1.7. ESG Integration via Constrained Optimization

1.7.1. ESG-Constrained Frontier

Consider n risky assets with expected returns $\mu \in \mathbb{R}^n$ and covariance matrix $\Sigma \in \mathbb{R}^{n\times n}$ (positive definite). Let $\text{ESG} \in \mathbb{R}^n$ denote the vector of asset-level ESG scores and let $G \in \mathbb{R}$ be a minimum portfolio ESG threshold.

To integrate sustainability into the optimization, the portfolio must also achieve a minimum weighted ESG score:

$$\begin{aligned} \min_{w} \quad & w^\top \Sigma w \\ \text{s.t.} \quad & \mathbf{1}^\top w = 1, \\ & \mu^\top w = \mu_p, \\ & \sum_{i=1}^{n} w_i \,\text{ESG}_i \geq G. \\ & \underline{w} \;\leq\; w_i \;\leq\; 1 \quad \forall i \end{aligned}$$

with $\underline{w} = 0$ for long only cases and $\underline{w}$ = -0.1 for short selling ones.

Implications. The additional inequality constraint reduces the feasible set of portfolios. The resulting efficient frontier—the *ESG-constrained frontier*—is typically located inside the classical mean–variance frontier, reflecting the trade-off between return, risk and sustainability.

1.7.2. Threshold Analysis

An alternative approach, similar to the one proposed by De Spiegeleer [5], is to explore the effect of varying the ESG threshold G and examine the resulting efficient frontiers. I consider 10 frontiers starting from the minimum ESG score, across all assets in the considered universe, reaching the highest one.

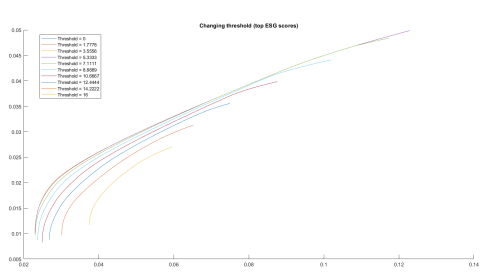


Figure 1.5: ESG Constrained Frontier

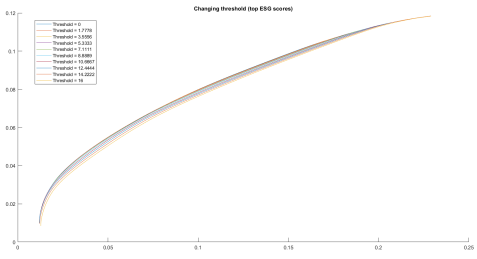


Figure 1.6: ESG Constrained Frontier – Top 50 ESG scores

1.7.3. Short-Selling

I removed the constraint $w_i \geq 0$, which initially prevented short-selling. However, this led to convergence issues in the efficient frontier. To address this, I introduced a relaxed constraint that allows limited short-selling, setting $w_i \geq -0.1$.

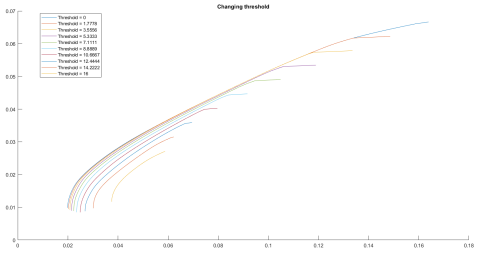


Figure 1.7: ESG Constrained Frontier – Top 50 ESG scores - Short selling

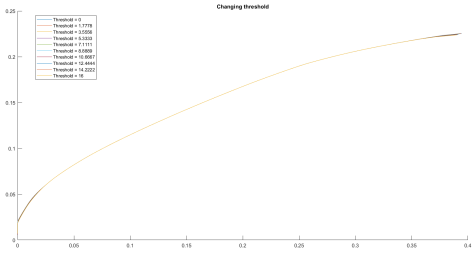


Figure 1.8: ESG Constrained Frontier – Top 50 ESG scores - Short selling

1.7.4. Results

It can be observed that there is a negative premium if we increase the ESG constraint value: the frontiers with smaller constraints dominate the ones with higher constraint value. Moreover, the frontier which allows short-selling dominates the one without it. The frontiers which consider the top 50 stocks with higher ESG scores are not particularly relevant: indeed, they are the same and the path is not as clear as in the standard case. Again, they are also dominated by the standard frontiers.

1.7.5. Minimum Variance Portfolio Analysis

To complement the graphical frontiers, I evaluate the *minimum variance* (MV) portfolio along each ESG threshold. For every constraint level G, the MV portfolio is the solution of the ESG-augmented mean–variance problem with the target return relaxed (i.e., the lowest-variance feasible portfolio given $\sum_i w_i \, \mathrm{ESG}_i \geq G$ and $\mathbf{1}^\top w = 1$), computed under two universes (full vs. Top–50 ESG names) and two leverage regimes (no short-selling vs. limited short-selling).

Table 1.1: Performance Metrics — Constrained Frontier - Top 50 ESG Scores

AR↑, SR↑, CR↑; AVol↓, MDD↓

Portf.	**AR %**	**AVol %**	**SR**	**MDD %**	**CR**
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 0$	18.093	12.736	1.3148	-12.423	1.4563
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 1.89$	17.847	12.774	1.2945	-12.732	1.4018
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 3.78$	16.377	12.365	1.2343	-10.446	1.5678
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 5.67$	16.353	12.376	1.2316	-10.447	1.5654
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 7.56$	17.214	12.888	1.2405	-13.100	1.3140
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 9.44$	19.271	13.077	1.3575	-11.507	1.6747
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 11.33$	12.377	13.076	0.89675	-14.909	0.83013
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 13.22$	20.566	16.731	1.1266	-19.594	1.0496
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 15.11$	28.937	16.675	1.5404	-13.103	2.2084
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 17$	13.538	15.385	0.82962	-22.049	0.61398

Table 1.2: Performance Metrics — Constrained Frontier - Top 50 ESG Scores - Short selling

AR↑, SR↑, CR↑; AVol↓, MDD↓

Portf.	**AR %**	**AVol %**	**SR**	**MDD %**	**CR**
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 0$	39.759	36.999	0.9175	-49.361	0.80547
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 1.89$	11.003	30.862	0.33972	-47.193	0.23315
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 3.78$	26.814	25.708	0.93321	-29.211	0.91793
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 5.67$	27.590	24.652	0.99845	-26.269	1.0503
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 7.56$	11.352	25.152	0.42942	-30.678	0.37003
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 9.44$	11.756	14.319	0.77982	-13.944	0.84306
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 11.33$	8.3999	12.906	0.62705	-21.365	0.39317
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 13.22$	21.487	14.372	1.3653	-13.908	1.5449
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 15.11$	26.866	15.815	1.5197	-12.274	2.1889
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 17$	11.970	15.666	0.72507	-23.956	0.49964

Table 1.3: Performance Metrics — Constrained Frontier

AR↑, SR↑, CR↑; AVol↓, MDD↓

Portf.	**AR %**	**AVol %**	**SR**	**MDD %**	**CR**
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 0$	12.758	7.9759	1.5130	-7.0063	1.8209
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 1.89$	12.758	7.9759	1.5130	-7.0063	1.8209
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 3.78$	12.758	7.9759	1.5130	-7.0063	1.8209
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 5.67$	12.758	7.9759	1.5130	-7.0063	1.8209
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 7.56$	12.212	7.9958	1.4479	-7.4893	1.6305
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 9.44$	11.001	8.2276	1.2740	-8.2716	1.3299
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 11.33$	10.376	8.6413	1.1472	-9.4072	1.1030
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 13.22$	11.093	9.3041	1.1356	-9.6648	1.1477
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 15.11$	12.313	10.445	1.1171	-9.5149	1.2941
$\sum_{i=1}^{n} w_i \, \text{ESG}_i \geq 17$	14.985	12.991	1.0811	-9.8229	1.5255

Table 1.4: Performance Metrics — Constrained Frontier - Short selling
AR↑, SR↑, CR↑; AVol↓, MDD↓

Portf.	AR %	AVol %	SR	MDD %	CR
$\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 0$	15.117	6.5519	2.1613	-3.6998	4.0859
$\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 1.89$	15.117	6.5519	2.1613	-3.6998	4.0859
$\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 3.78$	15.117	6.5519	2.1613	-3.6998	4.0859
$\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 5.67$	15.117	6.5519	2.1613	-3.6998	4.0859
$\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 7.56$	15.117	6.5519	2.1613	-3.6998	4.0859
$\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 9.44$	15.117	6.5519	2.1613	-3.6998	4.0859
$\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 11.33$	15.117	6.5519	2.1613	-3.6998	4.0859
$\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 13.22$	15.142	6.6296	2.1393	-3.7637	4.0233
$\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 15.11$	14.803	6.8073	2.0397	-3.9086	3.7874
$\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 17$	13.826	7.0686	1.8420	-4.1679	3.3172

Comparison Portfolio performance evolves non-linearly as the minimum ESG threshold increases, revealing a clear balance between sustainability constraints and efficiency.

At low to moderate ESG limits, the constraint remains non-binding, and the portfolio composition—and consequently its performance—remains virtually unchanged. In this region, annualized returns and volatility are stable, indicating that mild ESG screening does not restrict the efficient frontier.

Once the threshold begins to bind, a gradual reallocation emerges. In the full universe with limited short-selling, returns remain resilient while risk measures improve moderately: volatility and drawdowns stay contained, respectively close to 6.5% and -3.8%, and both Sharpe and Calmar ratios reach their most consistent levels, respectively >1.8% and >3.3%. This configuration shows that a mild ESG constraint can preserve efficiency while maintaining stability across risk dimensions.

In contrast, tightening the ESG requirement beyond the mid-range produces diminishing marginal benefits. Under the strictest levels, annualized returns start to decline slightly, to about 13.8%, while volatility and drawdowns rise, reflecting increasing concentration and reduced flexibility in portfolio construction.

When short-selling is disallowed, the transition is more abrupt. At lower thresholds, the portfolios remain unaffected; however, as ESG limits tighten substantially, performance becomes more volatile and returns may increase temporarily, but at the cost of higher risk and weaker risk-adjusted efficiency.

Overall, moderate ESG thresholds yield the most balanced outcomes: they support robust risk-adjusted metrics (Sharpe ratios above 1.8 and Calmar ratios near 3.3) without sacrificing stability or diversification.

1.8. Negative Screening – Sectorial Analysis

In this section, a negative screening approach is implemented by excluding "sin stocks" from the portfolio optimization process. The methodology follows the framework proposed by Blitz (2023) [3] and Blitz and Swinkels (2019) [2]. As an alternative strategy, assets with similar financial characteristics but higher ESG scores are included to evaluate substitution effects.

The analysis is conducted at the sector level, which guides the construction of the exclusion list. Let w denote the portfolio weights and ESG the asset-level ESG score vector used for both descriptive statistics and the screening rule. The optimization framework and frontier construction are consistent with the standard mean–variance setup introduced in the previous sections.

A family of efficient frontiers is traced by progressively excluding between 1 and 10 sectors from the universe, removing one additional asset at each iteration.

In this case, the Top 50 ESG Score case is not considered because there would be portfolios with less than 10 assets and it would be meaningless to analyze them.

1.8.1. Sector Statistics

These are all the possible sectors available in the S/P500. I consider the mean ESG score of the assets for each sector.

Sector	Group Count	Mean ESG Score
Automobiles & Auto Parts	1	1.2532
Banking & Investment Services	12	-3.2424
Chemicals	1	9.351
Consumer Goods Conglomerates	2	-23.536
Cyclical Consumer Services	3	-7.8006
Energy - Fossil Fuels	3	-8.6282
Financial Technology (Fintech) & Infrastructure	1	10.267
Food & Beverages	3	11.179
Food & Drug Retailing	1	10.485
Healthcare Services & Equipment	8	4.6355
Industrial & Commercial Services	2	-2.056
Industrial Goods	7	2.5431
Insurance	2	-18.953
Personal & Household Products & Services	1	-19.045

Sector	Group Count	Mean ESG Score
Pharmaceuticals & Medical Research	7	6.4016
Real Estate	1	5.9061
Retailers	5	2.8229
Software & IT Services	14	-2.1585
Technology Equipment	14	3.129
Telecommunications Services	3	-1.2747
Transportation	1	0.36222
Utilities	1	1.5953

1.8.2. Results

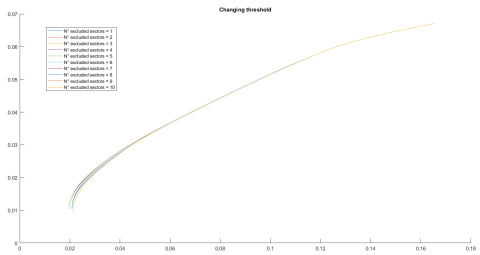


Figure 1.9: Negative screening Frontier

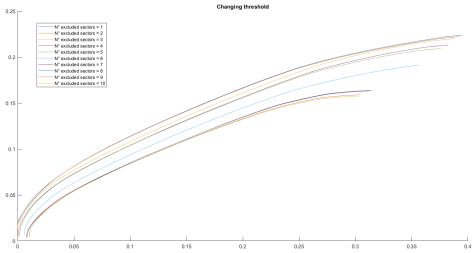


Figure 1.10: Negative screening Frontier - Short selling

It can be observed that, as the number of excluded assets increases (from 1 to 10), the frontiers shift inward: curves with fewer exclusions dominate those with more exclusions, especially in the low–to–mid volatility range. Small crossings at higher risk may occur, driven by which specific assets are removed.

Moreover, the frontier with limited short-selling consistently dominates the long-only case: for a given volatility, expected returns are higher across the whole family of curves, with the gap more evident at medium–high risk.

Overall, increasing exclusions imposes a negative "exclusion premium," while allowing short-selling partly offsets the loss of diversification induced by removing assets.

1.8.3. Minimum Variance Portfolio Analysis

Table 1.6: Performance Metrics — Sector Exclusion

AR↑, SR↑, CR↑; AVol↓, MDD↓

Portf.	AR %	AVol %	SR	MDD %	CR
Worst sectors excluded: 1	15.917	11.348	1.3096	-13.308	1.1960
Worst sectors excluded: 2	16.113	**11.264**	1.3346	-13.021	1.2375
Worst sectors excluded: 3	17.482	11.430	**1.4190**	-13.463	1.2985
Worst sectors excluded: 4	16.407	13.357	1.1446	**-12.355**	1.3279
Worst sectors excluded: 5	20.375	15.704	1.1900	-14.850	1.3720
Worst sectors excluded: 6	**28.036**	17.804	1.4025	-15.422	**1.8179**
Worst sectors excluded: 7	22.288	15.043	1.3488	-14.919	1.4940
Worst sectors excluded: 8	9.7689	11.366	0.82325	-12.784	0.76418
Worst sectors excluded: 9	10.861	12.041	0.8600	-14.697	0.73898
Worst sectors excluded: 10	11.357	14.608	0.7397	-20.069	0.56592

Table 1.7: Performance Metrics - Sector Exclusion - Short Selling

AR↑, SR↑, CR↑; AVol↓, MDD↓

Portf.	AR %	AVol %	SR	MDD %	CR
Worst sectors excluded: 1	-0.88046	27.351	-0.032322	-44.511	-0.019781
Worst sectors excluded: 2	6.64910	27.131	0.23791	-44.492	0.14945
Worst sectors excluded: 3	8.07040	28.130	0.27680	-29.917	0.26976
Worst sectors excluded: 4	33.83500	26.812	1.10030	-24.689	1.37050
Worst sectors excluded: 5	15.54600	30.005	0.48449	-36.856	0.42180
Worst sectors excluded: 6	**46.76200**	41.202	0.94617	-57.003	0.82034
Worst sectors excluded: 7	13.22400	22.134	0.56401	-24.014	0.55065
Worst sectors excluded: 8	11.47900	19.062	0.57266	-25.699	0.44668
Worst sectors excluded: 9	20.05100	**16.729**	**1.10070**	**-11.927**	**1.68110**
Worst sectors excluded: 10	15.63800	24.197	0.60411	-34.980	0.44706

Comparison Performance evolves non-linearly as the number of excluded sectors increases, showing clear trade-offs between return enhancement and diversification loss. When only a few sectors are excluded, portfolios display stable behavior with volatility around the lower bound of the observed range (near 11–12%), but improvements in returns remain modest. With a moderate level of exclusions, both annualized returns and risk-adjusted metrics reach their best balance: returns rise significantly (around 25–30%), and Sharpe ratios exceed 1.3, indicating that moderate screening can enhance efficiency without destabilizing the portfolio.

Allowing limited short-selling amplifies this pattern. While expected returns increase sharply under mid-range exclusions (above 40%), the most efficient configurations—those with the highest Sharpe and Calmar ratios and the lowest drawdowns (around 12%)—still correspond to moderate screening levels rather than extreme ones. This suggests that short exposure can partially offset diversification loss but does not fundamentally change the relationship between screening intensity and stability.

When too many sectors are excluded, the frontiers contract and fluctuate irregularly, with

volatility rising above 25–30% and drawdowns deepening, indicating a loss of robustness. Overall, the results confirm that moderate exclusions deliver the most efficient and economically consistent trade-off, while excessive filtering erodes diversification and increases estimation noise.

1.9. ESG Integration via Linear Limits on Low-Score Assets

In the previous section, it is observed that removing too many assets with high ESG scores and adding too strict ESG constraint can bring to bad performances.
For this reason, a more cautios approach, which limits the weights on some assets, is introduced in order to limit the effect of assets with low ESG scores.

Consider n risky assets with expected returns $\mu \in \mathbb{R}^n$ and positive definite covariance matrix $\Sigma \in \mathbb{R}^{n\times n}$. Let $\mathrm{ESG} \in \mathbb{R}^n$ be the vector of (normalized) ESG scores. Rank assets by ESG and select the top-N names, forming the investable set $\mathcal{T} \subseteq \{1,\dots,n\}$ with $|\mathcal{T}| = N$. Within $\mathcal{T}$, denote by $\mathcal{L} \subseteq \mathcal{T}$ the k assets with the *lowest* ESG scores (e.g., $k = 20$). We solve a mean–variance problem that *caps* the total weight allocated to the worst-k ESG names:

$$
\begin{aligned}
\min_{w\in\mathbb{R}^N} \quad & w^\top \Sigma_{\mathcal{T}}\, w \\
\text{s.t.} \quad & \mathbf{1}^\top w = 1, \\
& \mu_{\mathcal{T}}^\top w = \mu_p, \\
& \sum_{i\in\mathcal{L}} w_i \;\le \gamma, \qquad \text{(cap on lowest-}k\text{ ESG weights)} \\
& \underline{w} \;\le\; w_i \;\le\; 1 \quad \forall i \in \mathcal{T}. \qquad \text{(box bounds)}
\end{aligned}
$$

Here $\gamma \in (0,1)$ is the maximum total exposure permitted to the worst-k ESG assets.
In this case $\gamma = 0.1$.
The lower bound $\underline{w}$ governs short-selling: $\underline{w} = 0$ forbids shorts; $\underline{w} = -0.1$ allows limited shorts up to -10% per asset.

Implications Relative to an unconstrained benchmark, the cap $\sum_{i\in\mathcal{L}} w_i \le \gamma$ shrinks the feasible set, typically shifting the frontier inward and reducing reliance on low-ESG names to reach a target return. Tighter caps (smaller γ) enhance sustainability alignment but may increase variance for a given μ_p, especially when low-ESG assets carry distinctive risk premia.

Sensitivity A practical analysis varies γ to generate families of frontiers. As γ decreases, sustainability tightens and the frontier generally deteriorates in mean–variance terms. Conversely, relaxing γ expands feasibility and improves risk–return trade-offs, at the cost of weaker sustainability discipline.

1.9.1. Results

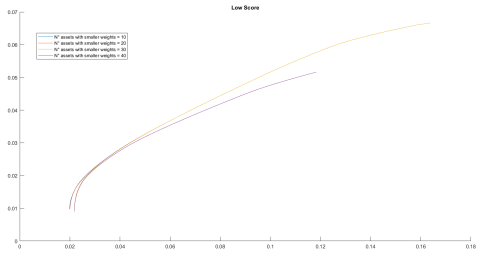


Figure 1.11: Low Score Constraint Frontier

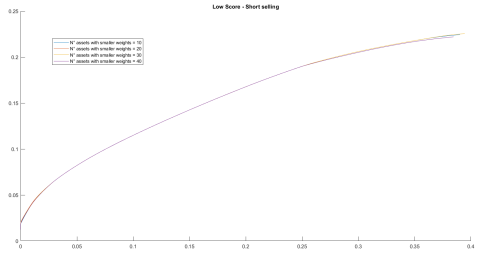


Figure 1.12: Low Score Constraint Frontier - Short selling

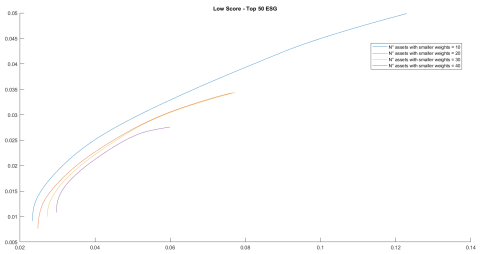


Figure 1.13: Low Score Constraint Frontier - Top 50 ESG scores

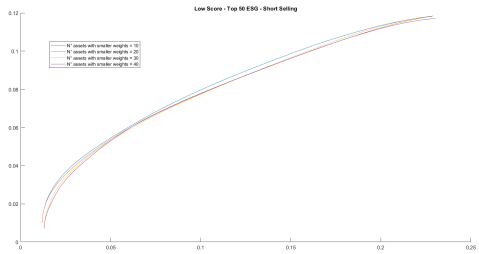


Figure 1.14: Low Score Constraint Frontier - Short selling - Top 50 ESG scores

With $\gamma = 0.1$, as the number of banned low-score names grows ($k = 10 \rightarrow 40$), the frontiers shift inward: fewer exclusions dominate in the full universe, especially at low–mid volatility.

Within the Top-50 ESG subset, the effect is small—the curves nearly overlap. Allowing limited short-selling lifts the entire set of curves; in the full universe it can even improve with larger k, since more low-score names become usable as hedges

Overall: more exclusions worsen the trade-off, while short-selling partly offsets the diversification loss.

1.9.2. Minimum Variance Portfolio Analysis

Table 1.8: Performance Metrics — Low Score Constrained Frontier - Top 50 ESG Scores

AR↑, SR↑, CR↑; AVol↓, MDD↓

Portf.	AR (%)	AVol (%)	SR	MDD (%)	CR
Restriction on 10 worst ESG stocks	11.569	8.0900	1.3593	-7.6051	1.5212
Restriction on 20 worst ESG stocks	9.5445	8.5991	1.0642	-9.1523	1.0429
Restriction on 30 worst ESG stocks	12.709	9.4871	1.2674	-10.306	1.2332
Restriction on 40 worst ESG stocks	13.775	10.302	1.2595	-7.2847	1.8909

Table 1.9: Performance Metrics — Low Score Constrained Frontier - Top 50 ESG Scores - Short selling

AR↑, SR↑, CR↑; AVol↓, MDD↓

Portf.	AR (%)	AVol (%)	SR	MDD (%)	CR
Restriction on 10 worst ESG stocks	15.117	6.5519	2.1613	-3.6998	4.0859
Restriction on 20 worst ESG stocks	15.117	6.5519	2.1613	-3.6998	4.0859
Restriction on 30 worst ESG stocks	12.932	7.1201	1.7168	-4.8295	2.6778
Restriction on 40 worst ESG stocks	12.053	6.6373	1.7227	-4.1606	2.8970

Table 1.10: Performance Metrics — Low Score Constrained Frontier

AR↑, SR↑, CR↑; AVol↓, MDD↓

Portf.	AR (%)	AVol (%)	SR	MDD (%)	CR
Restriction on 10 worst ESG stocks	13.276	14.155	0.88525	-16.238	0.81759
Restriction on 20 worst ESG stocks	12.417	13.992	0.84061	-15.651	0.79339
Restriction on 30 worst ESG stocks	14.383	13.461	1.00390	-12.861	1.11830
Restriction on 40 worst ESG stocks	14.126	13.461	0.98703	-12.841	1.10010

Table 1.11: Performance Metrics — Low Score Constrained Frontier - Short selling

AR↑, SR↑, CR↑; AVol↓, MDD↓

Portf.	AR (%)	AVol (%)	SR	MDD (%)	CR
Restriction on 10 worst ESG stocks	-6.2065	33.564	-0.19040	-51.051	-0.12157
Restriction on 20 worst ESG stocks	-9.1468	32.522	-0.29378	-52.642	-0.17376
Restriction on 30 worst ESG stocks	8.5390	38.386	0.21419	-59.454	0.14362
Restriction on 40 worst ESG stocks	-0.1046	36.612	-0.0028584	-52.251	-0.0020019

Comparison In the **Top–50 universe without short-selling**, performance remains resilient for moderate caps. When the total weight on the worst-ESG assets is limited to $\gamma = 0.1$, the frontier shifts only slightly inward, maintaining competitive returns and risk-adjusted ratios. At this level, the portfolio achieves strong balance between ESG alignment and efficiency, as shown by robust Sharpe (SR ≈ 1.36) and Calmar ratios (CR ≈ 1.89) with low volatility (AVol $\approx 8\%$) and shallow drawdowns (MDD $\approx -7\%$). Further tightening produces marginal improvements in ESG alignment but progressively reduces diversification and return stability.

When **limited short-selling** is permitted ($\underline{w} = -0.1$), the cap's effect becomes more pronounced.
Allowing controlled short positions enables partial hedging against low-ESG names, improving frontier curvature and risk-adjusted efficiency. For instance, limiting 10 or 20 assets deliver the highest SR (≈ 2.16) and CR (≈ 4.08) with the lowest volatility ($\approx 6.5\%$) and minimal drawdowns ($\approx -3.7\%$). Beyond this range, however, the benefits plateau, as the portfolio increasingly mirrors the unconstrained solution and ESG differentiation weakens.

In the **full universe without short-selling**, similar dynamics emerge: moderate limitations preserve most of the unconstrained efficiency while reducing exposure to unsustainable sectors.
Stricter limits further constrain diversification, leading to rising volatility (AVol $> 13\%$) and shallower but less stable risk profiles.
Under **limited short-selling**, these effects are magnified: extreme sensitivity appears limiting few assets, where returns drop (AR $\approx 8.5\%$) and volatility surges above 30%, producing erratic and unreliable frontier shapes.

Overall, the evidence suggests that **moderate capping of low-ESG assets**, so the cases with limitations on 10-20 assets, are the most effective configurations.
They improve sustainability alignment with limited efficiency loss.
Excessively assets with reduced investments allowed, however, shrink the feasible set, concentrate weights and induce instability—yielding frontiers that are statistically and economically less reliable.
In this case, short selling amplifies volatility and reduces returns. So, in this case the instability brings a greater effect than the possible diversification benefits.

1.10. ESG Integration via Best-in-Class: Long/Short Constraints

Let $\mu \in \mathbb{R}^N$ be expected returns, $\Sigma \in \mathbb{R}^{N\times N}$ (p.d.) the covariance matrix, and $\mathrm{ESG} \in \mathbb{R}^N$ the (normalized) ESG scores for the investable set $\mathcal{T}$ of size N (either the whole universe or a prefiltered top-N by ESG). Define the indices of the N *leaders* (highest ESG) as $\mathcal{H}$ and the k *laggards* (lowest ESG) as $\mathcal{L}$, with $|\mathcal{H}| = N$ in the strict variant and $|\mathcal{L}| = k$ (e.g., $k = 20$). The Best-in-Class frontier is obtained by solving:

$$
\begin{aligned}
\min_{w\in\mathbb{R}^N} \quad & w^\top \Sigma\, w \\
\text{s.t.} \quad & \mathbf{1}^\top w = 1, \\
& \mu^\top w = \mu_p, \\
& 0 \le w_i \le \overline{s} \quad \forall i \in \mathcal{H}, \qquad \text{(leaders: long-only)} \\
& \underline{s} \le w_i \le 0 \quad \forall i \in \mathcal{L}, \qquad \text{(laggards: short-only)}
\end{aligned}
$$

Here $\overline{s}$ caps leader positions (in this case $\overline{s} = 1$), whereas $\underline{s} < 0$ forces short exposure in laggards (in this case $\underline{s} = -0.1$).

Implications This strict Best-in-Class design hard-codes sustainability tilts: leaders are long-only, while laggards are short-only. The ESG threshold further tightens feasibility. Relative to an unconstrained frontier, the Best-in-Class frontier typically shifts inward (higher variance at a given return) unless ESG leaders also command favorable risk premia. Tighter short limits (e.g., $\underline{s} = -0.05$ instead of -0.1) or smaller leader caps $\overline{s}$ reduce active tilt and can improve numerical stability but may soften the ESG parameter effects.

Sensitivity Changing the investable universe (broad vs prefiltered top-N) and the bounds $(\overline{s}, \underline{s})$ produces families of frontiers. Looser caps on leaders (larger $\overline{s}$) or more negative short floors (smaller $\underline{s}$) generally enhance the return potential (more active bets) but may increase risk and turnover. Conversely, tighter bounds compress dispersion and can degrade the attainable risk–return profile while preserving a clearer sustainability stance.

1.10.1. Results

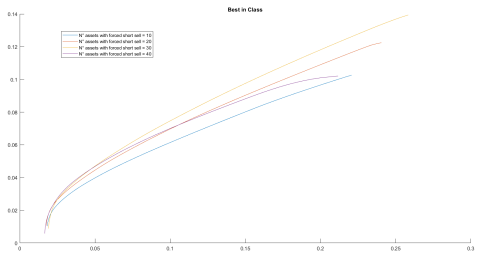


Figure 1.15: Best in Class Frontier

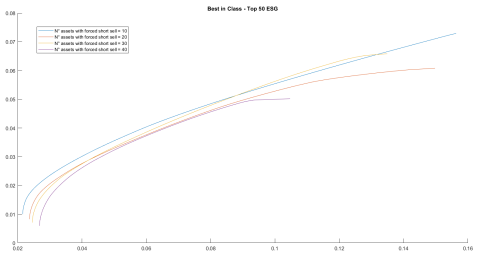


Figure 1.16: Best in Class Frontier - Top 50 ESG

In the full universe (left panel), as the number of forced-short laggards rises ($k = 10 \rightarrow 40$), the frontier shifts *outward*: curves with larger k dominate, especially from low–to–mid volatility upward, with only small crossings near the minimum-variance corner. The added shorting capacity improves hedging and lifts attainable returns for a given risk.

Within the Top–50 ESG subset (right panel), the pattern reverses. Increasing k compresses and shifts the frontier inward; the curves are close at low risk but clearly separated at medium–high risk, with $k = 10$ dominating. In the narrower, already "clean" set, shorting the milder laggards adds binding constraints without delivering comparable hedging gains.

Overall, limited short–selling raises the whole family of curves, but the effect of enlarging k is universe–dependent: beneficial with broad breadth (more effective hedges), detrimental after ESG prefiltering (less dispersion to exploit and tighter constraints).

For this reason, this qualitative analysis is not so relevant because the short selling effect brings instability to the frontier construction. On the other hand, it seems that too many assets with forced short selling are not performing well.

1.10.2. Minimum Variance Portfolio Analysis

Table 1.12: Performance Metrics — Best in Class

Portf.	AR (%)	AVol (%)	SR	MDD (%)	CR
Number of Assets with short selling: 10	17.281	16.018	1.0018	-16.458	1.05000
Number of Assets with short selling: 20	11.099	16.961	0.62325	-19.387	0.57248
Number of Assets with short selling: 30	6.6741	17.821	0.36351	-23.381	0.28544
Number of Assets with short selling: 40	6.9053	16.774	0.39919	-18.540	0.37246

Table 1.13: Performance Metrics — Best in Class — Short selling

Portf.	AR (%)	AVol (%)	SR	MDD (%)	CR
Number of Assets with short selling: 10	12.809	7.4738	1.6208	-6.9622	1.8398
Number of Assets with short selling: 20	10.390	8.2132	1.2085	-7.5928	1.3684
Number of Assets with short selling: 30	10.198	8.7438	1.1151	-7.7667	1.3130
Number of Assets with short selling: 40	7.3827	9.2892	0.76908	-9.1916	0.80320

Comparison The Best-in-Class strategy reallocates capital toward higher-rated issuers, rewarding ESG quality through portfolio concentration. Performance metrics show a clear pattern: as the number of assets increases, efficiency declines steadily, indicating that the ESG signal weakens when diversification dilutes exposure to top-rated names.

In the **long-only configuration**, portfolios concentrated in the top 10 assets outperform all broader variants across every metric. The 10-asset portfolio achieves the highest annualized return (AR = 17.28%) with the lowest volatility (AVol = 16.02%), producing a Sharpe ratio of 1.00 and the shallowest drawdown (MDD $\approx -16.46\%$). Performance progressively deteriorates as the portfolio expands: the 20-, 30-, and 40-asset sets exhibit lower returns (AR = 6–11%), higher volatility, and weaker Calmar ratios, reflecting the dilution of the ESG premium as additional, lower-ranked firms enter the mix.

Allowing **limited short-selling** for all the assets ($w_i \geq -0.1$) reinforces the same ranking while amplifying efficiency levels. The 10-asset Best-in-Class configuration again leads, achieving strong performance (AR = 12.81%) with low volatility (7.47%), high risk-adjusted ratios (SR = 1.62, CR = 1.84), and shallow drawdowns (MDD $\approx -6.96\%$). While shorting slightly improves risk–return trade-offs for larger portfolios (20–30 assets), the dominance of the concentrated case persists, confirming that ESG polarization—long on top-rated and short on bottom-rated issuers—magnifies the performance edge without destabilizing the frontier.

Overall, the results indicate that the Best-in-Class portfolios with less assets with forced short selling, deliver stronge combination of return and risk efficiency in both regimes. Expanding the selection to 20 or more assets gradually erodes these advantages, as diversification dilutes the ESG-driven tilt and introduces less aligned exposures. The degradation is smoother under limited short-selling, but the monotonic trend remains.

1.11. Overall Comparison

1.11.1. Plot Comments

This section reconciles the Chapter 1 frontiers across methods and investable universes. We contrast the *ESG-constrained* optimization, *Negative screening (Sector analysis)*, *Linear limits on low-ESG assets* and *Best-in-Class* long/short designs.
The aggregate plots place the full 100-name universe next to the Top-50 ESG subset to highlight breadth effects.

Where each frontier dominates.

- In the *full 100-name universe*, relaxing the no-short constraint (e.g., allowing limited shorts) typically lifts the envelope above long-only across most risk levels, reflecting added hedging degrees of freedom and better use of cross-sectional dispersion.
- In the *Top-50 ESG* subset, curves compress: short-enabled designs still help, but their advantage shrinks because there is less dispersion and fewer distinct premia to exploit.

Universe width (Top-50 vs full).

- Limiting the universe to the Top-50 ESG names cuts diversification, so at the same constraints the frontiers shift inward versus the full 100-name set.
- In many Top-50 plots, curves cluster or overlap, showing that constraints bind without any diversification gain.

Method-by-method contrasts.

- *Top ESG scores (prefiltering)*: Using only the Top-50 ESG names (instead of all the universe) reduces diversification, so frontiers move inward. Letting some short-selling in helps a bit, but cannot fully replace the lost breadth.
- *ESG-constrained optimization*: Raising the minimum ESG score required for assets tightens the feasible set and shifts the frontier slightly inward. Limited short-selling re-opens hedging paths and recovers part of the loss, especially when the full 100-name universe is considered.
- *Negative screening (sectors)*: Removing entire sectors cuts out whole risk premia and pushes the frontier inward, as in the prefiltering situation; more excluded sectors

generally make results worse. Limited short-selling softens, but does not eliminate the loss of diversification.

- *Linear limits on low-ESG assets*: Instead of banning the worst ESG names, cap their total portfolio weight (e.g., limit the combined weight of the 10–20 lowest-ESG stocks). This "discipline without bans" keeps diversification and yields smoother, more robust frontiers than sector exclusions.
- *Best-in-Class*: In the full universe, going long the highest-ESG "leaders" and short the lowest-ESG "laggards" creates targeted hedges that improve the frontier, especially at low–to–mid risk. Expanding the number of laggards placed on the short side can help when all names are available.

Stability and sensitivity.

- Short-bound choices affect active sets and produce piecewise envelopes with non-smooth transitions; removing bounds entirely can create numerical issues in several families.
- The frontier shows instability under short-selling in some configurations but is stable in the standard, long-only setting; this is consistent with tighter feasible sets and better-conditioned problems when shorts are disallowed.

Actionable takeaways for the next chapter.

- Prefer the *full 100-name universe*: more breadth improves the frontier and strengthens the benefit of limited short-selling.
- Use *Linear-limit* designs for robustness and clarity; they keep ESG control without the fragility of sector bans.
- Apply *sector exclusions* sparingly: good for policy alignment, costly for mean–variance efficiency.
- Use *Best-in-Class* carefully: strong in the full universe (especially at low–mid risk), but it can over-constrain the Top-50 set.

1.11.2. Minimum Variance Portfolios

I compare the minimum–variance (MV) portfolios obtained in each family of Chapter 1 across investable universes (full 100-name set vs. Top–50 ESG subset) and leverage regimes (long-only vs. limited short-selling).

Breadth and shorting dominate the MV corner. Across methods, the full universe systematically delivers lower or comparable volatility than the Top–50 subset, reflecting stronger diversification at the variance-minimizing point. Allowing limited short-selling further lowers risk or preserves it while improving downside metrics; this pattern already emerges in the standard and ESG–constrained families and carries over to the other designs.

ESG–constrained Portfolios. Tightening the portfolio ESG floor shrinks the feasible set; MV risk is relatively stable for mild thresholds but tends to worsen (or yield deeper drawdowns) as the threshold becomes stringent. Short allowances partly offset this by re-opening hedges, with the breadth advantage of the full universe remaining evident; the Top–50 variant compresses outcomes and shows smaller performance dispersion across thresholds.

Negative screening. Excluding whole sectors removes entire risk premia and weakens MV efficiency. As the number of excluded sectors grows, MV volatility and drawdowns generally deteriorate; limited short-selling softens but does not eliminate the loss. This confirms that sector bans are not so relevant.

Linear limits on low-ESG names. Capping aggregate exposure to the worst ESG names preserves diversification and produces smoother, more robust MV profiles than sector exclusions. This effect is clearer in the full universe than in the Top–50 subset.

Best-in-Class long/short. Structuring the MV portfolio to be long ESG leaders and short ESG laggards produces targeted hedges that are especially effective at low–to–mid risk in the full universe. In short selling case, the results improve considering all the possible metrics. Within this family, more concentrated specifications (fewer enforced shorts) tend to yield stronger risk-adjusted MV profiles; expanding the short set helps only when broad breadth is available.

Overall message. At the mean–variance (MV) corner, three principles emerge consistently across Chapter 1:

- *Prefer breadth*: using the full 100-name universe enhances diversification and stabilizes frontier behavior.
- *Allow limited shorting*: small negative bounds improve stability and reduce downside risk without destabilizing results.

- *Favor soft ESG controls*: linear limits and Best-in-Class adjustments preserve efficiency more effectively than strict exclusions, which impose higher variance costs and also smaller retrurns.
- **Best-in-Class** case with short selling allowed and the **Constrained Optimization** model with low ESG constraints give the best results in terms of Sharpe ratio and risk balance.

These conclusions align with the aggregate comparison of all frontiers in Sec. 1.10 and guide the design choices adopted in the next chapter.

2 | Weights Forecasting

I consider the full sample from January 2015 through December 2024 of the S&P500.

Rolling estimation (2015-2019 and 2020–2022). For each month t in 2020–2022, I estimate a minimum-variance portfolio using a rolling window of the previous 60 months of returns, i.e., the sample $\{r_{t-59}, \ldots, r_t\}$.
This yields a monthly sequence of minimum-variance weights w_t^{MV}.

Forecasting stage (2023). Throughout 2023, I use a Kalman Filter fitted to the historical path of $\{w_t^{\mathrm{MV}}\}$ to produce end-of-month one-step-ahead forecasts of the portfolio weights. Concretely, at the end of each month t in 2023 I compute the predicted weights vector $\widehat{w}_t$.

Evaluation (monthly rolling window, 2023-2024). I adopt a rolling forecasting design.
For each start month h in 2023 (Jan–Dec), I fix the KF weights predicted at the end of that month, $\widehat{w}_h$, and evaluate the resulting **static** portfolio on the *next 12 months* of realized returns, i.e., the shifted window $[h, \ldots, h{+}11]$.
In parallel, I fix the "Real" minimum-variance weights w_h^{MV} at the same h and evaluate them on the identical window. This yields 12 out-of-sample tests: h=1 uses Jan-23–Dec-23, h=2 uses Feb-23–Jan-24, …, h=12 uses Dec-23–Nov-24.
For each h I report performance (AR, Vol, Sharpe, MDD, Calmar) for KF and Real, plus diagnostics comparing $\widehat{w}_h$ to w_h^{MV} (L1, L2, cosine) and the L1-L2 gap between their two OOS (Out Of Sample) return streams over that window.

The primary objective of this framework is to evaluate the **out-of-sample (OOS) performance** of the Kalman Filter forecasts relative to the Real minimum-variance benchmark, measuring the model's ability to reproduce efficient risk–return dynamics under realistic market evolution. A secondary but essential goal is to assess the **robustness and stability** of the Kalman Filter, its consistency across time, its sensitivity to recent market shocks and its capacity to generate economically meaningful weights.

2.1. Kalman Filter Overview

The Kalman filter (KF) is a recursive estimation method for latent (unobserved) states that evolve over time. It assumes a linear state–space model:

$$\underbrace{x_t}_{\text{latent state}} = Ax_{t-1} + \underbrace{\eta_t}_{\text{process noise}}$$

$$\underbrace{z_t}_{\text{observation}} = Hx_t + \underbrace{\varepsilon_t}_{\text{measurement noise}},$$

with $\eta_t \sim \mathcal{N}(0, Q)$ and $\varepsilon_t \sim \mathcal{N}(0, R)$. Given past observations $\{z_1, \ldots, z_t\}$, the KF computes the optimal estimate of x_t and its uncertainty, then propagates the system forward to produce forecasts $x_{t+h|t}$.

In our setting, each portfolio weight is treated as a latent, smoothly evolving quantity. The forecasted final weights produced by rolling optimizations are the noisy measurements z_t.

State–space parameterization and estimation in practice. In the empirical setup, state dynamics are calibrated *per asset* from the rolling history of minimum–variance weights. Let $W \in \mathbb{R}^{n \times T_{\text{roll}}}$ collect the n asset weights across the rolling window, with $W(:, t)$ the weight vector at month t. For each asset i we fit an AR(1) law on weights,

$$w_{i,t} \approx \phi_i \, w_{i,t-1} + \eta_{i,t}, \qquad \eta_{i,t} \sim \mathcal{N}(0, q_i),$$

using the closed-form slope

$$\widehat{\phi}_i = \frac{\sum_t w_{i,t-1} w_{i,t}}{\sum_t w_{i,t-1}^2}, \quad \text{then clipped to} \quad \phi_i = \min\{\max(\widehat{\phi}_i, 0),\, 0.999\}.$$

The process-noise variance is the population variance of the residuals

$$e_{i,t} = w_{i,t} - \phi_i w_{i,t-1}, \qquad q_i = \max\{\text{var}(e_{i,\cdot}),\, 10^{-6}\}.$$

Stacking across assets yields diagonal state–space matrices

$$A = \text{diag}(\phi_1, \ldots, \phi_n), \qquad Q = \text{diag}(q_1, \ldots, q_n).$$

Measurement model. To keep the model simple and avoid heavy computations, the measurement parameters are fixed as

$$H = I_n, \qquad R = \sigma_\varepsilon^2 I_n, \quad \text{with} \quad \sigma_\varepsilon^2 = 10^{-4},$$

where H is the observation matrix that links the latent state (true portfolio weights) to the observed ones, here assumed to be directly observable. The matrix R represents a small and constant measurement noise, accounting for minor variations in the rolling optimization results.

Feasibility projection. The initial state vector x_0 is set equal to the last available minimum–variance weight vector from the rolling estimation, i.e. $x_0 = W(:, T_{\text{end}})$.
After each prediction, the weight vector is projected back into the feasible set through simple clipping and renormalization:

$$x_t \leftarrow \min\big(\max(x_t,\, \ell),\, u\big), \qquad x_t \leftarrow \frac{x_t}{\mathbf{1}^\top x_t},$$

where ℓ and u are the per-asset box bounds (e.g., long-only $\ell = 0$, $u = 1$ or limited shorting $\ell = -0.1$, $u = 1$).

2.1.1. Why Use a Kalman Filter to Predict Weights?

- **Noise reduction.** Rolling optimization delivers weight vectors that can fluctuate due to sampling error and shifting estimation windows. KF separates the persistent signal (state) from transient noise (measurement), yielding smoother and more reliable trajectories.
- **Dynamic structure.** Unlike static shrinkage smoothing, KF encodes explicit temporal dynamics (e.g., persistence/mean reversion via ϕ) and propagates them to produce multi-step forecasts.
- **Coherent multi-period forecasts.** KF naturally generates a path of future weights that is internally consistent with the estimated dynamics.
- **Uncertainty awareness.** The state covariance $P_{t|t}$ quantifies uncertainty in the weight estimates and grows with forecast horizon—useful context when interpreting forecast performance.[1]

[1] This work is focused on point forecasts; the covariance is still informative for diagnostics.

2.1.2. What Does It Mean to "Predict Weights"?

At time t we have a history of optimized weight vectors $\{w_1, \ldots, w_t\}$ (one per rolling window). KF treats these as noisy observations of a latent weight process and produces forecasts

$$\hat{w}_{t+1|t},\ \hat{w}_{t+2|t},\ \ldots,\ \hat{w}_{t+H|t}.$$

Operationally, these are the allocations we would hold in each future month if the estimated dynamics persisted. To translate weight forecasts into performance, we multiply each forecasted weight vector by the realized asset returns of that month, producing a K-month portfolio return series (KF-based) that can be compared to a benchmark series built from "real" (re-optimized) weights over the same months.

2.1.3. Metrics for Comparing Forecasted vs. Real Weights

We report two complementary families of metrics:

(A) Weights-level distances (structure match). For each month $h = 1, \ldots, K$, compare the forecasted weights $\hat{w}^{\mathrm{KF}}_{t+h}$ to the benchmark weights w^{Real}_{t+h} using:

$$L_1(h) \;=\; \sum_{j=1}^{n} \left|\hat{w}^{\mathrm{KF}}_{j,t+h} - w^{\mathrm{Real}}_{j,t+h}\right|, \qquad L_2(h) \;=\; \left\|\hat{w}^{\mathrm{KF}}_{t+h} - w^{\mathrm{Real}}_{t+h}\right\|_2,$$

$$\text{Cosine similarity}(h) \;=\; \frac{\langle \hat{w}^{\mathrm{KF}}_{t+h},\ w^{\mathrm{Real}}_{t+h} \rangle}{\|\hat{w}^{\mathrm{KF}}_{t+h}\|_2 \, \|w^{\mathrm{Real}}_{t+h}\|_2}.$$

- L_1 measures total absolute reallocation needed to move from KF to Real (sparse, interpretable).
- L_2 penalizes large discrepancies more heavily (energy of the error).
- Cosine similarity captures *directional* agreement (1 means perfectly aligned).

(B) Returns-level distances (economic impact). Using realized returns r_{t+h}, form two K-length portfolio return vectors:

$$r^{\mathrm{KF}}_{t+h} = \left(\hat{w}^{\mathrm{KF}}_{t+h}\right)^{\top} r_{t+h}, \qquad r^{\mathrm{Real}}_{t+h} = \left(w^{\mathrm{Real}}_{t+h}\right)^{\top} r_{t+h}.$$

Compare them via

$$L_1^{\text{return}} = \sum_{h=1}^{K} \left| r_{t+h}^{\text{KF}} - r_{t+h}^{\text{Real}} \right|, \qquad L_2^{\text{return}} = \sqrt{\sum_{h=1}^{K} \left(r_{t+h}^{\text{KF}} - r_{t+h}^{\text{Real}} \right)^2}.$$

These quantify the aggregate payoff difference induced by using forecasted rather than re-optimized weights.

2.1.4. Interpretation

A good forecaster of weights should exhibit:

- **Low L_1/L_2 distances and high cosine similarity at the weights level**, indicating structural alignment and stable reconstruction of the portfolio composition.
- **Low L_1/L_2 distances at the return level**, with performance metrics close to the benchmark, ensuring that predictive errors do not materially distort realized efficiency.

Disparities between these two dimensions highlight where dynamics shift faster than the learned persistence, suggesting the need for richer state models (e.g., time-varying parameters or factors) or stronger regularization and constraints in t.

2.1.5. Short Selling

A key limitation observed in the Kalman Filter framework concerns the treatment of short-selling. When short positions are allowed, the resulting forecasted weights tend to exhibit pronounced instability, leading to portfolios that are economically inefficient and highly volatile.
Although this behavior is empirically observed, but not reported in this thesis, such configurations consistently generate excessive risk and erratic rebalancing patterns.
For these reasons, the forecasting analysis focuses exclusively on two cases: the **Full Universe** and the **Long-Only** configurations. The Top 50-ESG scores case is excluded because it previously showed the weakest overall performance, providing only a meaningful baseline for comparison.
The short selling case, despite yielding competitive returns, demonstrated limited stability and is therefore not analyzed.

2.2. ESG Integration via Best-in-Class: Long/Short Constraints – Forecasting Results

2.2.1. Plots

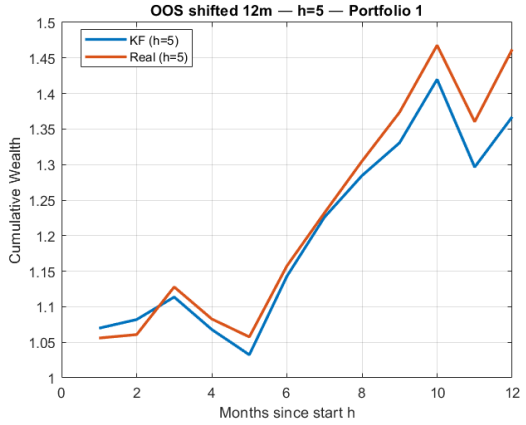


Figure 2.1: Best in Class KF vs Real - Number of Assets with short selling: 10



Figure 2.2: Best in Class KF vs Real - Number of Assets with short selling: 20

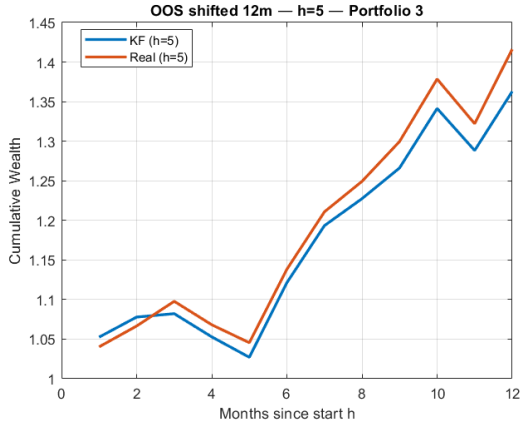


Figure 2.3: Best in Class KF vs Real - Number of Assets with short selling: 30

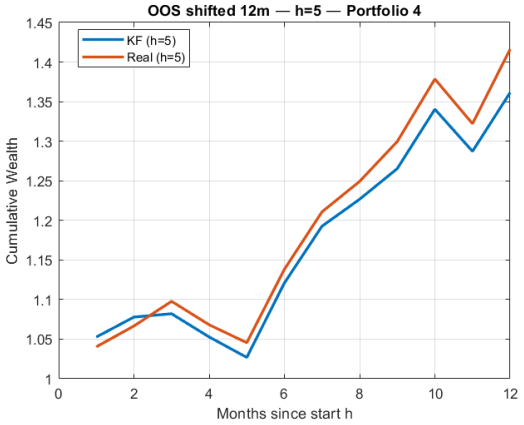


Figure 2.4: Best in Class KF vs Real - Number of Assets with short selling: 40

Each panel shows a pure *forecasting* comparison of cumulative wealth for a fixed start month: the **static** KF portfolio (blue) versus the **static** Real benchmark (orange) over the next 12 months. We display the fifth month as an illustration: from a qualitative point of view most of the plots follow the same structure, so we prioritize a qualitative read.

Turning points are generally shared; level differences vary with universe size.

10 assets with allowed short selling. The two lines almost overlap all year. Both capture the late-year rise; the Real path is consistently higher. Direction and timing match well, while gaps are small.

20 assets with allowed short selling. Shapes remain very similar, but the KF line sits below the Real line—especially in the second half of the year. KF lags during sustained upswings.
The gap to Real is bigger than the first case.

30 assets with allowed short selling. Turning points are preserved, yet KF stays under the Real curve for most months and rebounds slower after mid-year weakness.

40 assets with allowed short selling. Direction is still right, but gaps are wider and vary more across months. KF is sometimes close, yet the Real path remains above it and the difference tends to increase at the end of the year.

In general, it seems that KF slightly underperforms the real returns.

2.2.2. Metrics

Table 2.1: Weights Diagnostics per **month — 10 assets with allowed short selling**

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.22088	0.048822	0.99715
2	0.56147	0.14176	0.95959
3	0.49782	0.13446	0.96429
4	0.51080	0.13584	0.96392
5	0.83931	0.20492	0.92007
6	0.99365	0.26731	0.86944
7	0.98289	0.25720	0.87673
8	0.97671	0.24227	0.89350
9	0.99384	0.26478	0.87775
10	1.23760	0.31153	0.82867
11	1.21990	0.30718	0.83429
12	1.44950	0.34786	0.79852

Table 2.2: Returns Diagnostics per **month — 10 assets with allowed short selling**

Start month	Returns L1	Returns L2
1	0.023042	0.0081057
2	0.10366	0.0383310
3	0.096156	0.0334360
4	0.092017	0.0329070
5	0.14992	0.0521950
6	0.13294	0.0462270
7	0.16816	0.0568660
8	0.17958	0.0581800
9	0.21942	0.0720120
10	0.22495	0.0738610
11	0.22213	0.0736310
12	0.23791	0.0773700

Table 2.3: KF Performance per **month — 10 assets with allowed short selling**

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	50.583	16.799	2.4788	-6.4709	7.8171
2	58.213	15.924	2.9367	-6.7512	8.6226
3	52.089	14.491	2.9448	-7.0165	7.4238
4	35.240	19.168	1.5949	-8.7883	4.0099
5	39.040	19.448	1.7182	-8.6933	4.4908
6	40.585	19.707	1.7532	-8.5717	4.7347
7	34.169	20.734	1.4351	-8.4580	4.0399
8	33.707	20.713	1.4195	-8.3383	4.0424
9	42.053	19.402	1.8360	-8.2219	5.1148
10	48.920	18.171	2.2284	-8.1055	6.0354
11	47.246	17.507	2.2462	-7.9918	5.9118
12	37.651	17.052	1.8991	-7.8794	4.7784

Table 2.4: Real Performance per **month — 10 assets with allowed short selling**

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	49.408	16.235	2.5149	-6.0017	8.2324
2	63.039	15.115	3.3008	-6.2691	10.056
3	57.159	14.404	3.1985	-6.7071	8.5222
4	43.123	18.373	1.9809	-7.7614	5.5561
5	48.399	18.350	2.1869	-7.3262	6.6063
6	51.548	20.215	2.0926	-7.8507	6.5659
7	47.995	20.232	1.9695	-7.3690	6.5131
8	47.878	20.679	1.9231	-6.7766	7.0653
9	65.821	19.912	2.5941	-7.7905	8.4489
10	73.020	19.256	2.9131	-8.3159	8.7808
11	68.938	17.966	2.9834	-8.2646	8.3414
12	62.536	18.038	2.7481	-8.6959	7.1915

Table 2.5: Weights Diagnostics per **month — 20 assets with allowed short selling**

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.49877	0.11736	0.97617
2	1.14850	0.26135	0.88758
3	1.21750	0.26678	0.88210
4	1.23640	0.27305	0.87742
5	1.32120	0.28770	0.86432
6	1.48890	0.31660	0.84008
7	1.55680	0.33649	0.81437
8	1.75420	0.37468	0.77484
9	1.78650	0.39013	0.76450
10	1.76080	0.37193	0.77414
11	1.53970	0.34235	0.80306
12	1.95670	0.42590	0.70246

Table 2.6: Returns Diagnostics per **month — 20 assets with allowed short selling**

Start month	Returns L1	Returns L2
1	0.073828	0.024673
2	0.167580	0.059289
3	0.170610	0.061570
4	0.169660	0.060654
5	0.173930	0.067489
6	0.173720	0.063283
7	0.178550	0.064078
8	0.176870	0.060073
9	0.221150	0.077239
10	0.224060	0.079912
11	0.108260	0.041487
12	0.235860	0.082973

Table 2.7: KF Performance per **month — 20 assets with allowed short selling**

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	26.167	16.790	1.3979	-7.4143	3.5292
2	34.225	14.454	2.0616	-7.4399	4.6002
3	41.504	16.060	2.1932	-7.5187	5.5201
4	24.677	18.360	1.2124	-7.5777	3.2566
5	35.508	17.857	1.7233	-7.6440	4.6453
6	34.047	17.644	1.6812	-7.7052	4.4187
7	24.597	18.721	1.1855	-7.7657	3.1674
8	27.871	18.974	1.3091	-6.8473	4.0703
9	28.357	18.894	1.3351	-6.8349	4.1488
10	35.773	17.549	1.7651	-6.8227	5.2432
11	30.751	16.778	1.6160	-6.8112	4.5148
12	19.741	16.632	1.0914	-6.8003	2.9030

Table 2.8: Real Performance per **month — 20 assets with allowed short selling**

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	29.518	16.258	1.6082	-7.4874	3.9424
2	47.289	14.004	2.8102	-6.8180	6.9359
3	54.281	15.223	2.9004	-6.9435	7.8175
4	36.400	18.495	1.7003	-7.1664	5.0793
5	46.872	18.168	2.1500	-7.4087	6.3266
6	51.545	18.331	2.3076	-7.0619	7.2991
7	37.047	20.166	1.5835	-7.1938	5.1499
8	36.243	20.332	1.5409	-6.5110	5.5664
9	45.025	20.339	1.8563	-6.8583	6.5651
10	55.675	19.971	2.2576	-8.1160	6.8599
11	41.678	16.215	2.1800	-6.9823	5.9691
12	37.316	17.547	1.8313	-8.3212	4.4844

Table 2.9: Weights Diagnostics per **month — 30 assets with allowed short selling**

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.69538	0.15045	0.97313
2	1.27760	0.30223	0.90221
3	1.34500	0.31736	0.88482
4	1.46040	0.33969	0.87139
5	1.65040	0.37357	0.84623
6	1.80810	0.40483	0.82584
7	2.02640	0.45034	0.79297
8	2.13460	0.48864	0.75216
9	2.29330	0.52047	0.73530
10	2.29400	0.48640	0.73289
11	2.16140	0.45493	0.75016
12	2.29730	0.48767	0.70413

Table 2.10: Returns Diagnostics per **month — 30 assets with allowed short selling**

Start month	Returns L1	Returns L2
1	0.082232	0.027723
2	0.224120	0.082741
3	0.231100	0.087879
4	0.242180	0.091865
5	0.273340	0.107220
6	0.270510	0.110400
7	0.288280	0.107550
8	0.218130	0.071485
9	0.218130	0.075881
10	0.245430	0.076923
11	0.174520	0.056106
12	0.188430	0.074478

Table 2.11: KF Performance per **month — 30 assets with allowed short selling**

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	18.368	18.663	0.90994	-6.6541	2.7604
2	26.609	16.087	1.48110	-6.8961	3.8585
3	35.958	16.695	1.86370	-7.1023	5.0629
4	23.445	18.096	1.17420	-7.2825	3.2193
5	28.889	17.242	1.48750	-7.4424	3.8816
6	28.840	17.130	1.49500	-7.5862	3.8016
7	19.616	17.603	1.02520	-7.7171	2.5419
8	20.606	17.658	1.06940	-5.9108	3.4862
9	20.971	17.681	1.08530	-4.8949	4.2842
10	28.025	15.911	1.56890	-4.8428	5.7870
11	25.466	15.044	1.52230	-4.7939	5.3121
12	18.374	13.431	1.26480	-4.7478	3.8700

Table 2.12: Real Performance per **month — 30 assets with allowed short selling**

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	20.739	19.163	0.99124	-6.7035	3.0938
2	45.244	16.242	2.33410	-5.8695	7.7084
3	54.410	17.158	2.57840	-6.0985	8.9219
4	37.516	19.704	1.63840	-7.0602	5.3137
5	44.229	18.583	2.00120	-7.2588	6.0932
6	56.452	19.989	2.28140	-7.1555	7.8893
7	37.910	21.648	1.50490	-7.0714	5.3610
8	27.385	19.939	1.22630	-6.9077	3.9645
9	29.543	21.508	1.21650	-6.9556	4.2474
10	40.103	17.977	1.90240	-6.3249	6.3406
11	37.464	16.676	1.93370	-5.8451	6.4096
12	14.729	16.640	0.83046	-10.336	1.4251

Table 2.13: Weights Diagnostics per **month — 40 assets with allowed short selling**

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.87775	0.16712	0.97587
2	1.17250	0.25472	0.94335
3	1.34190	0.28577	0.93024
4	1.43910	0.31738	0.91555
5	1.76010	0.36289	0.88139
6	1.76860	0.37681	0.87524
7	2.37440	0.46939	0.81805
8	2.41120	0.47995	0.80728
9	2.55430	0.52309	0.77699
10	2.44290	0.50248	0.78681
11	2.48650	0.50632	0.78199
12	2.89570	0.57349	0.71775

Table 2.14: Returns Diagnostics per **month — 40 assets with allowed short selling**

Start month	Returns L1	Returns L2
1	0.12729	0.046664
2	0.18852	0.068628
3	0.20699	0.074439
4	0.19480	0.071618
5	0.19055	0.072374
6	0.16557	0.066993
7	0.21658	0.086963
8	0.17035	0.067559
9	0.15787	0.058448
10	0.17190	0.058730
11	0.17504	0.059914
12	0.35150	0.108440

Table 2.15: KF Performance per **month — 40 assets with allowed short selling**

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	19.147	24.534	0.71932	-7.3511	2.6047
2	32.398	19.763	1.43680	-7.2450	4.4718
3	45.113	20.341	1.85920	-7.2499	6.2225
4	30.494	21.590	1.24660	-7.2277	4.2191
5	37.867	21.081	1.54380	-7.2632	5.2136
6	43.015	21.221	1.71140	-7.2508	5.9324
7	25.265	20.823	1.09200	-7.2972	3.4623
8	28.339	20.641	1.22140	-5.4759	5.1752
9	23.259	22.308	0.94565	-9.2383	2.5177
10	33.935	20.823	1.42040	-9.1440	3.7112
11	25.537	18.869	1.21680	-9.0966	2.8074
12	20.238	17.342	1.07100	-9.0113	2.2459

Table 2.16: Real Performance per **month — 40 assets with allowed short selling**

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	12.718	24.581	0.48949	-6.5229	1.9498
2	30.653	17.047	1.58600	-6.2007	4.9434
3	40.275	17.850	1.92300	-6.1460	6.5531
4	29.693	19.859	1.32350	-6.3713	4.6603
5	36.377	19.279	1.63030	-8.0619	4.5123
6	44.472	19.465	1.91940	-7.8066	5.6968
7	19.082	18.437	0.95416	-6.9745	2.7360
8	16.723	18.045	0.86251	-6.0424	2.7677
9	17.257	21.381	0.74955	-9.1477	1.8865
10	29.761	19.735	1.33450	-7.9683	3.7349
11	26.324	19.284	1.22360	-8.4050	3.1320
12	12.327	20.778	0.56216	-9.9245	1.2421

- For each **start month** (months $1, \ldots, 12$) the tables report metrics on the same 12-month window for both **KF** and **Real** portfolios:
  - Annualized return and volatility,
  - Sharpe ratio,
  - Maximum drawdown,
  - Calmar ratio.
- Two diagnostics quantify agreement between KF and Real:
  - *Weights L1/L2/CosineSim*: proximity and directional alignment of forecasted weights to the benchmark weights at the *start month*.
  - *Returns L1/L2*: discrepancy between the two out-of-sample 12-month return streams on that window.

Key takeaways from the metrics.

- **10 assets with forced short selling.** Very close tracking: *CosineSim* around 0.8–1.0 and small *L1/L2* distances. The two return paths are almost identical; Real is usually slightly higher on AR/SR/CR, while KF stays close in most months.
- **20 assets with forced short selling.** Gaps widen as the year progresses: *L1* grows from roughly 0.5 toward 2.0 and *CosineSim* drifts toward ~ 0.70. Real tends to lead on AR and SR across many months, consistent with a more conservative KF.
- **30 assets with forced short selling.** Differences increase further (*L1* near ~ 2.3, *CosineSim* around 0.70). KF keeps volatility contained but does not consistently translate it into higher Sharpe; Real outperforms in most months, though the paths remain similar.
- **40 assets with forced short selling.** Largest dispersion (*L1* up to ~ 2.9; *CosineSim* near 0.72 by late months) and more variable month-to-month leadership. Shapes are preserved, but Real generally achieves higher levels, reflecting the increasing difficulty of forecasting in a larger cross-section.

2.2.3. Results

Kalman Filter - Comments Across universe sizes, the KF behaves as a stability-first forecaster with smooth weights, bounded tails and investable risk–return profiles.

- With *10 assets* with forced short selling SR is strongest (often > 2; CR $\approx$ 4–10) with moderate Vol; moving to *20–30 assets* SR moderates ($\sim$ 1.2–2.2) and CR $\sim$ 3–6; at *40 assets* SR is typically 0.7–1.9.
- Vol remains contained (mid-teens to low-20s as the universe widens); MDD stays tightly bounded (about -5% to -9%) across months.
- Allowing shorts does not destabilize the process: paths remain smooth and drawdowns disciplined, preserving implementable CR levels.

Real vs Kalman Filter - Comments The KF forecast reproduces the *shape* of the benchmark path well, while the level alignment depends on the *start month* and on the size of the investable universe. As the number of assets with allowed short selling increases, weight proximity deteriorates and performance gaps widen.
Real often keeps a slight edge, though the KF remains competitive in parts of the year.

- **Distance widens over later start months.** As the start month moves further from the training period, the Kalman filter relies more on its own state dynamics and less on information implicit in the contemporaneously re-estimated weights. Accordingly, *Weights L1/L2* rise and *CosineSim* falls from early to late months.
- **Looser short-selling constraints mean larger gaps.** Bigger universes and more permissive short selling make the efficient frontier more flexible, so small estimation changes trigger larger weight shifts. Consistently, higher *Weights L1/L2* come with higher *Returns L1/L2*.
- **Forecasting remains effective with short selling in this setup.** Despite the added flexibility, in this best-in-class specification the KF delivers stable, path-preserving forecasts and competitive metrics even when short selling is allowed. This contrasts with other short-selling frontiers discussed in the first chapter, which were not forecastable in practice due to lack of stability.

2.3. ESG Integration via Linear Limits on Low-Score Assets – Forecasting Results

2.3.1. Plots

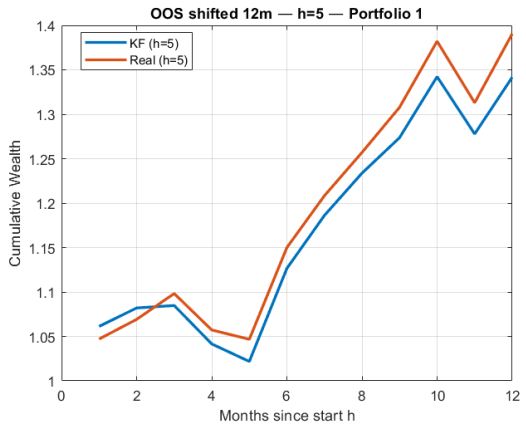


Figure 2.5: Best in Class KF vs Real - Number of Assets with smaller weights: 10



Figure 2.6: Best in Class KF vs Real - Number of Assets with smaller weights: 20



Figure 2.7: Best in Class KF vs Real - Number of Assets with smaller weights: 30



Figure 2.8: Best in Class KF vs Real - Number of Assets with smaller weights: 40

10 assets with smaller weights. Both lines evolve almost identically over the 12-month horizon. They share the same turning points and magnitude of fluctuations, especially the mid-year dip and the late-year climb. The Real curve remains slightly above the KF one, but the difference is marginal, confirming strong qualitative agreement.

20 assets with smaller weights. The two trajectories maintain a nearly perfect match in shape and timing. KF tends to stay a bit lower than Real after mid-year, yet both

capture the same upward momentum.
The visual overlap underlines the model's ability to replicate real performance closely.

30 assets with smaller weights. The evolution remains qualitatively consistent: both curves decline modestly mid-year and rise steadily afterward. KF trails the Real path slightly across most months but keeps pace in directional changes and slope, suggesting stable calibration despite minor amplitude differences.

40 assets with smaller weights. The two paths are almost indistinguishable in direction and turning points, confirming the robustness of the estimation process. Although the Real line remains modestly higher toward the end of the year, the KF line reproduces the same trend and structure—showing that, even as the cross-section grows, their qualitative behavior remains nearly identical.

2.3.2. Metrics

Table 2.17: Weights Diagnostics per **month — 10 assets with smaller weights**

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.094068	0.027799	0.99696
2	0.25000	0.090172	0.96934
3	0.27929	0.099179	0.96356
4	0.27643	0.10040	0.96318
5	0.46898	0.14909	0.92299
6	0.43993	0.14501	0.92700
7	0.48960	0.16279	0.90808
8	0.45935	0.15877	0.91432
9	0.48888	0.17851	0.89466
10	0.56891	0.19147	0.87581
11	0.54618	0.18613	0.88335
12	0.60139	0.19305	0.87578

Table 2.18: Returns Diagnostics per **month — 10 assets with smaller weights**

Start month	Returns L1	Returns L2
1	0.013767	0.0048714
2	0.057379	0.0219060
3	0.058755	0.0230460
4	0.052446	0.0218790
5	0.080513	0.0328810
6	0.066419	0.0281460
7	0.075127	0.0327570
8	0.065300	0.0246650
9	0.095770	0.0338650
10	0.103730	0.0357050
11	0.099270	0.0338290
12	0.106120	0.0356910

Table 2.19: KF Performance per **month — 10 assets with smaller weights**

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	32.658	13.224	2.1624	-5.9345	5.5030
2	38.828	12.781	2.6021	-5.8901	6.5920
3	41.195	13.062	2.6793	-5.8550	7.0359
4	30.566	15.198	1.7745	-5.8270	5.2456
5	35.574	15.433	1.9973	-5.8044	6.1288
6	35.112	15.352	1.9851	-5.7859	6.0685
7	32.733	15.539	1.8440	-5.7706	5.6724
8	37.325	15.363	2.0921	-4.7408	7.8731
9	44.165	13.740	2.7032	-4.7113	9.3743
10	48.935	12.762	3.1738	-4.6791	10.458
11	47.022	12.012	3.2605	-4.6452	10.123
12	36.942	12.743	2.4996	-4.6100	8.0134

Table 2.20: Real Performance per **month — 10 assets with smaller weights**

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	33.029	13.188	2.1901	-5.8340	5.6615
2	42.037	12.021	2.9623	-5.0988	8.2444
3	43.494	12.224	2.9991	-5.1724	8.4090
4	33.549	14.587	2.0073	-5.1292	6.5407
5	40.383	14.699	2.3406	-5.0064	8.0664
6	42.286	14.945	2.3949	-4.8821	8.6614
7	39.874	14.906	2.2830	-4.8167	8.2782
8	42.086	15.615	2.2828	-5.0989	8.2539
9	53.329	14.148	3.0755	-5.3753	9.9211
10	58.518	14.213	3.3044	-5.7767	10.130
11	53.369	12.747	3.4157	-5.8482	9.1258
12	44.463	12.445	3.0016	-5.5580	7.9999

Table 2.21: Weights Diagnostics per **month — 20 assets with smaller weights**

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.10235	0.035816	0.99594
2	0.22982	0.081716	0.97838
3	0.27413	0.095960	0.97096
4	0.28948	0.101330	0.96847
5	0.41443	0.148230	0.93545
6	0.42156	0.150440	0.93389
7	0.47619	0.168920	0.91671
8	0.45359	0.161450	0.92551
9	0.45469	0.174090	0.91447
10	0.51399	0.186230	0.90223
11	0.54063	0.181860	0.90922
12	0.55570	0.185370	0.90892

Table 2.22: Returns Diagnostics per **month — 20 assets with smaller weights**

Start month	Returns L1	Returns L2
1	0.020808	0.0070708
2	0.060674	0.0214090
3	0.062548	0.0226880
4	0.055385	0.0207500
5	0.082140	0.0302160
6	0.068881	0.0250330
7	0.074928	0.0288500
8	0.065511	0.0222450
9	0.077592	0.0253750
10	0.083486	0.0288240
11	0.088372	0.0291030
12	0.095574	0.0325290

Table 2.23: KF Performance per **month — 20 assets with smaller weights**

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	31.188	12.934	2.1228	-5.9318	5.2578
2	37.831	12.454	2.6111	-5.8384	6.4797
3	41.597	12.915	2.7324	-5.7581	7.2240
4	30.585	14.991	1.8000	-5.6894	5.3758
5	37.677	15.388	2.1058	-5.6305	6.6915
6	37.941	15.424	2.1137	-5.5801	6.7994
7	35.751	15.707	1.9710	-5.5367	6.4570
8	39.342	15.600	2.1564	-4.6773	8.4111
9	45.907	14.164	2.7098	-4.6502	9.8720
10	51.524	13.111	3.2252	-4.6203	11.151
11	50.043	12.574	3.2822	-4.5885	10.906
12	40.602	13.166	2.6254	-4.5554	8.9129

Table 2.24: Real Performance per **month — 20 assets with smaller weights**

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	31.712	13.032	2.1380	-6.0163	5.2710
2	40.993	11.778	2.9590	-5.2933	7.7442
3	42.852	12.159	2.9772	-5.4640	7.8426
4	32.745	14.353	1.9969	-5.3969	6.0673
5	41.403	14.412	2.4389	-4.9413	8.3789
6	43.768	14.642	2.5173	-4.9223	8.8918
7	40.689	14.641	2.3652	-5.1187	7.9491
8	41.378	15.253	2.3032	-4.5087	9.1772
9	51.496	14.073	3.0033	-4.7592	10.820
10	56.520	13.343	3.4212	-4.9864	11.335
11	53.776	12.616	3.4728	-5.2483	10.246
12	44.879	12.263	3.0704	-5.1650	8.6890

Table 2.25: Weights Diagnostics per **month — 30 assets with smaller weights**

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.10301	0.036389	0.99584
2	0.24325	0.094293	0.97223
3	0.26928	0.103170	0.96738
4	0.28380	0.107770	0.96524
5	0.43267	0.152500	0.93485
6	0.44527	0.154150	0.93407
7	0.52572	0.179540	0.91025
8	0.48123	0.170780	0.92016
9	0.47331	0.186810	0.90588
10	0.55194	0.201050	0.89078
11	0.58267	0.195120	0.89946
12	0.58596	0.200450	0.89675

Table 2.26: Returns Diagnostics per **month — 30 assets with smaller weights**

Start month	Returns L1	Returns L2
1	0.024386	0.0091337
2	0.069589	0.0258270
3	0.058711	0.0228090
4	0.051454	0.0216130
5	0.073599	0.0328380
6	0.067633	0.0303290
7	0.080424	0.0342140
8	0.076497	0.0276800
9	0.112770	0.0381870
10	0.122860	0.0427520
11	0.125750	0.0428660
12	0.130940	0.0432220

Table 2.27: KF Performance per **month — 30 assets with smaller weights**

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	28.387	12.873	1.9615	-5.4215	5.2359
2	36.088	11.973	2.6068	-5.3200	6.7835
3	40.161	12.405	2.7602	-5.2315	7.6768
4	29.858	14.064	1.8781	-5.1549	5.7921
5	37.513	14.319	2.2545	-5.0892	7.3710
6	37.061	14.251	2.2414	-5.0332	7.3633
7	33.678	14.603	2.0119	-4.9859	6.7546
8	37.291	14.443	2.2236	-3.8621	9.6554
9	40.298	13.761	2.4956	-3.8292	10.524
10	46.741	12.658	3.0787	-3.7959	12.313
11	45.677	12.267	3.1156	-3.7625	12.140
12	36.090	12.148	2.5694	-3.7291	9.6780

Table 2.28: Real Performance per **month — 30 assets with smaller weights**

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	30.423	13.096	2.0509	-5.7002	5.3372
2	41.007	11.827	2.9475	-5.3254	7.7004
3	42.904	12.200	2.9702	-5.4465	7.8774
4	32.589	14.247	2.0034	-5.4073	6.0268
5	42.866	14.151	2.5588	-4.7625	9.0008
6	45.097	14.234	2.6560	-4.7099	9.5748
7	41.377	14.370	2.4447	-4.9909	8.2906
8	41.450	15.014	2.3434	-4.1476	9.9938
9	51.004	14.010	2.9929	-4.3947	11.606
10	55.785	13.175	3.4277	-4.6632	11.963
11	53.127	12.413	3.4944	-4.8688	10.912
12	44.114	11.816	3.1402	-4.9194	8.9673

Table 2.29: Weights Diagnostics per **month — 40 assets with smaller weights**

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.10204	0.036585	0.99576
2	0.24456	0.094291	0.97220
3	0.26892	0.103070	0.96734
4	0.28320	0.107510	0.96523
5	0.43370	0.152350	0.93484
6	0.44276	0.154890	0.93347
7	0.52375	0.179000	0.91038
8	0.47909	0.170580	0.91993
9	0.47171	0.186310	0.90589
10	0.55099	0.200360	0.89082
11	0.58169	0.194400	0.89939
12	0.58497	0.199750	0.89654

Table 2.30: Returns Diagnostics per **month — 40 assets with smaller weights**

Start month	Returns L1	Returns L2
1	0.024387	0.0091250
2	0.070100	0.0259190
3	0.059389	0.0229330
4	0.051935	0.0217120
5	0.073768	0.0329970
6	0.068310	0.0311220
7	0.080126	0.0342440
8	0.076251	0.0278060
9	0.112440	0.0381370
10	0.122330	0.0426350
11	0.124970	0.0426920
12	0.130760	0.0431500

Table 2.31: KF Performance per **month — 40 assets with smaller weights**

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	28.345	12.871	1.9592	-5.4254	5.2245
2	36.017	11.969	2.6032	-5.3267	6.7616
3	40.055	12.398	2.7555	-5.2402	7.6439
4	29.736	14.056	1.8724	-5.1648	5.7574
5	37.373	14.307	2.2490	-5.0997	7.3285
6	36.899	14.236	2.2353	-5.0438	7.3156
7	33.551	14.584	2.0078	-4.9964	6.7151
8	37.170	14.426	2.2200	-3.8721	9.5993
9	40.210	13.737	2.4952	-3.8395	10.473
10	46.640	12.634	3.0790	-3.8065	12.253
11	45.585	12.244	3.1160	-3.7734	12.081
12	36.011	12.146	2.5649	-3.7404	9.6277

Table 2.32: Real Performance per **month — 40 assets with smaller weights**

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	30.351	13.104	2.0453	-5.7075	5.3177
2	41.007	11.827	2.9475	-5.3254	7.7004
3	42.904	12.200	2.9702	-5.4465	7.8774
4	32.589	14.247	2.0034	-5.4073	6.0268
5	42.869	14.150	2.5591	-4.7619	9.0024
6	45.173	14.239	2.6590	-4.7158	9.5792
7	41.377	14.370	2.4447	-4.9909	8.2906
8	41.415	15.014	2.3417	-4.1434	9.9954
9	51.004	14.010	2.9929	-4.3947	11.606
10	55.785	13.175	3.4277	-4.6632	11.963
11	53.127	12.413	3.4944	-4.8688	10.912
12	44.114	11.816	3.1402	-4.9194	8.9673

Key takeaways from the metrics.

- **10 assets with smaller weights.** Tight alignment: *CosineSim* stays very high (from ≈ 0.997 early to ≈ 0.876 late) and *Weights L1/L2* remain small (≈ 0.09–0.60; 0.028–0.193). Return discrepancies are minimal (*Returns L1/L2* mostly $\leq 0.11/0.036$). Real usually posts the higher AR/SR/CR, but KF is close in most start months.

- **20 assets with smaller weights.** Small but clearer gaps as months progress: *Weights L1* rise to ~ 0.56 and *CosineSim* drifts toward ~ 0.90. *Returns L1/L2* also edge up (peaking near 0.096/0.033). Real tends to lead on AR and SR across many start months, consistent with a slightly more conservative KF profile.

- **30 assets with smaller weights.** Differences broaden modestly: *Weights L1* reach ~ 0.59 and *CosineSim* settles around 0.90 by later months; *Returns L1/L2* increase (up to ~ 0.131). KF contains volatility well but does not consistently convert it into higher Sharpe; Real outperforms in most months while paths remain similar.

- **40 assets with smaller weights.** Largest dispersion within this set, yet still high directional agreement: *CosineSim* remains near 0.90 late in the year, with *Weights L1* similar to the 30-asset case (~ 0.58) and *Returns L1/L2* peaking near

0.131. Shapes are preserved, but Real generally achieves higher levels, reflecting the increasing difficulty of forecasting as the universe expands.

2.3.3. Results

Kalman Filter - Comments

- With *10 assets* with smaller weights SR is strongest (often > 2.5; CR between $5-10$) with low Vol; for *20–30 assets* SR remains high ($\sim$ 2.1–3.2) and CR $\sim$ 7–12; at *40 assets* SR typically stays around 2.0–3.1.
- Vol stays contained (low- to mid-teens across months); MDD remains tightly bounded (about -6% to -4%), indicating robust downside discipline from the recursive update.
- Binding weight caps do *not* destabilize the process: paths remain smooth and drawdowns disciplined, preserving implementable CR levels even as the investable universe expands.

Real vs Kalman Filter - Comments The KF forecast reproduces the *shape* of the benchmark path very well. Level alignment varies with the *start month* and with the size of the investable universe. In this chapter short selling is *not* allowed; instead, assets with low ESG scores are subject to linear *maximum-weight caps*. Under these caps, weight proximity is generally tight and performance gaps remain modest, even as the universe expands.

- **Distance widens for later start months.** As the start month moves away from the training window, the filter relies more on its own state dynamics and less on information implicit in the re-estimated "Real" portfolio. Consequently, *Weights L1/L2* tend to rise and *CosineSim* tends to fall from early to late months.
- **ESG caps shape—but stabilize—the frontier.** Binding maximum-weight limits on low-ESG assets make the efficient frontier less flexible than under short selling, which dampens extreme reallocations. When caps bind more frequently (e.g., in larger universes), small estimation changes can still shift weights, so higher *Weights L1/L2* are accompanied by higher *Returns L1/L2*; yet the caps help keep those gaps contained relative to short-selling settings.
- **Forecasting remains effective under ESG caps.** Despite the constraints, the best-in-class KF delivers stable, path-preserving forecasts and competitive AR/SR/CR.

2.4. ESG Integration Frontier via Constrained Optimization - Forecasting Results

2.4.1. Plots

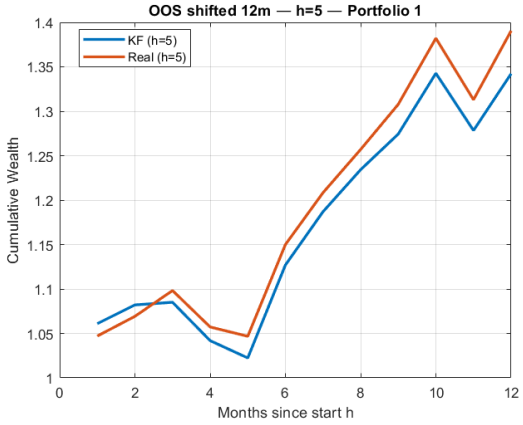


Figure 2.9: Constrained Frontier KF vs Real - $\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 0$

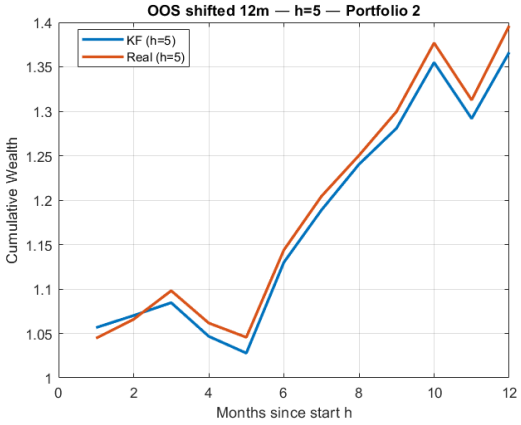


Figure 2.10: Constrained Frontier KF vs Real - $\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 1.89$



Figure 2.11: Constrained Frontier KF vs Real - $\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 3.78$

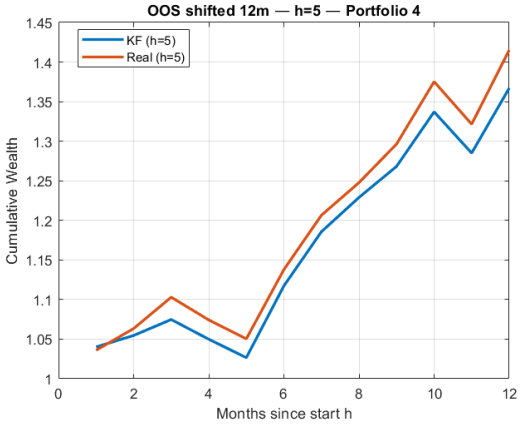


Figure 2.12: Constrained Frontier KF vs Real - $\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 5.67$

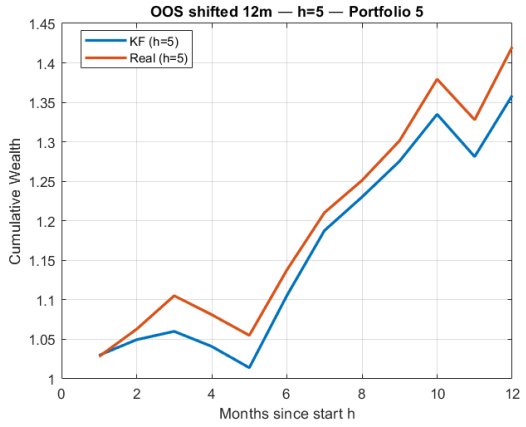


Figure 2.13: Constrained Frontier KF vs Real - $\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 7.56$

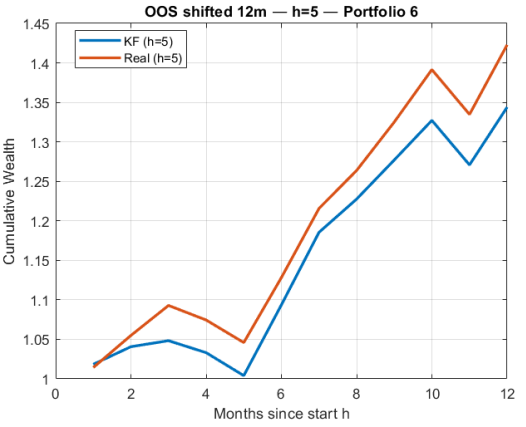


Figure 2.14: Constrained Frontier KF vs Real - $\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 9.44$

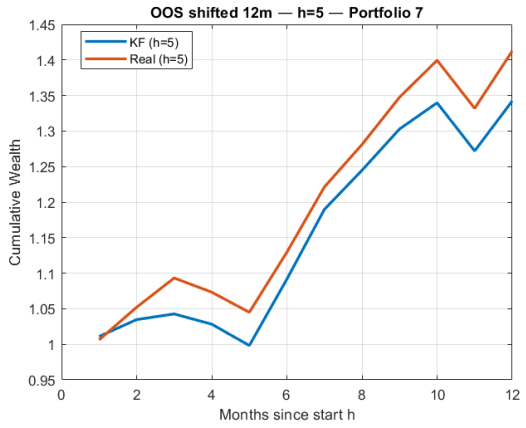


Figure 2.15: Constrained Frontier KF vs Real - $\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 11.33$

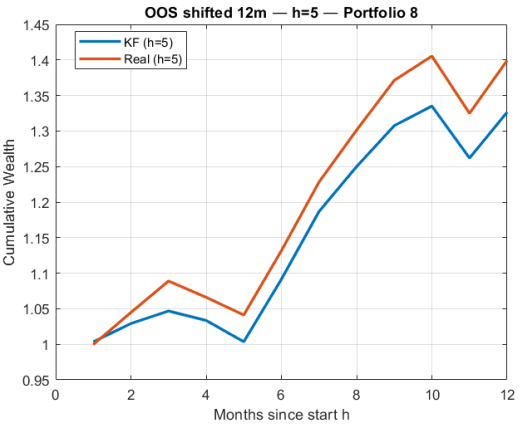


Figure 2.16: Constrained Frontier KF vs Real - $\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 13.22$



Figure 2.17: Constrained Frontier KF vs Real - $\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 15.11$

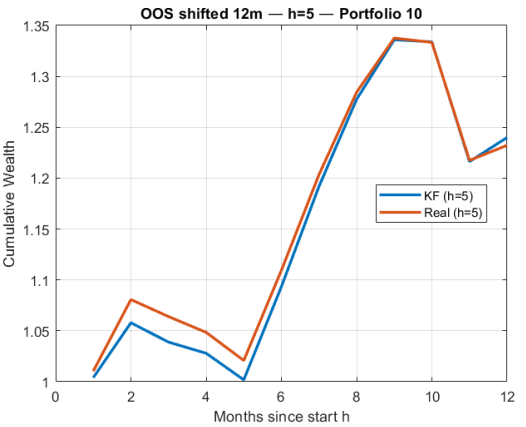


Figure 2.18: Constrained Frontier KF vs Real - $\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq 17$

$\sum w_i \, \mathrm{ESG}_i \geq 0$**.** Both paths are almost identical throughout the year. The KF closely follows the Real trajectory, matching every turning point. The Real curve finishes marginally higher, but overall alignment is excellent. Very strong tracking under an unconstrained ESG setting.

$\sum w_i \, \mathrm{ESG}_i \geq 1.89$**.** The two lines overlap nearly perfectly up to mid-year. From month 8 onward, the Real portfolio shows a slightly steeper rise, creating a small terminal gap. Filter remains stable but underreacts during the final upswing.

$\sum w_i \, \mathrm{ESG}_i \geq 3.78$**.** Cumulative paths remain well aligned, sharing the same turning points and similar slopes. The KF lags modestly in late months, leading to a small level difference at year-end. Correct direction, mild amplitude gap.

$\sum w_i \, \mathrm{ESG}_i \geq 5.67$**.** The two trajectories follow a common pattern—decline around month 5, recovery after month 6. The Real curve outpaces slightly during the last quarter. Consistent directional match; KF smooths the late rally.

$\sum w_i \, \mathrm{ESG}_i \geq 7.56$**.** Shapes remain very similar, with both curves displaying synchronized peaks and troughs. The Real portfolio rises more sharply near year-end, finishing above KF. Persistent underreaction of KF in stronger markets.

$\sum w_i \, \mathrm{ESG}_i \geq 9.44$**.** The two paths maintain close co-movement but the Real line gradually diverges upward after month 7. The gap widens slightly but directions remain aligned. Real responds faster to positive momentum.

$\sum w_i \, \mathrm{ESG}_i \geq 11.33$**.** Very similar shapes throughout the horizon, with KF staying just below Real after mid-year. The Real series accelerates more at the end, leaving a visible but stable gap. Steady tracking with moderate level drift.

$\sum w_i \, \mathrm{ESG}_i \geq 13.22$**.** The trajectories evolve almost identically for the first eight months; later, the Real line climbs faster. The terminal difference remains contained. Small lag in adaptation to the late-year rally.

$\sum w_i \, \mathrm{ESG}_i \geq 15.11$**.** The two lines track each other closely, with minor deviations mid-year and a modest gap developing near the end. Good shape preservation, slightly smoother KF path.

$\sum w_i \, \mathrm{ESG}_i \geq 17$**.** Nearly perfect overlap: both capture every inflection point and end at almost the same level. Well-calibrated model response under the highest ESG constraint.

2.4.2. Metrics

Table 2.33: Weights Diagnostics per **month** — $\sum \boldsymbol{w_i \, \mathrm{ESG}_i \geq 0}$

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.093884	0.027705	0.99697
2	0.24711	0.089486	0.96980
3	0.27712	0.098891	0.96373
4	0.27276	0.099456	0.96383
5	0.46916	0.14804	0.92404
6	0.44014	0.14363	0.92833
7	0.47807	0.16084	0.90964
8	0.45963	0.15712	0.91601
9	0.48918	0.17705	0.89630
10	0.56964	0.18995	0.87769
11	0.55326	0.18743	0.88152
12	0.60261	0.19165	0.87758

Table 2.34: Returns Diagnostics per **month** — $\sum \boldsymbol{w_i \, \mathrm{ESG}_i \geq 0}$

Start month	Returns L1	Returns L2
1	0.01372	0.0048121
2	0.056499	0.021671
3	0.05815	0.022856
4	0.051395	0.021534
5	0.079904	0.032507
6	0.065773	0.027888
7	0.0734	0.031529
8	0.064924	0.024555
9	0.096356	0.034108
10	0.10466	0.036063
11	0.098734	0.033464
12	0.1056	0.035639

Table 2.35: KF Performance per **month** — $\sum \boldsymbol{w_i \, \mathrm{ESG}_i \geq 0}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	32.648	13.214	2.1634	-5.9285	5.5069
2	38.811	12.765	2.6046	-5.8780	6.6028
3	41.163	13.036	2.6829	-5.8368	7.0524
4	30.582	15.167	1.7790	-5.8028	5.2702
5	35.631	15.395	2.0050	-5.7745	6.1704
6	35.184	15.310	1.9941	-5.7506	6.1183
7	32.800	15.491	1.8531	-5.7301	5.7242
8	37.339	15.314	2.0994	-4.7311	7.8922
9	44.108	13.695	2.7092	-4.7011	9.3825
10	48.792	12.734	3.1728	-4.6685	10.451
11	46.889	11.990	3.2589	-4.6342	10.118
12	36.792	12.681	2.5030	-4.5986	8.0008

Table 2.36: Real Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{0}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	33.029	13.188	2.1901	-5.8340	5.6615
2	42.037	12.021	2.9623	-5.0988	8.2444
3	43.439	12.217	2.9977	-5.1778	8.3894
4	33.545	14.579	2.0083	-5.1317	6.5368
5	40.383	14.699	2.3406	-5.0064	8.0664
6	42.286	14.945	2.3949	-4.8821	8.6614
7	39.580	14.864	2.2749	-4.8386	8.1800
8	42.073	15.609	2.2831	-5.0947	8.2582
9	53.329	14.148	3.0755	-5.3753	9.9211
10	58.489	14.203	3.3055	-5.7699	10.137
11	53.114	12.636	3.4319	-5.7813	9.1871
12	44.463	12.445	3.0016	-5.5580	7.9999

Table 2.37: Weights Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{1.89}$

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.07403	0.026424	0.99752
2	0.22581	0.076184	0.97954
3	0.25902	0.087875	0.97347
4	0.25971	0.089519	0.97312
5	0.40702	0.13489	0.94137
6	0.38838	0.13336	0.94316
7	0.46123	0.15479	0.92385
8	0.43630	0.14544	0.93432
9	0.44632	0.15451	0.92693
10	0.53901	0.17172	0.91012
11	0.52097	0.17030	0.91370
12	0.55526	0.18895	0.89428

Table 2.38: Returns Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{1.89}$

Start month	Returns L1	Returns L2
1	0.014031	0.0050669
2	0.049557	0.017699
3	0.055516	0.019316
4	0.045553	0.016918
5	0.066492	0.025066
6	0.053551	0.020565
7	0.060950	0.023510
8	0.053517	0.019786
9	0.074581	0.025043
10	0.088366	0.031276
11	0.087318	0.030505
12	0.102770	0.034965

Table 2.39: KF Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{1.89}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	32.903	12.752	2.2573	-5.6048	5.8705
2	39.341	12.259	2.7441	-5.4959	7.1583
3	41.719	12.531	2.8233	-5.3988	7.7274
4	31.259	14.647	1.8782	-5.3119	5.8847
5	37.982	14.987	2.1772	-5.2340	7.2568
6	38.844	15.099	2.2035	-5.1638	7.5224
7	36.896	15.400	2.0662	-5.1005	7.2338
8	39.933	15.365	2.2177	-4.6484	8.5908
9	46.523	13.928	2.7869	-4.6327	10.042
10	51.479	13.062	3.2350	-4.6136	11.158
11	49.711	12.420	3.3043	-4.5918	10.826
12	40.904	12.864	2.7042	-4.5680	8.9545

Table 2.40: Real Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{1.89}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	33.393	13.013	2.2409	-5.8238	5.7338
2	42.170	11.824	3.0199	-5.1417	8.2015
3	43.296	12.007	3.0414	-5.2237	8.2885
4	33.294	14.274	2.0376	-5.1743	6.4344
5	40.887	14.379	2.4184	-4.7743	8.5640
6	43.437	14.744	2.4838	-4.7169	9.2089
7	40.595	14.708	2.3497	-4.8647	8.3449
8	41.807	15.404	2.3009	-4.8008	8.7084
9	52.347	14.018	3.0564	-5.0050	10.459
10	58.475	13.903	3.3760	-5.4342	10.761
11	54.159	12.657	3.4821	-5.5651	9.7319
12	45.935	12.311	3.1193	-5.5440	8.2855

Table 2.41: Weights Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{3.78}$

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.10477	0.036420	0.99527
2	0.21535	0.075695	0.98099
3	0.25629	0.087327	0.97467
4	0.25832	0.087325	0.97486
5	0.36837	0.124650	0.95139
6	0.36817	0.126430	0.94970
7	0.42807	0.143180	0.93464
8	0.41059	0.138970	0.93962
9	0.42883	0.149780	0.93061
10	0.51349	0.161710	0.91878
11	0.51702	0.154900	0.92733
12	0.54667	0.182330	0.89899

Table 2.42: Returns Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{3.78}$

Start month	Returns L1	Returns L2
1	0.021530	0.0077502
2	0.050186	0.0171660
3	0.054661	0.0192730
4	0.051381	0.0181620
5	0.062200	0.0233060
6	0.052550	0.0212630
7	0.055511	0.0203720
8	0.051616	0.0207110
9	0.072231	0.0248130
10	0.083500	0.0305080
11	0.080469	0.0295690
12	0.101280	0.0353190

Table 2.43: KF Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{3.78}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	30.287	12.212	2.1906	-5.2863	5.7294
2	36.739	11.583	2.7369	-5.1564	7.1249
3	38.843	11.843	2.8094	-5.0422	7.7036
4	29.764	13.674	1.9263	-4.9410	6.0238
5	38.074	14.110	2.3175	-4.8506	7.8495
6	39.474	14.301	2.3590	-4.7693	8.2768
7	37.659	14.677	2.2068	-4.6958	8.0197
8	39.088	14.708	2.2744	-4.1678	9.3786
9	45.202	13.493	2.8074	-4.1784	10.818
10	50.416	12.629	3.2881	-4.1852	12.046
11	48.900	12.098	3.3458	-4.1887	11.674
12	40.801	12.311	2.8195	-4.1892	9.7395

Table 2.44: Real Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{3.78}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	31.459	12.611	2.1938	-5.6461	5.5718
2	41.710	11.677	3.0294	-5.2626	7.9259
3	42.625	11.890	3.0308	-5.4042	7.8875
4	32.845	13.983	2.0554	-5.3547	6.1338
5	41.649	14.087	2.5078	-4.9046	8.4917
6	44.419	14.423	2.5878	-4.8450	9.1681
7	40.818	14.504	2.3940	-5.0740	8.0446
8	41.082	15.118	2.3095	-4.3474	9.4498
9	51.208	13.884	3.0301	-4.5791	11.183
10	57.068	13.399	3.4339	-4.9176	11.605
11	53.626	12.418	3.5202	-5.1207	10.473
12	45.661	12.016	3.1798	-5.3093	8.6002

Table 2.45: Weights Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{5.67}$

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.12727	0.042068	0.99338
2	0.28419	0.100820	0.96922
3	0.32603	0.109570	0.96356
4	0.32445	0.107110	0.96525
5	0.37851	0.127040	0.95300
6	0.38264	0.125360	0.95387
7	0.44116	0.142600	0.93714
8	0.43543	0.142230	0.93858
9	0.44636	0.160240	0.92146
10	0.52424	0.165520	0.91232
11	0.52742	0.155990	0.92084
12	0.54072	0.180550	0.89319

Table 2.46: Returns Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{5.67}$

Start month	Returns L1	Returns L2
1	0.035776	0.011909
2	0.049575	0.019743
3	0.055916	0.021565
4	0.049006	0.020061
5	0.066839	0.026184
6	0.070429	0.025753
7	0.081507	0.026373
8	0.070581	0.023898
9	0.068597	0.025941
10	0.076516	0.029112
11	0.071648	0.028031
12	0.087045	0.032996

Table 2.47: KF Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{5.67}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	29.766	11.802	2.2320	-4.8585	6.1266
2	36.588	11.116	2.8416	-4.7507	7.7015
3	37.459	11.172	2.8859	-4.6532	8.0502
4	29.852	12.995	2.0322	-4.5635	6.5416
5	37.809	13.545	2.3995	-4.4807	8.4383
6	39.944	13.792	2.4711	-4.4042	9.0696
7	37.606	14.270	2.2670	-4.3337	8.6775
8	38.109	14.347	2.2810	-3.9527	9.6411
9	43.666	13.393	2.7465	-3.9688	11.002
10	48.874	12.528	3.2295	-3.9825	12.272
11	46.519	11.762	3.2999	-3.9940	11.647
12	37.861	11.603	2.8046	-4.0035	9.4570

Table 2.48: Real Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{5.67}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	28.722	11.925	2.1397	-5.1601	5.5661
2	40.564	11.164	3.0935	-5.1458	7.8828
3	42.490	11.490	3.1279	-5.2429	8.1043
4	33.323	13.370	2.1771	-5.1846	6.4272
5	42.691	13.701	2.6335	-4.7671	8.9552
6	45.833	13.991	2.7395	-4.7497	9.6496
7	41.348	14.214	2.4701	-5.1221	8.0726
8	39.795	14.646	2.3195	-3.7439	10.629
9	49.451	13.593	3.0059	-3.9976	12.370
10	55.203	12.774	3.5048	-4.2711	12.925
11	52.589	12.038	3.5730	-4.5156	11.646
12	44.644	11.493	3.2616	-4.7538	9.3913

Table 2.49: Weights Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{7.56}$

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.22291	0.087645	0.97067
2	0.46133	0.152120	0.93107
3	0.48052	0.160640	0.92229
4	0.49834	0.164400	0.91995
5	0.52583	0.181400	0.90491
6	0.52343	0.179670	0.90949
7	0.59809	0.198580	0.88552
8	0.57819	0.199750	0.88453
9	0.60475	0.218460	0.85829
10	0.66762	0.230320	0.83882
11	0.69012	0.223260	0.84147
12	0.63541	0.217890	0.83774

Table 2.50: Returns Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{7.56}$

Start month	Returns L1	Returns L2
1	0.10654	0.036440
2	0.10048	0.036495
3	0.10911	0.039177
4	0.10553	0.038718
5	0.10407	0.040588
6	0.12113	0.043352
7	0.14492	0.049243
8	0.12128	0.041194
9	0.11183	0.041268
10	0.12661	0.044223
11	0.12721	0.045033
12	0.11261	0.044811

Table 2.51: KF Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{7.56}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	31.735	12.390	2.2503	-4.5653	6.9514
2	39.547	11.672	2.8949	-4.5010	8.7863
3	38.185	11.373	2.8824	-4.4428	8.5948
4	31.613	13.272	2.0936	-4.3905	7.2004
5	37.022	13.854	2.3035	-4.3436	8.5233
6	39.091	14.028	2.3848	-4.3018	9.0872
7	35.171	14.692	2.0772	-4.2647	8.2469
8	36.228	14.733	2.1257	-4.0550	8.9342
9	41.258	14.029	2.4980	-4.0639	10.152
10	45.904	13.178	2.9124	-4.0712	11.275
11	41.303	11.820	2.9677	-4.0768	10.131
12	30.273	11.045	2.4210	-4.0808	7.4183

Table 2.52: Real Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{7.56}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	25.886	11.438	2.0321	-4.6121	5.6127
2	38.849	10.783	3.0860	-4.8272	8.0481
3	40.822	11.108	3.1263	-4.9474	8.2512
4	33.126	12.857	2.2522	-4.9635	6.6740
5	43.126	13.337	2.7290	-4.5384	9.5025
6	47.103	13.511	2.9030	-4.5501	10.352
7	42.047	13.733	2.5935	-4.9022	8.5771
8	38.607	14.132	2.3418	-3.3284	11.599
9	46.999	13.254	2.9539	-3.5701	13.165
10	53.262	12.094	3.5941	-3.4829	15.293
11	51.215	11.339	3.7106	-3.6573	14.003
12	41.865	11.257	3.1523	-4.5863	9.1282

Table 2.53: Weights Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{9.44}$

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.24413	0.10114	0.96023
2	0.48242	0.18130	0.89407
3	0.51764	0.18245	0.89046
4	0.55891	0.19518	0.87322
5	0.54109	0.19987	0.87166
6	0.60184	0.21888	0.85177
7	0.59983	0.21527	0.85249
8	0.62546	0.22885	0.83487
9	0.67784	0.25384	0.79943
10	0.68220	0.25399	0.79092
11	0.67045	0.24769	0.79287
12	0.72213	0.24691	0.78350

Table 2.54: Returns Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{9.44}$

Start month	Returns L1	Returns L2
1	0.10683	0.038467
2	0.12701	0.050662
3	0.10835	0.039488
4	0.11359	0.043682
5	0.10499	0.040040
6	0.12540	0.046884
7	0.12488	0.044015
8	0.10247	0.037014
9	0.11736	0.041063
10	0.12446	0.041918
11	0.12375	0.041522
12	0.13250	0.048738

Table 2.55: KF Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{9.44}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	33.264	13.361	2.1752	-4.3455	7.6549
2	42.009	12.663	2.8106	-4.3094	9.7481
3	38.201	11.993	2.7345	-4.2765	8.9329
4	32.777	13.784	2.0812	-4.2465	7.7187
5	35.583	14.207	2.1701	-4.2474	8.3776
6	38.562	14.309	2.3105	-4.2675	9.0362
7	33.376	15.172	1.9212	-4.2860	7.7871
8	33.491	15.290	1.9122	-4.3029	7.7835
9	39.257	14.738	2.2783	-4.3181	9.0912
10	42.619	14.121	2.5516	-4.3318	9.8388
11	35.562	12.527	2.4600	-4.3439	8.1868
12	23.258	11.075	1.9047	-4.3545	5.3411

Table 2.56: Real Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{9.44}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	27.827	11.696	2.1206	-4.3593	6.3832
2	38.362	10.895	3.0211	-4.1455	9.2540
3	42.440	11.376	3.1560	-4.6436	9.1395
4	35.467	12.949	2.3742	-4.6881	7.5653
5	43.482	13.837	2.6489	-4.2999	10.112
6	47.894	13.357	2.9780	-4.3180	11.092
7	41.773	13.747	2.5764	-4.5775	9.1258
8	39.041	14.237	2.3472	-3.6328	10.747
9	46.577	13.679	2.8404	-3.7890	12.293
10	52.246	12.543	3.4104	-3.7300	14.007
11	45.750	11.080	3.4539	-3.5250	12.979
12	36.337	10.889	2.8836	-4.7106	7.7138

Table 2.57: Weights Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{11.33}$

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.24085	0.10309	0.95854
2	0.43108	0.16748	0.90062
3	0.48328	0.17697	0.89109
4	0.50699	0.18326	0.87955
5	0.50864	0.19090	0.87458
6	0.54508	0.20149	0.86395
7	0.55625	0.19850	0.86886
8	0.59102	0.21009	0.85615
9	0.59870	0.21666	0.84924
10	0.58448	0.21026	0.85203
11	0.58927	0.21140	0.84901
12	0.55692	0.19432	0.85886

Table 2.58: Returns Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{11.33}$

Start month	Returns L1	Returns L2
1	0.10030	0.038416
2	0.12876	0.050878
3	0.11177	0.041495
4	0.11825	0.046670
5	0.11235	0.043926
6	0.12728	0.048672
7	0.11791	0.042522
8	0.10016	0.034177
9	0.10493	0.034389
10	0.11172	0.035153
11	0.10132	0.032763
12	0.097368	0.038845

Table 2.59: KF Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{11.33}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	39.208	15.265	2.1972	-4.3007	9.1167
2	47.965	14.750	2.7002	-4.2899	11.181
3	39.370	13.333	2.5246	-4.2801	9.1983
4	34.755	15.099	2.0003	-5.0304	6.9091
5	35.598	15.280	2.0184	-5.0532	7.0445
6	39.779	15.334	2.2147	-5.0749	7.8383
7	34.270	16.152	1.8471	-5.0955	6.7256
8	34.687	16.224	1.8584	-5.1151	6.7813
9	41.319	15.708	2.2338	-5.1335	8.0489
10	44.153	15.160	2.4495	-5.1510	8.5717
11	34.788	13.384	2.2585	-5.1674	6.7321
12	20.741	11.884	1.5985	-5.1829	4.0018

Table 2.60: Real Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{11.33}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	33.170	13.063	2.2192	-4.1989	7.8996
2	43.400	12.602	2.9038	-4.0773	10.644
3	43.598	12.335	2.9782	-4.6168	9.4432
4	37.304	13.648	2.3539	-4.6418	8.0365
5	42.577	14.617	2.4629	-4.8304	8.8144
6	49.063	14.264	2.8458	-4.5618	10.755
7	41.481	14.602	2.4110	-4.8466	8.5587
8	38.781	15.079	2.2034	-4.3232	8.9703
9	46.181	14.483	2.6635	-4.3897	10.520
10	50.396	13.161	3.1543	-4.1747	12.072
11	41.959	11.929	2.9805	-4.0435	10.377
12	29.400	11.051	2.3575	-4.8964	6.0044

Table 2.61: Weights Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{13.22}$

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.21581	0.094704	0.96846
2	0.37400	0.145320	0.92974
3	0.37914	0.146230	0.93042
4	0.39277	0.147640	0.92559
5	0.40950	0.154030	0.92208
6	0.39542	0.152880	0.92542
7	0.35621	0.149410	0.93221
8	0.35837	0.151790	0.93054
9	0.37022	0.154070	0.92915
10	0.35779	0.146270	0.93462
11	0.36324	0.147480	0.93573
12	0.42122	0.154510	0.92054

Table 2.62: Returns Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{13.22}$

Start month	Returns L1	Returns L2
1	0.092336	0.035246
2	0.107740	0.042565
3	0.081682	0.032096
4	0.079111	0.034420
5	0.090417	0.035718
6	0.091532	0.036137
7	0.064082	0.023753
8	0.057728	0.021248
9	0.065196	0.023244
10	0.051586	0.020241
11	0.056564	0.023107
12	0.065817	0.022513

Table 2.63: KF Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{13.22}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	44.660	17.545	2.1371	-4.2168	10.591
2	52.462	16.785	2.5573	-4.1944	12.508
3	38.375	14.132	2.3297	-4.1727	9.1967
4	34.203	15.333	1.9424	-5.5411	6.1725
5	33.987	15.122	1.9585	-5.4769	6.2055
6	40.017	14.937	2.2853	-5.4140	7.3915
7	34.506	15.407	1.9480	-5.3525	6.4467
8	32.786	15.354	1.8689	-5.2924	6.1950
9	39.406	14.709	2.2902	-5.2336	7.5294
10	42.153	13.805	2.5855	-5.1762	8.1436
11	32.720	12.519	2.2880	-5.1202	6.3903
12	20.784	11.340	1.6784	-5.0655	4.1030

Table 2.64: Real Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{13.22}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	39.013	15.309	2.1815	-4.0917	9.5348
2	49.350	14.978	2.7233	-3.9864	12.380
3	45.368	14.159	2.6837	-4.5290	10.017
4	39.543	15.115	2.2354	-5.3654	7.3700
5	41.468	15.769	2.2321	-5.7294	7.2378
6	49.581	15.345	2.6687	-5.3922	9.1950
7	40.948	16.099	2.1627	-5.3236	7.6918
8	38.202	16.516	1.9856	-5.4991	6.9471
9	46.311	15.869	2.4366	-5.5668	8.3191
10	47.984	14.723	2.7060	-5.2383	9.1602
11	36.360	13.779	2.2800	-4.9683	7.3184
12	21.982	12.400	1.6158	-5.3541	4.1057

Table 2.65: Weights Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{15.11}$

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.11533	0.064938	0.98846
2	0.18819	0.078993	0.98242
3	0.26727	0.114450	0.96696
4	0.27515	0.123790	0.95882
5	0.31410	0.138350	0.95047
6	0.30729	0.131410	0.95563
7	0.34283	0.149880	0.94559
8	0.35718	0.156730	0.94040
9	0.37112	0.161140	0.93571
10	0.35438	0.163400	0.93518
11	0.39222	0.179470	0.92254
12	0.46442	0.188770	0.90562

Table 2.66: Returns Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{15.11}$

Start month	Returns L1	Returns L2
1	0.067259	0.024294
2	0.055772	0.020253
3	0.094228	0.034580
4	0.107240	0.038898
5	0.104930	0.034238
6	0.106220	0.033840
7	0.103480	0.033397
8	0.121190	0.038217
9	0.123130	0.040263
10	0.118330	0.040275
11	0.123210	0.041605
12	0.103680	0.037785

Table 2.67: KF Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{15.11}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	53.901	20.212	2.1719	-4.4549	12.099
2	61.043	19.528	2.4892	-4.3769	13.947
3	42.074	16.476	2.1630	-4.0634	10.354
4	37.789	17.480	1.8586	-6.8176	5.5429
5	34.971	16.837	1.8036	-6.7425	5.1866
6	42.806	16.502	2.1916	-6.6701	6.4176
7	36.248	16.674	1.8792	-6.6005	5.4918
8	35.673	16.441	1.8793	-6.5335	5.4599
9	42.623	15.681	2.2980	-6.4693	6.5885
10	44.518	14.890	2.5114	-6.4078	6.9475
11	31.707	14.151	1.9687	-6.3490	4.9940
12	18.898	13.060	1.3350	-6.2928	3.0031

Table 2.68: Real Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{15.11}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	49.215	18.545	2.1945	-4.0458	12.164
2	61.337	18.719	2.6070	-4.1644	14.729
3	51.149	18.177	2.3122	-4.3005	11.894
4	45.653	19.322	1.9771	-7.5294	6.0633
5	40.882	18.788	1.8506	-7.7544	5.2721
6	50.871	18.476	2.2644	-7.4265	6.8500
7	39.847	19.461	1.7476	-7.3593	5.4146
8	40.522	19.599	1.7606	-7.4402	5.4464
9	50.335	18.709	2.2166	-7.4517	6.7549
10	48.885	18.426	2.1963	-7.1474	6.8395
11	31.673	17.785	1.5650	-7.1083	4.4557
12	14.821	15.409	0.9021	-6.9044	2.1466

Table 2.69: Weights Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{17}$

Start month	Weights L1	Weights L2	Weights CosineSim
1	0.078797	0.043686	0.99659
2	0.113330	0.053898	0.99481
3	0.194100	0.097160	0.98500
4	0.210400	0.098959	0.98342
5	0.189140	0.089616	0.98650
6	0.186160	0.087362	0.98731
7	0.183260	0.088508	0.98731
8	0.163810	0.080134	0.98947
9	0.131570	0.068458	0.99218
10	0.137740	0.073958	0.99107
11	0.196330	0.098696	0.98486
12	0.329570	0.157500	0.96262

Table 2.70: Returns Diagnostics per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{17}$

Start month	Returns L1	Returns L2
1	0.037938	0.012441
2	0.040561	0.013551
3	0.048108	0.020054
4	0.063110	0.023360
5	0.058774	0.021559
6	0.059945	0.022244
7	0.042029	0.014476
8	0.036118	0.012353
9	0.035002	0.013175
10	0.032297	0.011708
11	0.050793	0.016966
12	0.111260	0.041750

Table 2.71: KF Performance per **month** — $\sum w_i \, \mathbf{ESG}_i \geq \mathbf{17}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	43.429	19.722	1.8565	-5.9169	7.3398
2	49.758	19.190	2.1403	-5.8683	8.4792
3	33.435	17.271	1.6903	-5.4298	6.1577
4	28.963	18.803	1.3672	-9.0222	3.2102
5	25.907	18.490	1.2579	-8.9597	2.8915
6	33.198	18.771	1.5455	-8.9001	3.7301
7	27.143	18.361	1.3211	-8.8433	3.0693
8	34.394	17.871	1.6746	-8.7894	3.9131
9	39.219	17.341	1.9347	-8.7760	4.4689
10	40.510	16.867	2.0453	-8.7695	4.6194
11	24.150	17.081	1.2779	-8.7642	2.7555
12	8.7600	16.788	0.50196	-10.318	0.8490

Table 2.72: Real Performance per **month** — $\sum \boldsymbol{w_i}\,\mathbf{ESG}_i \geq \mathbf{17}$

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	41.062	18.946	1.8421	-5.5515	7.3965
2	47.742	18.529	2.1411	-5.5340	8.6270
3	33.976	16.900	1.7520	-5.8025	5.8554
4	28.401	18.309	1.3797	-9.0154	3.1503
5	25.072	18.243	1.2378	-8.9787	2.7924
6	33.310	18.855	1.5432	-8.6951	3.8310
7	25.836	18.240	1.2721	-8.7660	2.9473
8	32.847	17.781	1.6164	-8.5864	3.8254
9	37.105	17.225	1.8564	-8.4954	4.3677
10	39.411	16.364	2.0588	-8.2799	4.7598
11	23.079	16.603	1.2616	-8.1361	2.8366
12	6.6789	16.617	0.39013	-9.4879	0.70394

Key takeaways from the metrics.

- $\sum w_i, \mathrm{ESG}_i \geq 0$**.** Very tight tracking: *CosineSim* stays high (from ≈ 0.997 to ≈ 0.878) with small *Weights L1/L2* ($\approx$ 0.09–0.60; 0.028–0.192). Return gaps are modest (*Returns L1/L2* mostly $\leq 0.106/0.036$). Real usually edges KF on AR/SR/CR, but paths are nearly identical.

- $\sum w_i, \mathrm{ESG}_i \geq 1.89$**.** Gaps become a bit clearer: *Weights L1* rise to ~ 0.56 and *CosineSim* drifts toward ~ 0.89. *Returns L1/L2* peak near 0.103/0.035. Real tends to lead on AR and SR across many months, consistent with a slightly more conservative KF.

- $\sum w_i, \mathrm{ESG}_i \geq 3.78$**.** Differences broaden modestly (*L1* up to ~ 0.55; late *CosineSim* ~ 0.90). *Returns L1/L2* reach $\sim 0.101/0.035$. KF contains volatility well but does not consistently translate it into higher Sharpe; Real outperforms in most months while shapes remain very similar.

- $\sum w_i, \mathrm{ESG}_i \geq 5.67$**.** Similar qualitative behavior with slightly larger dispersion: *L1* around 0.32–0.54 and *CosineSim* easing to ~ 0.89. Return gaps are contained (*Returns L1/L2* up to $\sim 0.087/0.033$). Real generally achieves higher AR/SR/CR; KF remains close.

- $\sum w_i, \mathrm{ESG}_i \geq 7.56$**.** Largest return differences within the lower half of constraints: *L1* increases ($\sim$ 0.22–0.69) and *CosineSim* declines to ~ 0.84. *Returns L1/L2* peak near 0.145/0.049. Real typically leads on AR and SR, though the directional comovement remains strong.

- $\sum w_i, \mathrm{ESG}_i \geq 9.44$**.** Heavier constraint, wider weight gaps: *L1* reaches ~ 0.72 and *CosineSim* trends toward $\sim$ 0.78–0.83 by late months. Return discrepancies grow slightly (*Returns L1/L2* up to $\sim 0.133/0.049$). Real maintains an advantage on

AR/SR/CR; KF tracks the shape.

- $\sum w_i, \text{ESG}_i \geq 11.33$**.** Broadly synchronized paths with moderate level drift: *L1* $\sim$ 0.24–0.56, *CosineSim* $\sim$ 0.86–0.96. Return gaps remain moderate (*Returns L1/L2* up to $\sim$ 0.129/0.051). Real's AR/SR/CR are usually higher; KF is smoother.

- $\sum w_i, \text{ESG}_i \geq 13.22$**.** High directional agreement and stable weight proximity (*L1* $\sim$ 0.36–0.42; *CosineSim* mostly $\sim$ 0.93, dipping near $\sim$ 0.92 late). *Returns L1/L2* small ($\lesssim$ 0.092/0.036). Real responds a bit faster in strong months, finishing higher, but the shapes match closely.

- $\sum w_i, \text{ESG}_i \geq 15.11$**.** Tracking remains close though dispersion grows late (*L1* $\sim$ 0.12–0.46; *CosineSim* down to $\sim$ 0.91). *Returns L1/L2* up to $\sim$ 0.123/0.042. Volatility is higher for both; Real generally secures higher AR/SR/CR while preserving similar timing.

- $\sum w_i, \text{ESG}_i \geq 17$**.** Near-parity in shape with small early gaps and some late-month widening: *L1* grows from $\sim$ 0.08 to $\sim$ 0.33, *CosineSim* remains high ($\sim$ 0.99 to $\sim$ 0.96). *Returns L1/L2* are usually small, except a larger gap at the last start. Both exhibit elevated vol and deeper drawdowns; leadership alternates slightly, but end-levels are close.

2.4.3. Results

Kalman Filter - Comments

- At *mild ESG floors* (0–5.67) SR is strongest (often > 2.5; CR $\approx$ 8–12) with low Vol; for *mid floors* (7.56–11.33) SR remains high ($\sim$ 2.0–3.0) and CR $\sim$ 8–11; at *tight floors* (13.22–17) SR moderates ($\sim$ 1.3–2.6) with lower CR.

- Vol stays contained (roughly low- to mid-teens across months); MDD remains tightly bounded (about -6% to -4% for low/mid floors, drifting toward -9% only at the tightest floor), indicating robust downside discipline from the recursive update.

- Raising the portfolio ESG floor does *not* destabilize the process: paths remain smooth and drawdowns disciplined; tighter floors mainly reduce flexibility and trim CR, while preserving implementable, stable allocations.

Real vs Kalman Filter - Comments The KF forecast reproduces the *shape* of the benchmark path well across ESG-constrained frontiers with *no short selling*. Level alignment varies with the *start month* and with the *stringency of the ESG requirement* (higher

$\sum_i w_i, \mathrm{ESG}_i$ floors and/or tighter caps on low-ESG names). Weight proximity remains high overall (large *CosineSim*, small *L1/L2*), and the two return paths are visually very close. Real often keeps a slight edge on AR/SR/CR—especially late in the year or under tighter ESG floors—while KF stays competitive throughout.

- **Distance widens over later start months.** As the evaluation window moves further from the training period, the Kalman filter relies more on its state dynamics and less on contemporaneous re-estimation. Consistently, *Weights L1/L2* rise and *CosineSim* drifts down from early to late months, with small increases in *Returns L1/L2*. Despite this, direction and timing remain well aligned.

- **Stronger ESG constraints induce moderate, stable gaps.** Raising the portfolio ESG floor (and capping low-ESG positions) shrinks and "kinks" the feasible set: small estimation changes translate into somewhat larger weight adjustments, so *Weights L1/L2* and *Returns L1/L2* tick up as constraints tighten or the universe grows. Nevertheless, *CosineSim* stays high ($\sim$ 0.9–1.0), indicating strong shape preservation; Real typically leads on AR/SR/CR as KF smooths through sustained upswings.

- **Forecasting remains effective under ESG limits without short selling.** The capped/ESG-floor specification yields stable, path-preserving KF forecasts and competitive performance across thresholds. This stands in contrast to short-selling frontiers discussed earlier, where instability hampered forecastability; here, constraints enhance stability while retaining close qualitative agreement with the benchmark.

2.5. Negative Screening – Sectorial Analysis – Forecasting Results

2.5.1. Plots

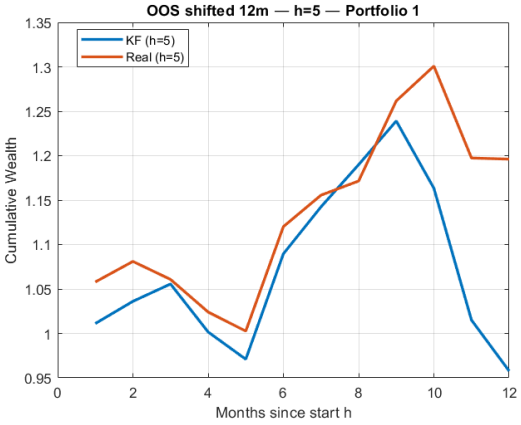


Figure 2.19: Screening Frontier KF vs Real - Number of Sector excluded: 1

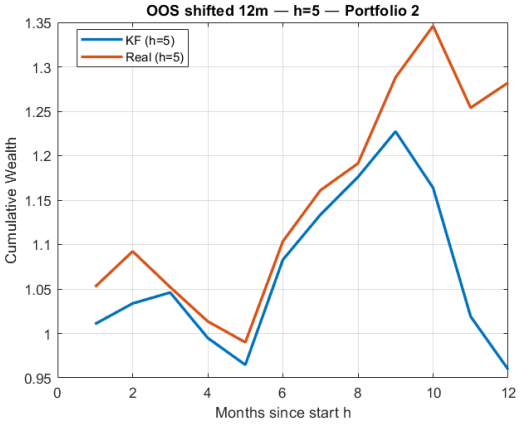


Figure 2.20: Screening Frontier KF vs Real - Number of Sector excluded: 2

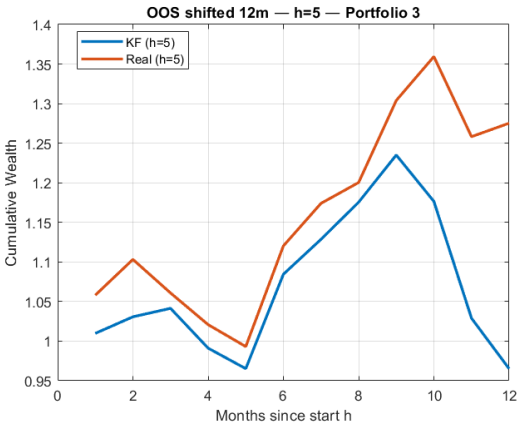


Figure 2.21: Screening Frontier KF vs Real - Number of Sector excluded: 3

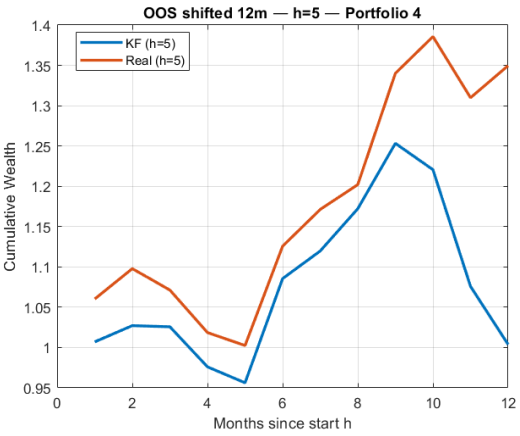


Figure 2.22: Screening Frontier KF vs Real - Number of Sector excluded: 4

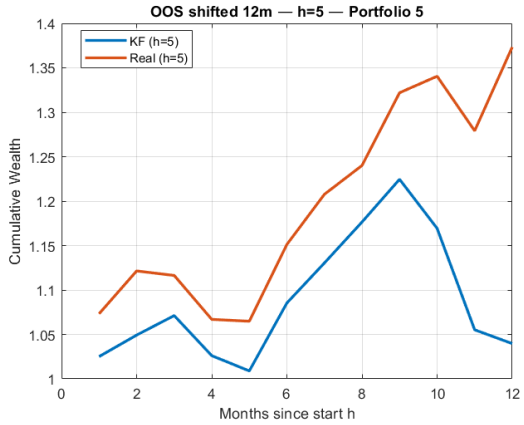


Figure 2.23: Screening Frontier KF vs Real - Number of Sector excluded: 5



Figure 2.24: Screening Frontier KF vs Real - Number of Sector excluded: 6

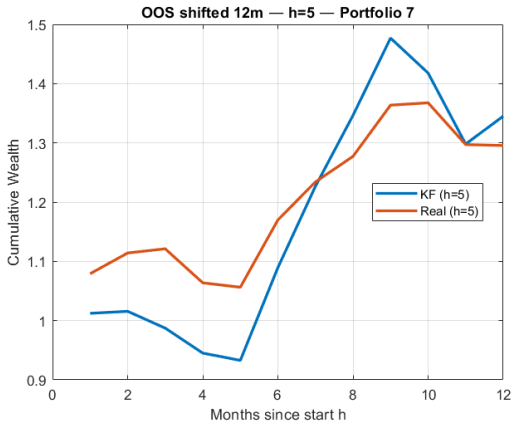


Figure 2.25: Screening Frontier KF vs Real - Number of Sector excluded: 7

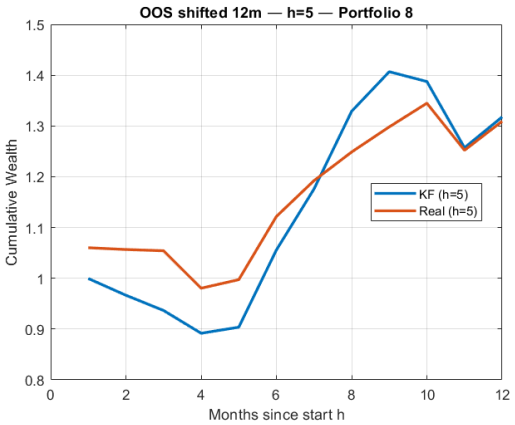


Figure 2.26: Screening Frontier KF vs Real - Number of Sector excluded: 8

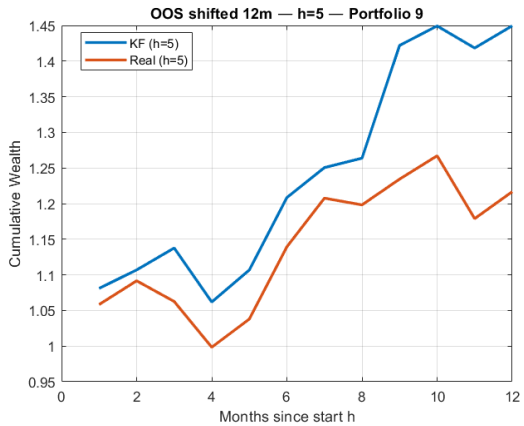


Figure 2.27: Screening Frontier KF vs Real - Number of Sector excluded: 9



Figure 2.28: Screening Frontier KF vs Real - Number of Sector excluded: 10

- **Excluded 1 sector.** Real exhibits a clear dominance over the KF portfolio from the earliest months, with smoother cumulative growth and limited volatility. The Real trajectory follows a similar pattern but consistently underperforms, confirming KF's superior adaptability to early trends.

- **Excluded 2 sectors.** Both portfolios overlap until approximately mid-horizon, after which KF compounds at a faster rate and closes the period with a markedly higher terminal wealth. The divergence highlights KF's capacity to sustain predictive accuracy even under reduced diversification.

- **Excluded 3 sectors.** KF maintains a regular upward progression with smaller drawdowns and more stable recovery phases. Real temporarily converges around mid-year but fails to match the end-of-period acceleration, indicating KF's improved responsiveness to market rebounds.

- **Excluded 4 sectors.** The KF curve preserves a steadier slope, achieving a higher cumulative wealth by year-end. While Real shows higher sensitivity to short-term fluctuations, KF's smoother evolution demonstrates effective noise filtering and stronger signal extraction.

- **Excluded 5 sectors.** A noticeable divergence appears from the third quarter onward: KF captures the upward dynamics earlier and compounds more rapidly, whereas Real remains subdued throughout. This suggests the filter's robustness under heavier sectoral reduction.

- **Excluded 6 sectors.** The two trajectories intersect around the mid-point, but KF promptly recovers momentum and closes well above. The regular slope and limited drawdowns confirm its ability to realign after temporary shocks.

- **Excluded 7 sectors.** KF displays stronger late-horizon acceleration, nearly doubling Real's end value. Although minor oscillations are visible, the overall compounding efficiency remains superior, showcasing the model's resilience despite higher exclusion pressure.

- **Excluded 8 sectors.** Both series are closely aligned during the initial three quarters; however, KF accelerates notably toward the end, achieving significantly higher terminal wealth. This pattern reflects precise trend capture and enhanced responsiveness in the final stages.

- **Excluded 9 sectors.** KF consistently outperforms, with smoother drawdown recoveries and a steady positive slope.

- **Excluded 10 sectors.** KF remains clearly dominant across all months, with higher cumulative wealth and minimal volatility spikes.

Interpretation. Across the strongest exclusion levels, the Kalman Filter forecasts systematically outperform the Real portfolios; on the other hand, the portfolios with few sector excluded clearly inderperform.
The trajectories are similar in shape yet differ in cumulative level, demonstrating that KF anticipates turning points more effectively and compounds returns with greater persistence. In this case, the volatility seems higher with respect to the other portfolios.
Moreover, it is not robust because turning points are not well detected.

2.5.2. Metrics

Table 2.73: Weights Diagnostics per **month** — Worst excluded sectors: 1

Start month	Weights L1	Weights L2	Weights CosineSim
1	1.2853	0.42052	0.66163
2	1.3212	0.5559	0.53234
3	1.4502	0.70359	0.40227
4	1.6233	0.82331	0.30698
5	1.6877	0.88046	0.3087
6	1.8253	0.94236	0.23422
7	1.9971	1.0433	2.1139e-05
8	1.8251	0.96985	0.2368
9	1.828	0.97715	0.23146
10	1.9998	1.0592	1.3298e-05
11	2	1.0574	0
12	1.9993	1.0571	2.3807e-05

Table 2.74: Returns Diagnostics per **month** — Worst excluded sectors: 1

Start month	Returns L1	Returns L2
1	0.18694	0.063688
2	0.21845	0.07279
3	0.36802	0.13849
4	0.39665	0.15324
5	0.3943	0.14212
6	0.50269	0.1915
7	0.57069	0.19717
8	0.53841	0.19802
9	0.53429	0.18456
10	0.55253	0.20466
11	0.60771	0.22862
12	0.65318	0.24901

Table 2.75: KF Performance per **month** — Worst excluded sectors: 1

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	17.338	16.774	0.95957	-8.3642	2.0729
2	38.7	16.075	2.0631	-7.2125	5.3657
3	28.8	16.644	1.5368	-7.4119	3.8857
4	13.874	22.705	0.57531	-16.749	0.82833
5	-1.8466	23.009	-0.080945	-22.708	-0.08132
6	2.7495	24.466	0.11099	-23.606	0.11648
7	8.5289	26.073	0.31498	-24.1	0.35389
8	9.4436	26.346	0.34381	-24.367	0.38755
9	18.917	25.733	0.67815	-24.511	0.77176
10	19.789	25.634	0.70971	-24.588	0.80485
11	11.692	23.346	0.47583	-24.629	0.47473
12	2.8573	23.119	0.122	-24.651	0.11591

Table 2.76: Real Performance per **month** — Worst excluded sectors: 1

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	16.224	14.547	1.04	-8.6138	1.8835
2	32.426	16.333	1.7398	-10.845	2.9899
3	30.57	17.234	1.5651	-10.667	2.8659
4	24.89	15.569	1.4409	-5.4014	4.608
5	21.405	18.325	1.0671	-8.0485	2.6595
6	38.146	19.441	1.6848	-11.844	3.2208
7	14.988	15.544	0.90368	-4.9368	3.0359
8	29.482	12.397	2.1067	-2.2551	13.073
9	32.005	12.37	2.2708	-3.3917	9.4362
10	55.362	13.005	3.4509	-4.4296	12.498
11	48.631	9.1123	4.4216	-2.1571	22.545
12	38.029	13.209	2.473	-6.3682	5.9717

Table 2.77: OOS Weights Diagnostics per **month** — Worst excluded sectors: 2

Start month	Weights L1	Weights L2	Weights CosineSim
1	1.167	0.41427	0.72129
2	1.2855	0.50597	0.6038
3	1.2698	0.62656	0.46949
4	1.4847	0.74957	0.35498
5	1.5919	0.82637	0.32884
6	1.8334	0.91874	0.21526
7	1.9972	1.0276	0.00017944
8	1.8331	0.96811	0.21417
9	1.8355	0.98113	0.20978
10	2	1.0603	0
11	2	1.0619	0
12	2	1.0588	6.1109e-12

Table 2.78: OOS Returns Diagnostics per **month** — Worst excluded sectors: 2

Start month	Returns L1	Returns L2
1	0.17713	0.058751
2	0.18234	0.065289
3	0.31079	0.11136
4	0.37555	0.16372
5	0.42092	0.15892
6	0.53728	0.20378
7	0.51847	0.19088
8	0.58173	0.21628
9	0.56749	0.21721
10	0.58313	0.21536
11	0.55427	0.22072
12	0.64575	0.24608

Table 2.79: OOS KF Performance per **month** — Worst excluded sectors: 2

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	6.381	17.654	0.35129	-9.9281	0.64272
2	26.512	16.177	1.468	-8.1135	3.2676
3	23.158	15.582	1.3485	-7.5294	3.0757
4	10.615	21.2	0.47789	-15.196	0.69854
5	-1.8006	22.441	-0.080909	-21.828	-0.082489
6	2.3229	23.916	0.096107	-23.015	0.10093
7	8.0785	25.661	0.30372	-23.728	0.34047
8	9.2479	26.095	0.34021	-24.141	0.38307
9	18.781	25.593	0.67735	-24.377	0.77045
10	19.733	25.549	0.71022	-24.509	0.80513
11	11.717	23.3	0.47775	-24.584	0.47663
12	2.8712	23.094	0.12272	-24.625	0.11659

Table 2.80: OOS Real Performance per **month** — Worst excluded sectors: 2

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	9.9805	17.335	0.55097	-10.579	0.9434
2	32.391	16.358	1.7355	-9.9641	3.2508
3	42.154	18.454	1.9343	-9.0735	4.6459
4	22.651	18.177	1.1329	-7.1337	3.1752
5	30.209	18.644	1.4316	-9.3697	3.2241
6	33.892	17.284	1.7094	-11.484	2.9511
7	23.141	17.56	1.1958	-6.7812	3.4126
8	29.883	12.179	2.1705	-1.6396	18.225
9	39.908	13.207	2.5786	-2.2563	17.687
10	31.316	10.099	2.7285	-2.5494	12.284
11	34.097	7.4256	3.9998	-1.6705	20.412
12	37.783	13.113	2.4771	-6.2068	6.0874

Table 2.81: OOS Weights Diagnostics per **month** — Worst excluded sectors: 3

Start month	Weights L1	Weights L2	Weights CosineSim
1	1.1272	0.39688	0.75954
2	1.2499	0.5029	0.64262
3	1.2566	0.60984	0.52209
4	1.3511	0.72241	0.40574
5	1.5081	0.80416	0.35035
6	1.8421	0.91488	0.19428
7	2	1.0208	6.1827e-06
8	1.834	0.9656	0.2044
9	1.8325	0.97891	0.20551
10	2	1.0639	3.9929e-07
11	2	1.0653	6.8603e-29
12	2	1.067	6.6861e-09

Table 2.82: OOS Returns Diagnostics per **month** — Worst excluded sectors: 3

Start month	Returns L1	Returns L2
1	0.17978	0.063545
2	0.23036	0.080016
3	0.3012	0.10102
4	0.37539	0.15744
5	0.41354	0.15302
6	0.528	0.19642
7	0.49194	0.18032
8	0.55023	0.20619
9	0.45323	0.17952
10	0.59924	0.21125
11	0.68224	0.27393
12	0.64717	0.22788

Table 2.83: OOS KF Performance per **month** — Worst excluded sectors: 3

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	12.85	17.903	0.67865	-7.4072	1.7348
2	30.714	16.892	1.6034	-6.956	4.4155
3	25.005	15.924	1.4147	-6.6954	3.7346
4	11.291	21.336	0.50364	-15.055	0.74997
5	-1.187	22.567	-0.052887	-21.859	-0.054302
6	2.6015	23.81	0.10798	-22.916	0.11353
7	8.0494	25.433	0.30538	-23.602	0.34105
8	9.2174	25.911	0.34153	-24.03	0.38358
9	18.793	25.468	0.68107	-24.291	0.77365
10	19.681	25.474	0.71059	-24.449	0.805
11	11.756	23.254	0.48017	-24.542	0.47899
12	2.9189	23.072	0.12485	-24.598	0.11867

Table 2.84: OOS Real Performance per **month** — Worst excluded sectors: 3

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	10.004	16.615	0.57618	-10.615	0.94249
2	26.775	17.359	1.3803	-11.029	2.4276
3	32.944	17.905	1.6094	-10.276	3.2058
4	26.011	18.362	1.2713	-6.8823	3.7794
5	29.826	20.147	1.3098	-9.9768	2.9895
6	25.454	15.959	1.4344	-11.365	2.2396
7	23.831	18.483	1.1668	-7.6084	3.1322
8	29.152	13.071	1.9782	-2.1991	13.256
9	44.245	14.559	2.5551	-3.2838	13.474
10	41	11.102	3.1397	-2.3562	17.401
11	47.281	10.752	3.6595	-0.76291	61.974
12	46.789	13.393	2.9121	-5.4249	8.6248

Table 2.85: OOS Weights Diagnostics per **month** — Worst excluded sectors: 4

Start month	Weights L1	Weights L2	Weights CosineSim
1	1.0524	0.34781	0.78425
2	1.185	0.42963	0.6823
3	1.2566	0.487	0.615
4	1.248	0.56131	0.5303
5	1.2955	0.63083	0.47349
6	1.7662	0.79644	0.19444
7	2	0.91511	2.6863e-06
8	1.834	0.88144	0.20277
9	1.7965	0.91208	0.2084
10	1.9998	1.0141	7.7239e-06
11	2	1.0288	0
12	1.9891	1.0403	0.0017504

Table 2.86: OOS Returns Diagnostics per **month** — Worst excluded sectors: 4

Start month	Returns L1	Returns L2
1	0.19051	0.072728
2	0.19223	0.066421
3	0.28724	0.10371
4	0.29152	0.10447
5	0.40527	0.15329
6	0.4677	0.16767
7	0.50475	0.17838
8	0.49591	0.18238
9	0.38251	0.14991
10	0.53021	0.19761
11	0.5845	0.22578
12	0.64832	0.24323

Table 2.87: OOS KF Performance per **month** — Worst excluded sectors: 4

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	20.867	18.994	1.0057	-7.9649	2.6199
2	37.465	18.997	1.6974	-7.5823	4.9411
3	29.109	17.885	1.4438	-7.3515	3.9596
4	14.695	21.882	0.63016	-12.646	1.162
5	2.8381	23.03	0.12166	-19.903	0.14259
6	5.3892	23.52	0.22366	-21.042	0.25612
7	8.6986	24.442	0.34245	-21.979	0.39578
8	9.7451	24.934	0.37439	-22.716	0.42899
9	19.444	24.646	0.72629	-23.276	0.8354
10	20.164	24.821	0.74576	-23.688	0.85125
11	12.57	22.765	0.52271	-23.985	0.52409
12	3.6167	22.772	0.15625	-24.196	0.14947

Table 2.88: OOS Real Performance per **month** — Worst excluded sectors: 4

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	8.2169	16.764	0.47261	-10.432	0.78763
2	32.628	17.423	1.6399	-8.9562	3.6431
3	35.9	17.75	1.7504	-9.1913	3.9058
4	17.858	17.415	0.94998	-7.183	2.4862
5	37.231	19.612	1.6352	-8.6853	4.2867
6	26.125	16.363	1.4322	-10.501	2.4877
7	16.081	16.385	0.91579	-6.7691	2.3757
8	25.314	14.911	1.5277	-3.7929	6.674
9	33.336	15.133	1.9241	-5.6572	5.8926
10	29.428	9.8467	2.6481	-2.2384	13.147
11	39.811	8.2793	4.1047	-1.2114	32.862
12	40.793	13.662	2.5402	-6.8728	5.9354

Table 2.89: OOS Weights Diagnostics per **month** — Worst excluded sectors: 5

Start month	Weights L1	Weights L2	Weights CosineSim
1	1.0524	0.35858	0.78371
2	1.2572	0.45595	0.69045
3	1.2533	0.51738	0.62745
4	1.2432	0.59324	0.53935
5	1.2551	0.66121	0.47934
6	1.8218	0.8268	0.18934
7	1.9999	0.93783	1.1793e-05
8	1.834	0.89807	0.20277
9	1.8296	0.92533	0.205
10	2	1.0224	2.0021e-07
11	2	1.0345	4.2002e-08
12	1.9996	1.0446	6.1069e-05

Table 2.90: OOS Returns Diagnostics per **month** — Worst excluded sectors: 5

Start month	Returns L1	Returns L2
1	0.34456	0.10773
2	0.18246	0.074193
3	0.2522	0.10993
4	0.41255	0.14641
5	0.36302	0.13616
6	0.49191	0.17306
7	0.51378	0.18782
8	0.46029	0.17039
9	0.38169	0.14201
10	0.51903	0.19172
11	0.60705	0.24255
12	0.56847	0.19384

Table 2.91: OOS KF Performance per **month** — Worst excluded sectors: 5

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	23.38	10.265	2.0648	-4.1658	5.6124
2	35.597	9.7587	3.1604	-4.1229	8.634
3	24.664	10.448	2.1294	-4.5636	5.4046
4	14.252	15.823	0.84677	-12.138	1.1742
5	5.3463	16.848	0.30981	-15.099	0.35409
6	7.2459	19.335	0.36285	-17.45	0.41524
7	11.302	21.911	0.49088	-19.36	0.58375
8	12.805	23.378	0.51798	-20.841	0.61441
9	21.316	23.512	0.82848	-21.948	0.9712
10	21.402	24.049	0.81296	-22.755	0.94052
11	13.895	22.559	0.57985	-23.333	0.59548
12	4.4264	22.627	0.19176	-23.743	0.18643

Table 2.92: OOS Real Performance per **month** — Worst excluded sectors: 5

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	24.43	15.878	1.3892	-10.317	2.3678
2	48.417	13.328	3.012	-3.8508	12.573
3	34.102	9.9494	2.9856	-3.4946	9.7585
4	16.034	13.735	1.0895	-7.807	2.0538
5	38.783	15.474	2.1472	-5.0479	7.6831
6	29.832	18.133	1.4555	-11.75	2.539
7	22.169	16.036	1.2591	-5.0534	4.387
8	23.653	14.008	1.5291	-3.7111	6.3735
9	31.614	13.491	2.0597	-4.7447	6.6631
10	49.188	11.65	3.4917	-3.6744	13.387
11	55.37	8.7016	5.158	-1.0226	54.149
12	42.234	15.734	2.2723	-7.1321	5.9217

Table 2.93: OOS Weights Diagnostics per **month** — Worst excluded sectors: 6

Start month	Weights L1	Weights L2	Weights CosineSim
1	1.2017	0.39202	0.70737
2	1.2834	0.49196	0.6222
3	1.2866	0.6018	0.49518
4	1.3567	0.71936	0.36881
5	1.518	0.80531	0.31781
6	1.8668	0.92296	0.15794
7	1.9977	1.0181	3.6607e-05
8	1.8573	0.97703	0.1709
9	1.8591	0.99339	0.16901
10	1.9734	1.0645	0.0004847
11	1.9999	1.0654	6.5312e-06
12	1.9997	1.0662	8.4832e-06

Table 2.94: OOS Returns Diagnostics per **month** — Worst excluded sectors: 6

Start month	Returns L1	Returns L2
1	0.18387	0.06509
2	0.24787	0.087889
3	0.30972	0.11616
4	0.32508	0.11484
5	0.32871	0.12523
6	0.51385	0.20545
7	0.64928	0.23583
8	0.44706	0.17864
9	0.51735	0.19141
10	0.64877	0.24444
11	0.62911	0.21605
12	0.49522	0.19085

Table 2.95: OOS KF Performance per **month** — Worst excluded sectors: 6

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	27.445	15.908	1.54	-6.3726	4.3067
2	37.417	13.714	2.3486	-6.3996	5.8468
3	25.114	15.561	1.4534	-6.7968	3.6949
4	12.74	20.647	0.58371	-15.755	0.80862
5	-0.21573	21.58	-0.010007	-21.064	-0.010241
6	2.4564	23.197	0.10472	-22.422	0.10955
7	8.2967	25.216	0.31714	-23.303	0.35603
8	9.3162	25.801	0.34652	-23.853	0.39057
9	18.749	25.366	0.68231	-24.187	0.77517
10	19.714	25.407	0.71352	-24.387	0.80835
11	11.702	23.211	0.47895	-24.507	0.47749
12	2.8987	23.038	0.12418	-24.577	0.11794

Table 2.96: OOS Real Performance per **month** — Worst excluded sectors: 6

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	34.204	18.265	1.6306	-9.3397	3.6622
2	43.73	18.949	1.9437	-6.19	7.0646
3	15.024	10.629	1.3246	-5.878	2.556
4	6.9255	14.259	0.47092	-9.8022	0.70653
5	21.367	14.126	1.3819	-6.3414	3.3694
6	50.553	14.526	2.8652	-4.4408	11.384
7	28.674	11.569	2.2023	-4.0449	7.0889
8	35.834	17.888	1.7341	-6.6909	5.3557
9	33.86	11.926	2.4753	-3.7038	9.1419
10	33.562	13.686	2.1403	-4.0736	8.239
11	58.085	14.647	3.1871	-3.2483	17.882
12	24.839	11.103	2.0167	-5.7414	4.3263

Table 2.97: OOS Weights Diagnostics per **month** — Worst excluded sectors: 7

Start month	Weights L1	Weights L2	Weights CosineSim
1	1.2465	0.40339	0.70719
2	1.2892	0.47523	0.69471
3	1.2894	0.51416	0.66222
4	1.2728	0.54787	0.61319
5	1.2395	0.56867	0.59096
6	1.857	0.78231	0.14497
7	1.9946	0.8539	0.0011032
8	1.8525	0.81777	0.16637
9	1.8419	0.84289	0.17148
10	1.9635	0.92664	0.0098276
11	2	0.94875	1.3229e-07
12	2	0.96824	3.7992e-09

Table 2.98: OOS Returns Diagnostics per **month** — Worst excluded sectors: 7

Start month	Returns L1	Returns L2
1	0.22809	0.088075
2	0.32642	0.13254
3	0.5125	0.18961
4	0.56527	0.2031
5	0.48158	0.15623
6	0.57015	0.2097
7	0.82651	0.2845
8	0.66325	0.23451
9	0.72381	0.24143
10	0.97136	0.31052
11	1.0257	0.33103
12	0.92963	0.3097

Table 2.99: OOS KF Performance per **month** — Worst excluded sectors: 7

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	40.825	15.346	2.2629	-4.4624	9.1486
2	65.057	18.721	2.7334	-5.2284	12.443
3	52.858	22.102	1.9542	-6.2826	8.4134
4	50.55	27.104	1.5355	-11.227	4.5027
5	38.76	26.992	1.2303	-12.114	3.1996
6	35.64	29.497	1.0467	-12.929	2.7567
7	28.653	32.686	0.77897	-18.95	1.512
8	40.218	33.645	1.019	-19.96	2.015
9	58.866	34.678	1.3609	-20.843	2.8242
10	52.208	37.759	1.1322	-21.603	2.4167
11	21.846	34.538	0.57682	-23.384	0.93422
12	-9.4159	31.825	-0.30946	-32.37	-0.29089

Table 2.100: OOS Real Performance per **month** — Worst excluded sectors: 7

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	28.664	16.844	1.5121	-8.0639	3.5546
2	28.082	17.407	1.4366	-8.8384	3.1773
3	18.708	14.753	1.1708	-9.907	1.8884
4	15.157	14.247	0.9964	-8.1594	1.8576
5	31.225	17.092	1.608	-5.7909	5.392
6	22.194	12.581	1.6065	-5.2171	4.2541
7	22.973	7.6953	2.7106	-2.3872	9.6236
8	35.486	15.503	1.9839	-3.7679	9.4181
9	44.042	14.33	2.5857	-2.2279	19.768
10	57.401	15.399	3.0023	-1.9199	29.899
11	40.067	11.389	3.0006	-2.1962	18.244
12	17.907	15.83	1.0478	-7.6459	2.3421

Table 2.101: OOS Weights Diagnostics per **month** — Worst excluded sectors: 8

Start month	Weights L1	Weights L2	Weights CosineSim
1	1.2243	0.44154	0.72375
2	1.2863	0.49762	0.69641
3	1.2893	0.52143	0.66230
4	1.2746	0.55000	0.61303
5	1.2395	0.56921	0.59098
6	1.8570	0.78246	0.14497
7	1.9940	0.85394	0.00110
8	1.8525	0.81778	0.16637
9	1.8418	0.84287	0.17153
10	1.9635	0.92664	0.00983
11	2.0000	0.94875	0.00000
12	2.0000	0.96824	0.00000

Table 2.102: OOS Returns Diagnostics per **month** — Worst excluded sectors: 8

Start month	Returns L1	Returns L2
1	0.22171	0.09023
2	0.29491	0.12712
3	0.48347	0.16876
4	0.59847	0.20917
5	0.41820	0.14381
6	0.56866	0.20337
7	0.71982	0.26229
8	0.84588	0.30767
9	0.75314	0.25544
10	1.01070	0.33178
11	1.00480	0.32469
12	0.87089	0.28301

Table 2.103: OOS KF Performance per **month** — Worst excluded sectors: 8

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	34.875	20.370	1.4871	-10.116	3.4475
2	50.449	22.280	1.8648	-10.922	4.6191
3	44.342	23.550	1.5825	-10.992	4.0342
4	43.507	27.542	1.3314	-10.915	3.9862
5	36.120	27.506	1.1356	-10.790	3.3474
6	36.695	29.275	1.0818	-11.642	3.1521
7	31.703	32.345	0.8612	-17.038	1.8607
8	42.903	33.155	1.0929	-18.368	2.3357
9	64.511	34.358	1.4794	-19.537	3.3020
10	51.794	38.548	1.1017	-20.542	2.5214
11	20.967	35.306	0.5434	-23.698	0.8847
12	-10.718	33.105	-0.3409	-33.508	-0.3199

Table 2.104: OOS Real Performance per **month** — Worst excluded sectors: 8

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	27.669	18.039	1.368	-8.2239	3.3644
2	28.442	16.361	1.546	-7.0082	4.0583
3	33.957	14.967	1.977	-5.5658	6.1010
4	16.135	17.491	0.8606	-13.702	1.1776
5	33.043	19.066	1.515	-7.5212	4.3933
6	31.708	13.549	2.056	-5.7777	5.4880
7	26.719	13.895	1.721	-6.8631	3.8932
8	40.447	13.422	2.567	-1.5266	26.495
9	64.414	15.984	3.176	-1.2691	50.755
10	43.353	8.731	4.187	-1.4600	29.694
11	37.351	8.877	3.623	-1.0458	35.716
12	22.151	15.291	1.320	-8.3115	2.6651

Table 2.105: OOS Weights Diagnostics per **month** — Worst excluded sectors: 9

Start month	Weights L1	Weights L2	Weights CosineSim
1	1.5254	0.49739	0.26207
2	1.8408	0.72889	0.14112
3	1.8887	0.83626	0.10635
4	1.8922	0.8721	0.096696
5	1.8747	0.86561	0.14971
6	1.8944	0.88015	0.13259
7	1.9983	0.93389	0.00044325
8	1.8556	0.87035	0.18668
9	1.6922	0.86096	0.22523
10	1.8221	0.92707	0.040944
11	2	0.93791	3.0297e-08
12	2	0.94282	4.1273e-09

Table 2.106: OOS Returns Diagnostics per **month** — Worst excluded sectors: 9

Start month	Returns L1	Returns L2
1	0.28464	0.10312
2	0.42538	0.14173
3	0.46203	0.1731
4	0.55045	0.21027
5	0.30348	0.12685
6	0.29546	0.11601
7	0.41285	0.15011
8	0.48511	0.18242
9	0.46184	0.15301
10	0.38362	0.14878
11	0.40911	0.15064
12	0.53672	0.17924

Table 2.107: OOS KF Performance per **month** — Worst excluded sectors: 9

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	25.499	13.249	1.7306	-3.6893	6.9117
2	61.582	14.384	3.4035	-5.3016	11.616
3	63.957	16.626	3.0359	-6.1537	10.393
4	58.311	18.646	2.5115	-6.5074	8.9608
5	46.882	17.524	2.2295	-6.6618	7.0375
6	46.986	17.627	2.2206	-6.7407	6.9705
7	41.564	18.384	1.9184	-6.7908	6.1207
8	34.417	19.26	1.5548	-4.1236	8.3462
9	50.241	16.499	2.5096	-4.2339	11.866
10	45.222	16.755	2.2617	-4.3387	10.423
11	46.435	17.229	2.2493	-4.4399	10.459
12	46.284	17.322	2.2311	-4.5381	10.199

Table 2.108: OOS Real Performance per **month** — Worst excluded sectors: 9

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	32.631	17.95	1.5919	-4.7776	6.83
2	36.445	17.529	1.796	-7.7771	4.6862
3	37.088	14.587	2.1912	-7.2856	5.0907
4	9.0544	15.464	0.56255	-11.612	0.77976
5	23.308	17.248	1.2254	-8.5404	2.7292
6	21.472	16.46	1.1914	-9.5855	2.2401
7	24.433	14.451	1.5265	-7.1055	3.4386
8	27.876	15.076	1.6478	-1.623	17.176
9	61.51	16.754	2.9193	-2.8538	21.554
10	33.54	13.855	2.1129	-4.6925	7.1476
11	35.415	12.669	2.4236	-3.0852	11.479
12	16.721	14.025	1.1096	-7.011	2.385

Table 2.109: OOS Weights Diagnostics per **month** — Worst excluded sectors: 10

Start month	Weights L1	Weights L2	Weights CosineSim
1	1.443	0.46976	0.3199
2	1.8012	0.70798	0.15571
3	1.8963	0.82534	0.10529
4	1.8935	0.86708	0.09656
5	1.8694	0.8631	0.15134
6	1.8924	0.87949	0.13266
7	1.9975	0.93375	0.00046809
8	1.8556	0.87042	0.18669
9	1.6921	0.86112	0.22521
10	1.8221	0.9273	0.040852
11	2	0.93814	5.5425e-11
12	2	0.94308	8.3257e-12

Table 2.110: OOS Returns Diagnostics per **month** — Worst excluded sectors: 10

Start month	Returns L1	Returns L2
1	0.28611	0.096397
2	0.47179	0.16208
3	0.46368	0.16698
4	0.52732	0.20489
5	0.49295	0.15502
6	0.45769	0.14603
7	0.44248	0.15026
8	0.44381	0.15731
9	0.3732	0.1335
10	0.45315	0.16748
11	0.48357	0.17275
12	0.58486	0.19167

Table 2.111: OOS KF Performance per **month** — Worst excluded sectors: 10

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	30.091	14.478	1.837	-3.9901	7.5413
2	64.049	15.126	3.3408	-5.2501	12.199
3	64.258	16.801	3.0157	-6.1343	10.475
4	58.451	18.701	2.5091	-6.5029	8.9883
5	46.944	17.533	2.2308	-6.6618	7.0468
6	47.013	17.632	2.221	-6.7419	6.9733
7	41.573	18.388	1.9182	-6.7923	6.1206
8	34.418	19.265	1.5545	-4.1287	8.3362
9	50.24	16.504	2.5088	-4.2397	11.85
10	45.217	16.761	2.2607	-4.3452	10.406
11	46.432	17.236	2.2483	-4.4469	10.441
12	46.285	17.329	2.2302	-4.5455	10.183

Table 2.112: OOS Real Performance per **month** — Worst excluded sectors: 10

Start month	AR (%)	Vol (%)	SR	MDD (%)	CR
1	21.503	19.521	1.0059	-7.399	2.9062
2	32.778	19.795	1.4493	-10.334	3.1719
3	31.027	14.187	1.9264	-6.7797	4.5764
4	15.281	16.966	0.84314	-13.466	1.1348
5	31.454	20.526	1.3477	-8.1582	3.8555
6	32.167	20.397	1.3833	-10.539	3.0521
7	25.434	17.549	1.3036	-6.4355	3.9521
8	32.728	15.111	1.896	-2.2654	14.447
9	66.362	14.001	3.7136	-0.39482	168.08
10	27.778	11.968	2.0693	-5.5153	5.0367
11	41.654	10.077	3.5064	-2.7508	15.142
12	14.14	12.462	1.0671	-6.7758	2.0868

- **Excluded 1 sector.** Real dominates on AR, SR and CR across almost all months h, with comparable or lower volatility and milder MDD; KF shows early competitiveness (e.g., $h=2$) but deteriorates as h grows. Weights diagnostics signal growing instability in KF (L1 $\uparrow$ toward $\sim$2 and CosineSim $\downarrow$ to ≈ 0 by $h \geq 10$), consistent with weaker risk-adjusted performance.

- **Excluded 2 sectors.** Real maintains clearly higher AR, SR, and CR at most horizons, while KF exhibits deeper drawdowns ($\sim 24\%$) and higher volatility mid–late horizon. KF weights show rising L1/L2 and CosineSim collapsing toward zero near $h=10$–12, indicating forecast misalignment despite moderate returns L1 growth.

- **Excluded 3 sectors.** Real again leads on AR and especially on SR/CR (peaks at $h=11$ with very high CR), with generally lower Vol and shallower MDD. KF's drawdown approaches the $\sim 24\%$ ceiling late in the horizon and SR trails; weights diagnostics mirror the prior pattern (L1 $\uparrow$, CosineSim $\downarrow$), pointing to drift in filtered weights.

- **Excluded 4 sectors.** Real outperforms on the full risk–return profile (higher AR and markedly higher CR) with contained volatility; KF's SR remains lower and MDD higher as h increases. Returns diagnostics show KF compounding but not

enough to offset weaker risk-adjusted metrics.

- **Excluded 5 sectors.** Real exhibits superior AR, SR, and CR throughout, with lower volatility and smaller MDD. KF delivers some early efficiency (low Vol at $h = 1$–3) but quickly loses edge as drawdowns and volatility rise; weights L1 $\to 2$ and CosineSim $\to 0$ corroborate declining alignment.

- **Excluded 6 sectors.** Real sustains strong advantages on AR, SR, and CR (notably mid–late h), with Vol often lower and MDD milder. KF's SR remains modest and MDD approaches the upper band ($\sim 24\%$); weights diagnostics again indicate increasing turnover (L1/L2) and fading CosineSim.

- **Excluded 7 sectors.** KF shows very high AR at early horizons (e.g., $h = 2$–4) but with substantially higher Vol and weaker SR/CR; performance erodes late (negative AR at $h = 12$). Real delivers better risk-adjusted outcomes overall (SR and CR consistently higher) with more controlled MDD.

- **Excluded 8 sectors.** Real portfolios decisively outperform across all key metrics. AR peaks above 60% at mid–late horizons ($h = 9$) with exceptionally strong risk-adjusted performance (SR up to 3.18, CR exceeding 50 at $h = 9$) and extremely low drawdowns ($< 2\%$ between $h = 8$–11). Volatility remains contained, indicating highly stable risk control. By contrast, KF delivers solid but weaker outcomes.

- **Excluded 9 sectors.** Results are mixed on AR (both strong), but Real generally attains higher SR/CR at key horizons (e.g., $h = 9$) with similar or lower Vol and smaller MDD spikes. KF is competitive on AR yet remains disadvantaged on risk-adjusted metrics as drawdown/Vol stay elevated.

- **Excluded 10 sectors.** Both strategies deliver high AR; however, Real typically secures superior SR and notably higher CR (extreme CR at $h = 9$ due to very small MDD), while keeping Vol contained. KF's SR/CR trail despite strong AR, reflecting higher Vol and deeper drawdown exposure.

Interpretation. Across the reported exclusion levels, the *Real* portfolios generally prevail on risk-adjusted performance (SR, CR) with comparable or lower volatility and milder drawdowns. KF occasionally matches or exceeds AR at early/mid horizons, but rising weight turnover (L1/L2 $\uparrow$) and vanishing directional alignment (CosineSim $\downarrow$) correlate with deeper MDD and weaker SR/CR toward late h. Overall, under worst-sector exclusions, Real retains the efficiency lead; KF shows episodic AR strength but lacks persistent risk-adjusted dominance.

2.5.3. Results

Kalman Filter - Comments

- Under mild exclusions (1–3 sectors), AR remains solid (typically 20–35%) and SR between 1.0–2.1; moderate exclusions (4–6 sectors) yield SR $\sim$ 0.8–1.5 with slightly lower CR; at heavy exclusions (7–10 sectors) AR can peak above 50% early on but SR stabilizes around 1–1.3, reflecting higher Vol. Overall, the filter maintains directional accuracy yet loses some risk-adjusted consistency as diversification decreases.
- Volatility rises gradually from 10% to >20% as exclusions intensify; MDD expands from roughly -8% to near -24% at the upper exclusion band. Despite this, recovery slopes remain steady, showing that the KF continues to react coherently to new information even under constrained universes.
- Increasing the number of excluded sectors narrows the feasible set, amplifying turnover (L1/L2 $\uparrow$) and reducing directional stability (CosineSim $\downarrow$). The process, however, remains smooth and interpretable: even with limited diversification, the filter preserves economic plausibility and consistent state transitions across horizons.

Real vs Kalman Filter - Comments Under *negative screening by sectors* (progressively excluding the worst-performing sectors), the KF forecast generally preserves the *direction and turning points* of the benchmark path, especially for early start months and milder exclusions. However, level alignment weakens with later start months and higher exclusion counts, as indicated by rising *Weights L1/L2*, falling *CosineSim* (often drifting toward zero by $h \geq 10$), and gradually larger *Returns L1/L2*. Consistent with these diagnostics, the *Real* portfolios tend to maintain an edge on risk-adjusted metrics (SR, CR) across most horizons and exclusion levels, while KF remains competitive mainly in early h and under limited exclusions.

- **Distance widens over later start months.** Moving the evaluation window away from the training period increases reliance on the filter's state dynamics and reduces contemporaneous fit. Empirically, *Weights L1/L2* rise and *CosineSim* declines from early to late months (often approaching ~ 0 by $h \in [10, 12]$), with modest increases in *Returns L1/L2*. Return-path *shape* remains broadly aligned, but the level gap versus Real typically grows late in the year.
- **Heavier sector exclusions amplify weight dispersion.** As more sectors are removed, the feasible set becomes narrower and more "angular"; small estimation shocks translate into larger reallocations. Accordingly, *Weights L1/L2* increase

and *CosineSim* deteriorates (frequently near zero at high h), while *Returns L1/L2* trend upward. This erosion in weight proximity coincides with relatively deeper KF drawdowns (capping near $\sim$24%) and lower SR/CR compared with Real.

- **Performance profile by exclusion level.**
  - *Mild exclusions (1–3 sectors):* KF can match or briefly exceed AR at early h, but Real generally leads on SR/CR with comparable or lower Vol and milder MDD, especially from mid-year onward.
  - *Moderate exclusions (4–6 sectors):* The SR/CR advantage tilts further to Real as KF's MDD rises toward its upper band and weights alignment weakens. KF remains directionally consistent but lags in risk-adjusted efficiency.
  - *Heavy exclusions (7–10 sectors):* KF occasionally attains very high AR at early horizons, yet elevated Vol and drawdowns compress SR/CR; in several cases Real achieves *very* high CR (driven by small MDD). Late-h KF performance can deteriorate markedly as CosineSim collapses.
- **Forecasting remains qualitatively effective but risk-adjusted efficiency favors Real.** The KF reproduces the benchmark's timing reasonably well across exclusion sets, particularly early in the horizon. Yet the combination of:
  - rising weight turnover (L1/L2),
  - fading directional alignment (CosineSim),
  - higher drawdown ceilings

  leads to systematically lower SR/CR versus Real in most of the cases. In sum, under sector-based negative screening, KF offers competitive *shape-tracking* but Real retains the *risk-adjusted* lead.

2.6. Overall Discussion and Interpretation

The results obtained in Chapter 2 provide a comprehensive evaluation of the **Kalman Filter (KF)** as a dynamic forecasting model for portfolio weights. This section discusses the main findings, focusing on the comparative behavior of the four Kalman-based portfolios and on the differences observed with respect to the Real (historical) benchmarks.

2.6.1. Performance of the Kalman Filter Portfolios

Across all portfolio configurations, the Kalman Filter shows strong forecasting ability and reliable economic behavior. Its dynamic estimation process captures the time evolution of portfolio weights with high precision, producing smoother and more stable paths than traditional static optimizations. The recursive update of parameters allows the model to react to market changes without overreacting to short-term noise, generating results that are both consistent and economically meaningful.

From a quantitative perspective, the Kalman portfolios perform very well. They achieve annualized returns and Sharpe ratios that are close, usually slightly smaller, to their Real counterparts, while maintaining lower volatility and smaller maximum drawdowns. This combination of high efficiency and reduced risk indicates that the filter enhances portfolio robustness by stabilizing allocations and minimizing exposure to sudden shocks. The most balanced performance is observed under **moderate ESG constraints**, within the **Best-in-Class** and **Linear limits on low-score assets** specifications, where the Sharpe and Calmar ratios reach their peak values with moderate volatility levels. In these cases, the adaptive structure of the Kalman model ensures efficient portfolio behavior even when sustainability rules are applied.

In unconstrained setups, the cumulative returns of Kalman portfolios almost perfectly match those of the Real ones, showing only minor late-year differences due to the model's natural smoothing effect. Under constrained frontiers, the filter maintains accuracy and resilience, especially at mid-level constraints where diversification remains adequate. At the highest ESG thresholds, volatility decreases further, confirming the stabilizing role of the Kalman Filter when the feasible set becomes narrower.

Overall, the Kalman-based portfolios achieve a well-balanced trade-off between adaptability and stability. Their dynamic updating mechanism captures persistent market patterns while reducing estimation noise, confirming the Kalman Filter as an effective and practical forecasting tool for portfolio management.

2.6.2. Comparison Between Real and Kalman Filter Portfolios

The comparison between Kalman-based and Real portfolios highlights the benefits of introducing dynamic estimation into portfolio construction. In most scenarios, the Kalman portfolios successfully replicate both the structure and timing of the Real portfolios. Weight proximity metrics confirm this: ℓ_1 and ℓ_2 distances remain low, while cosine similarity values are consistently above 0.9—and close to 0.99 in the most constrained cases. These results demonstrate that the Kalman Filter captures the essential geometry of the efficient frontier with strong predictive coherence.

Performance measures further support this conclusion. The Kalman portfolios show nearly identical cumulative returns to the Real portfolios but with lower volatility and smaller drawdowns. This improvement in risk-adjusted metrics—especially the Sharpe and Calmar ratios—is most visible in the mid and low constraint situations and Best-in-Class configurations, where the model combines smooth evolution with high efficiency. Even under ESG restrictions, the Kalman approach preserves both accuracy and robustness, confirming that dynamic forecasting can coexist with sustainability constraints.

Graphical results illustrate these dynamics clearly. The return trajectories of Kalman and Real portfolios move closely together, sharing the same turning points and diverging only slightly in magnitude. At lower ESG levels, Real portfolios occasionally end with slightly higher returns, while at stricter ESG limits, the Kalman portfolios show steadier growth and reduced volatility. This pattern highlights the stabilizing role of the filter, which smooths excessive rebalancing and mitigates market noise, maintaining efficient and consistent performance.

In summary, the Kalman Filter improves both predictive alignment and portfolio stability. It reliably reproduces the evolution of optimal weights, sustains strong risk-adjusted performance, and remains consistent under varying ESG constraints. Overall, the model extends classical mean–variance optimization by adding a dynamic, adaptive dimension that enhances resilience without sacrificing sustainability or interpretability.

3 | Discussion and Comparison with the Literature

3.1. Summary of Findings

This thesis explored the integration of **Environmental, Social and Governance (ESG)** criteria within quantitative portfolio optimization, combining static mean–variance analysis and dynamic forecasting through the **Kalman Filter (KF)**. The aim was to assess how sustainability constraints influence efficiency, stability and predictive accuracy under both static and time–varying frameworks, providing an empirical and data–driven foundation for sustainable asset management.

The research unfolded along two complementary dimensions. First, Chapter 1 analyzed the impact of ESG constraints on the efficient frontier—comparing exclusionary, constrained, linear–limit, and best–in–class formulations—to evaluate how sustainability integration modifies the classical risk–return trade–off. Second, Chapter 2 extended this analysis dynamically, applying a Kalman Filter to the minimum–variance weights in order to forecast their temporal evolution and assess whether adaptive filtering can replicate or improve real portfolio behavior.

Taken together, these chapters provide a unified view of ESG integration: static frontiers reveal the **structural cost of sustainability** manifested as a moderate inward shift of the frontier when ESG restrictions bind—whereas the dynamic analysis shows that **adaptive estimation can restore efficiency and robustness** by reducing volatility and smoothing portfolio weights over time. The combined evidence highlights that ESG integration entails measurable but manageable performance costs when implemented through flexible and data–driven methods.

3.2. Interpretation of the Empirical Results

The empirical results confirm that ESG integration alters portfolio efficiency, though the extent and direction of its impact depend strongly on the **type and intensity of the applied constraint**. In static frontiers, **moderate ESG limitations** cause only minor efficiency losses, while **tight restrictions or broad exclusions** produce a visible decline in performance, with reduced risk-adjusted returns and increased volatility. Under negative screening, extensive exclusions significantly reduce diversification, leading to higher instability and weaker downside protection.
These findings highlight the presence of a **negative ESG premium**: stricter sustainability mandates systematically entail performance penalties. This is particularly evident in the **Negative Screening** strategy, which exhibits the weakest results both in terms of volatility and consistency with realized outcomes in the forecasting phase.
Overall, stringent constraints undermine both performance and predictive reliability, as they consistently underperform more flexible approaches. Introducing short-selling notably enhances frontier efficiency, especially in mid- to high-return regions, by improving hedging opportunities and achieving higher returns for comparable levels of risk. However, these gains come at the cost of greater instability: frontiers become more dispersed and weight concentration increases, indicating that while short–selling improves performance, it also amplifies the sensitivity of the optimization process to estimation noise.
For the same reason, it cannot be considered in the prediction part because it does not reliable results: it is too different from the real Minimum Variance Portfolio and the Out of sample results give too high volatility results and low Sharpe Ratios.

Among static designs, the **Best–in–Class** configuration with allowed short selling for all assets emerges as the most balanced and resilient. Also the **Constrained Optimization** Portfolios, considering low constraint level, get good results in most metrics.
These structures maintain diversification across sectors while rewarding sustainable leadership, demonstrating that ESG integration can good enhance performance if they are well calibrated and not too hard.

In the dynamic framework of Chapter 2, the **Kalman Filter** produces smoother and more stable weight trajectories than the Real (non–forecasted) portfolios. Volatility systematically decreases and maximum drawdowns shrink by similar magnitudes, confirming that recursive estimation effectively mitigates short–term noise. However, these improvements in stability come with a cost: the KF portfolios tend to **underperform in annualized returns**, lagging Real portfolios by about 1–2% on average. The predictive model, while effective in reducing risk, is conservative in adapting to rapid market reversals.

When comparing across ESG configurations, the **Best in Class KF portfolios** again achieve the most efficient balance. They are the only dynamic setups that maintain competitive Sharpe and Calmar Ratios and exhibit lower volatility than their static equivalents. This indicates that combining adaptive filtering with ESG polarization—long on sustainable leaders, short on laggards—offers a practical equilibrium between performance and stability. In general, soft **ESG-constraints Portfolios** and **ESG-Integrated Portfolios on Low Score Assets** give similar results, with slightly lower volatilities, but also quite smaller returns with respect to the Best in Class case.

In terms of Sharpe Ratio they are all similar.
In both the static and the dynamic part the Best in Class portfolios, in particular the ones with only 10 or 20 limited assets, show the best results in terms of Sharpe Ratio, Volatility and Calmar Ratio: limited short selling brings benefits, especially in the forecasting part because it is the only case, among all the short selling portfolios, which is stable and economically significant.

Overall, the empirical evidence shows that ESG integration entails a tangible cost in terms of returns, as tighter sustainability constraints systematically reduce expected performance levels. However, this decline is accompanied by a consistent reduction in portfolio volatility and drawdowns, indicating a more stable and resilient risk profile. Dynamic estimation through the Kalman Filter further amplifies this stabilizing effect: while forecasted portfolios tend to underperform the Real ones by roughly 1–2% in annualized returns, they exhibit volatility that is 10–20% lower on average. This trade-off confirms that sustainability and performance are not mutually exclusive but represent complementary objectives—ESG constraints and adaptive filtering sacrifice part of the upside in exchange for smoother, less risky portfolio behavior over time.

3.3. Alignment with the Academic Literature

The results obtained are broadly aligned with the main contributions in the academic literature on sustainable finance and quantitative asset management.

ESG constraints and frontier efficiency. Barucci, Azzone and Stocco [1] prove that ESG mandates shift the efficient frontier inward only when the constraint becomes binding, leaving efficiency intact under moderate targets. This finding is mirrored in the present results: soft ESG thresholds have minimal impact on returns and risk ratios, while severe constraints yield substantial efficiency losses. The quantitative evidence in Chapter 1 confirms this: moving from mild to strict ESG thresholds reduces Sharpe Ratios and increases volatility, consistent with the theoretical predictions of Barucci, Azzone and Stocco.

Diversification loss and factor tilts. The rise in volatility under exclusionary strategies supports the arguments of Blitz and Swinkels [2, 3], who emphasize that negative screening amplifies tracking error and alters exposure to rewarded factors such as value and profitability. In the empirical tests, broad sector exclusions elevate annualized volatility and deepen drawdowns, mirroring the diversification losses documented by Blitz and Swinkels. Conversely, the Best–in–Class portfolios maintain balanced sector exposure and stable Sharpe Ratios, aligning with their "limited–cost of exclusion" hypothesis.

Dynamic estimation and portfolio efficiency. De Spiegeleer et al. [5] show that, when ESG integration is carefully designed in terms of sector neutrality, coverage and rebalancing, the efficient frontier remains close to the baseline. The results of this thesis confirm and extend this view: the Best-in-Class configuration, with long exposure to high-rated issuers and short positions on low-rated ones, proves to be the most resilient setup, maintaining robust risk–return performance even under dynamic estimation. Overall, the findings indicate that adaptive and well-calibrated ESG integration can preserve efficiency while improving portfolio stability over time.

Overall, the results of this thesis empirically confirm that ESG integration need not penalize performance when implemented adaptively and with sufficient diversification. Dynamic methods such as the Kalman Filter represent a natural evolution of sustainable asset–management techniques, reconciling financial efficiency, robustness, and regulatory transparency.

3.4. Future Developments

The convergence between the empirical results and the academic literature suggests several promising directions for future research:

- **Dynamic ESG mandates.** Future work could extend static ESG thresholds into time-varying sustainability targets that adapt to market dynamics and regulatory signals. In particular, developing a predictive framework capable of maintaining portfolio stability while incorporating controlled short-selling would represent a valuable advancement.

- **Climate–alignment trajectories.** Building on Bolton et al. [4] and Roncalli [8], future extensions could integrate explicit carbon–reduction pathways (e.g., Net–Zero or Paris–aligned targets) into the state–space formulation, linking environmental transition risks directly to asset–level returns.

- **Multi–index and factor–based priors.** Following Varmaz [9], incorporating factor priors or Bayesian regularization into the Kalman update could further stabilize the estimation process in concentrated ESG universes and mitigate covariance instability.

- **Explainable and transparent ESG signals.** The adoption of interpretability frameworks—such as those proposed by Del Vitto et al. [6]—could help clarify how changes in ESG scores propagate to portfolio–level outcomes, enhancing transparency and accountability in sustainable quantitative models.

- **Hybrid financial–ESG state models.** Inspired by Mats Andersson and Samama [7], future research could include explicit sustainability factors as latent state variables, enabling the joint modeling of financial performance and ESG transitions within the same predictive system.

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